Fokker–Planck equation: worked examples

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A running log of exactly solvable one-dimensional Fokker–Planck problems: setup, spectrum, propagator, long-time behaviour, first-passage quantities and figures. Notation follows Gardiner's *Handbook of Stochastic Methods*, 2nd ed. [1]; [G (5.2.100)] means equation (5.2.100) there. The figures are made by `scripts/make_figures.py`, which also checks every boxed formula numerically (Appendix A). The same examples can be run interactively in `sde-explorer`.

§	Domain, walls	Q_λ (backward)	λ	$t \to \infty$
Free diffusion, $A = 0$, $B = D$				
2	$(-\infty, \infty)$	e^{ikx}, $k \in \mathbb{R}$	$Dk^2/2$	Gaussian, width $\sqrt{Dt}$
3	$(0, \infty)$, refl. 0	$\cos kx$, $k \geq 0$	$Dk^2/2$	half-Gaussian, width $\sqrt{Dt}$
4	$(0, \infty)$, abs. 0	$\sin kx$, $k > 0$	$Dk^2/2$	$S \sim t^{-1/2}$; no QSD
5	$(0, L)$, reflecting	$\cos(n\pi x/L)$, $n \geq 0$	$n^2\pi^2 D/2L^2$	$p_s = 1/L$
6	$(0, L)$, absorbing	$\sin(n\pi x/L)$, $n \geq 1$	$n^2\pi^2 D/2L^2$	QSD $\propto \sin(\pi x/L)$
7	$(0, L)$, refl. 0, abs. L	$\cos \frac{(2n+1)\pi x}{2L}$, $n \geq 0$	$(2n+1)^2\pi^2 D/8L^2$	QSD $\propto \cos(\pi x/2L)$
8	circle $\mathbb{R}/L\mathbb{Z}$	$e^{\pm iq_m x}$, $q_m = 2\pi m/L$	$Dq_m^2/2$, doublets	$p_s = 1/L$
Constant drift, $A = v$, $B = D$; $\alpha = v/D$, $Q_\lambda = e^{-\alpha x}u_n$				
11	$(-\infty, \infty)$	$u = e^{ikx}$	$\frac{v^2}{2D} + \frac{Dk^2}{2}$	$g(x - x_0 - vt)$
12	$(0, \infty)$, refl. 0	$u = \cos kx + \frac{\alpha}{k}\sin kx$	$\{0\} \cup [\frac{v^2}{2D}, \infty)$ if $v < 0$	$p_s \propto e^{-2\lvert v\rvert x/D}$ if $v < 0$
13	$(0, \infty)$, abs. 0	$u = \sin kx$	$\frac{v^2}{2D} + \frac{Dk^2}{2}$	$v < 0$: QSD continuum
14	$(0, L)$, reflecting	$u_n = \cos k_n x + \frac{\alpha}{k_n}\sin k_n x$	$\frac{v^2}{2D} + \frac{n^2\pi^2 D}{2L^2}$, and 0	$p_s \propto e^{2vx/D}$
15	$(0, L)$, absorbing	$u_n = \sin k_n x$	$\frac{v^2}{2D} + \frac{n^2\pi^2 D}{2L^2}$	QSD $\propto e^{\alpha x}\sin k_1 x$
16	$(0, L)$, refl. 0, abs. L	$\cos k_n x + \frac{\alpha}{k_n}\sin k_n x$	$\frac{v^2}{2D} + \frac{Dk_n^2}{2}$, $\tan k_n L = -\frac{k_n}{\alpha}$	QSD; Arrhenius
17	circle	$e^{iq_m x}$, $m \in \mathbb{Z}$ (no gauge)	$\frac{Dq_m^2}{2} + ivq_m$	$p_s = 1/L$, $J_s = v/L$
Ornstein–Uhlenbeck, $A = -kx$, $B = D$; $\ell = \sqrt{D/k}$, $\xi = x/\ell$, walls measured from the bottom				
20	$(-\infty, \infty)$	$H_n(\xi)$	nk	$p_s \propto e^{-\xi^2}$, gap k
21	$(0, \infty)$, refl. 0	even H_n	$2mk$	half-Gaussian, gap $2k$
22	$(0, \infty)$, abs. 0	odd H_n	$(2m+1)k$	QSDs $\lambda \in (0, k]$; Rayleigh
23	$(0, L)$, reflecting	$M(-\frac{\nu}{2}, \frac{1}{2}, \xi^2)$	$\nu_n k$	truncated half-Gaussian
24	$(0, L)$, absorbing	$\xi M(\frac{1-\nu}{2}, \frac{3}{2}, \xi^2)$	$\nu_n k$	QSD: sine $\to$ Rayleigh
25	$(0, L)$, refl. 0, abs. L	$M(-\frac{\nu}{2}, \frac{1}{2}, \xi^2)$	$\nu_n k$	Kramers escape
26	circle, sawtooth A	sectors of §23, §24	$\nu_n^{\mathrm{A}} k$, $(\nu_n^{\mathrm{A}} + 1)k$	cusp; Lifson–Jackson
27	circle, $U \propto -\cos\frac{x}{R}$	Fourier recurrence	doublets, $\frac{m^2 k}{\kappa} \to nk$	von Mises; tilt: NESS
More solvable potentials, $A = -U'$, $B = D$ (Table 8); arbitrary potentials				
30	$x > 0$, $\frac{1}{2}kx^2 - c\ln x$	Laguerre $L_n^{(d/2-1)}$	$2nk$	chi; $d = 2$ Rayleigh
31	line, $a\ln\cosh\frac{x}{w}$	Jacobi	$\frac{D}{2w^2}n(2s-n)$, $n < s$	sech^{2s}; continuum
32	$\lvert x\rvert < \frac{\pi w}{2}$, $-a\ln\cos\frac{x}{w}$	Gegenbauer	$nk + \frac{n^2\pi^2 D}{2L^2}$	$\cos^{2s}$
35	line, double well	numerical	$\lambda_1 \simeq$ Kramers	bimodal; two-state
Interfaces and singular boundaries				
36	line, B jumps at 0	cos on each side	$\frac{1}{2}D_\pm k^2$	box: $p_s^+/p_s^- = K$
37	$y > 0$, $B = 4Dy$	Laguerre	$2nk$; $(2n+2-d)k$	gamma; exit if $d \leq 0$
38	line, $A = x^2 + I$	$Q_0 = 1$ only	—	current through ∞

Table 1: The examples in these notes, in section order. The third column gives Gardiner's backward eigenfunctions Q_λ, up to normalisation; the forward ones are $P_\lambda = p_s Q_\lambda$, except on the circle with a current and through infinity. Each section's box gives both, normalised (Section 1.4).

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1 Conventions and toolkit

1.1 Equation, current, boundary conditions

The forward equation and probability current are

$$\partial_t p(x,t) = -\partial_x\big[A(x)\,p\big] + \tfrac{1}{2}\,\partial_x^2\big[B(x)\,p\big] = -\partial_x J, \qquad J(x,t) = A\,p - \tfrac{1}{2}\,\partial_x(B\,p), \tag{1.1}$$

corresponding to the Itô SDE $\mathrm{d}x = A\,\mathrm{d}t + \sqrt{B}\,\mathrm{d}W$ [G (5.2.1)], [G (5.2.8)]. At a wall $x = a$ [G §5.2.1]:

- **reflecting:** $J(a,t) = 0$ [G (5.2.13)] (for $A = 0$, B constant: $\partial_x p = 0$);
- **absorbing:** $p(a,t) = 0$ [G (5.2.14)];
- **at infinity:** $J(\pm\infty, t) = 0$ [G (5.2.32)], or $J(+\infty, t) = J(-\infty, t)$ [G (5.2.33)] when what leaves at one end comes back at the other (Section 38);
- **periodic:** not a boundary condition in these notes. The domain is a circle, which has no boundary (Section 1.9);
- **at an interface** where A or B jumps: J continuous, and p continuous [G (5.2.15)]–[G (5.2.16)]. For a jump in B the second depends on where the noise comes from (Section 36);
- **where B vanishes or A diverges** the equation may decide by itself whether any condition is allowed: Feller's classification [G §5.2.1e] (Sections 37 and 39).

The backward equation, in the initial point, is [G (5.2.143)]

$$\partial_t p(x', t \,|\, x, 0) = A(x)\, \partial_x p + \tfrac{1}{2} B(x)\, \partial_x^2 p. \tag{1.2}$$

On the backward side the walls become $p = 0$ (absorbing, [G (5.2.67)]) and $\partial_x p = 0$ (reflecting, [G (5.2.69)]) in the initial point.

Factor of two. With $A = 0$, $B = D$ the equation is $\partial_t p = \frac{1}{2} D\, \partial_x^2 p$, so free motion has $\langle (\Delta x)^2 \rangle = Dt$. Many references (and `qsd-context.md`) write $\partial_t p = D' \partial_x^2 p$; then $D' = D/2$, and every rate below has $D/2 \to D'$. In `sde-explorer`, $D = \sigma^2$ where σ is the noise amplitude b.

Units. For an interval of length L the natural time is L^2/D. The figures use $L = D = 1$, so axes read x/L and Dt/L^2. Parts II and III use the scales set by the drift instead (Eqs. (10.1) and (19.1)).

1.2 Where the equation comes from: overdamped Brownian motion

The physical model. A Brownian particle of mass m in a potential $\mathcal{U}(x)$, with friction coefficient γ and a heat bath at temperature T, obeys Langevin's equations

$$\mathrm{d}x = v\, \mathrm{d}t, \qquad m\, \mathrm{d}v = -\gamma v\, \mathrm{d}t - \mathcal{U}'(x)\, \mathrm{d}t + \sqrt{2\gamma k_B T}\, \mathrm{d}W \tag{1.3}$$

[G (5.3.90)]–[G (5.3.91)]; Gardiner's β is our γ. The strength of the noise is fixed by the fluctuation–dissipation relation: it is the value for which the Maxwell–Boltzmann density $\propto e^{-(mv^2/2 + \mathcal{U})/k_B T}$ is stationary. The Fokker–Planck equation for $p(x, v, t)$ is Kramers' equation [G (5.3.92)], [G (5.3.97)], in two variables. The one-variable equation of these notes is its high-friction limit.

Eliminating the velocity [G §6.4]. The velocity forgets its past in the time m/γ [G (6.4.4)]. If that is shorter than every time on which the position changes, v is slaved to x. Setting $m\, \mathrm{d}v \approx 0$ gives $\gamma\, \mathrm{d}x = -\mathcal{U}'\, \mathrm{d}t + \sqrt{2\gamma k_B T}\, \mathrm{d}W$ [G (6.4.16)]–[G (6.4.17)], whose Fokker–Planck equation is Smoluchowski's [G (6.4.18)]. That is Eq. (1.1) with

$$A = -\frac{\mathcal{U}'}{\gamma}, \qquad B = D = \frac{2k_B T}{\gamma}, \qquad \text{so} \qquad U = \frac{\mathcal{U}}{\gamma} \quad \text{and} \quad e^{-2U/D} = e^{-\mathcal{U}/k_B T}. \tag{1.4}$$

- The potential U of these notes is the physical potential divided by the friction. $D/2 = k_B T/\gamma$ is Einstein's diffusion coefficient: that is the factor of two above. When the notes say that $D/2$ plays the role of $k_B T$, they mean $k_B T/\gamma$, in units where $\gamma = 1$. The stationary density is the Boltzmann factor.
- The examples translate as follows:

  - Part II is a constant force F, for instance gravity minus buoyancy in Smoluchowski's sedimentation problem (Section 12): $v = F/\gamma$, and the barometric profile $e^{-2x/\ell}$ is $e^{-|F|x/k_BT}$.
  - Part III is a harmonic trap of stiffness $\mathcal{K}$, such as an optical tweezer: $k = \mathcal{K}/\gamma$ (as $U = \mathcal{U}/\gamma$), $\tau = \gamma/\mathcal{K}$, and the stationary variance $\ell^2/2 = k_BT/\mathcal{K}$ is equipartition.
  - Kramers' factor $e^{2\Delta U/D}$ is Arrhenius' $e^{\Delta\mathcal{U}/k_BT}$.
  - The tilted cosine of Section 27 is the overdamped pendulum driven by a torque, or a current-biased Josephson junction.
- Walls survive the reduction. A reflecting wall for the full motion becomes $J = 0$ for the Smoluchowski equation [G (6.4.84)]–[G (6.4.89)].

When it holds: overdamped, not critically damped. Take $\mathcal{U} = \frac{1}{2}\mathcal{K}x^2$. The full model is then the damped oscillator, with natural frequency $\omega = \sqrt{\mathcal{K}/m}$ and damping ratio $\zeta = \gamma/2\sqrt{m\mathcal{K}}$. By the regression theorem its stationary position correlation obeys the deterministic equation $C'' + (\gamma/m)C' + \omega^2 C = 0$, so it is known exactly. The Smoluchowski equation predicts instead the OU form $(k_BT/\mathcal{K})e^{-\mathcal{K}t/\gamma}$ of Section 20 (Fig. 2).

- Underdamped ($\zeta < 1$): C oscillates, and no one-variable Markov description exists.
- Critically damped ($\zeta = 1$): the oscillation is gone, but inertia is not. $C \propto (1 + \omega t)e^{-\omega t}$ decays at ω, twice the Smoluchowski rate $\mathcal{K}/\gamma = \omega/2$. Critical damping is where the oscillator's two rates merge, not where the overdamped description begins.
- Overdamped ($\zeta \gg 1$): the slowest rate is $(\mathcal{K}/\gamma)(1 + 1/4\zeta^2 + \dots)$, so Smoluchowski is right up to a relative error of order $m\mathcal{K}/\gamma^2$. The other rate, about γ/m, is the velocity relaxation that was eliminated. Gardiner's corrected Smoluchowski equation [G (6.4.108)] multiplies the diffusion by $1 + m\mathcal{U}''/\gamma^2$, which reproduces the $1/4\zeta^2$.

So the condition is $m|\mathcal{U}''|/\gamma^2 \ll 1$ wherever the particle goes, barrier tops included: a damping ratio well above 1 everywhere. The description then holds for $t \gg m/\gamma$.

For escape over a barrier, Kramers' full result at moderate and high friction is

$$r = \frac{\omega_a}{2\pi\omega_b}\Big[\sqrt{\gamma^2/4m^2 + \omega_b^2} - \gamma/2m\Big]e^{-\Delta\mathcal{U}/k_BT}, \qquad \omega^2 = |\mathcal{U}''|/m.$$

It tends to the overdamped formula used throughout these notes when $\gamma \gg m\omega_b$. At low friction the rate is limited by energy diffusion instead [13].

Walls and domains, physically. With this dictionary every example in these notes is an overdamped particle in a potential $\mathcal{U}$, and the walls are features of the landscape (Fig. 1).

- A **reflecting wall** is a hard wall. $\mathcal{U}$ rises so steeply that the particle is turned back without loss: the floor of a container, or a membrane it cannot cross. Zero current says that nothing gets through.
- An **absorbing wall** is a cliff. $\mathcal{U}$ drops away so steeply that a particle reaching the edge falls off and never comes back. Any irreversible removal acts the same way: a perfectly reactive or sticky surface, a detector, an exit, a firing threshold, an extinction. The survival probability is the fraction not yet removed. A surface that removes arrivals only at a finite rate gives a mixed (Robin) condition instead, with the flux into the wall proportional to the density there; it is not treated here.
- A **wall at infinity** is a container much larger than every other length.
- A **circle** is a ring: a bead on a circular track, or an angle such as the orientation of a rotor or the phase of an oscillator.
- A **diverging potential** can make its own wall (Sections 30 and 32). A logarithmic one is often entropic, $-k_BT \ln(\text{number of available states})$.

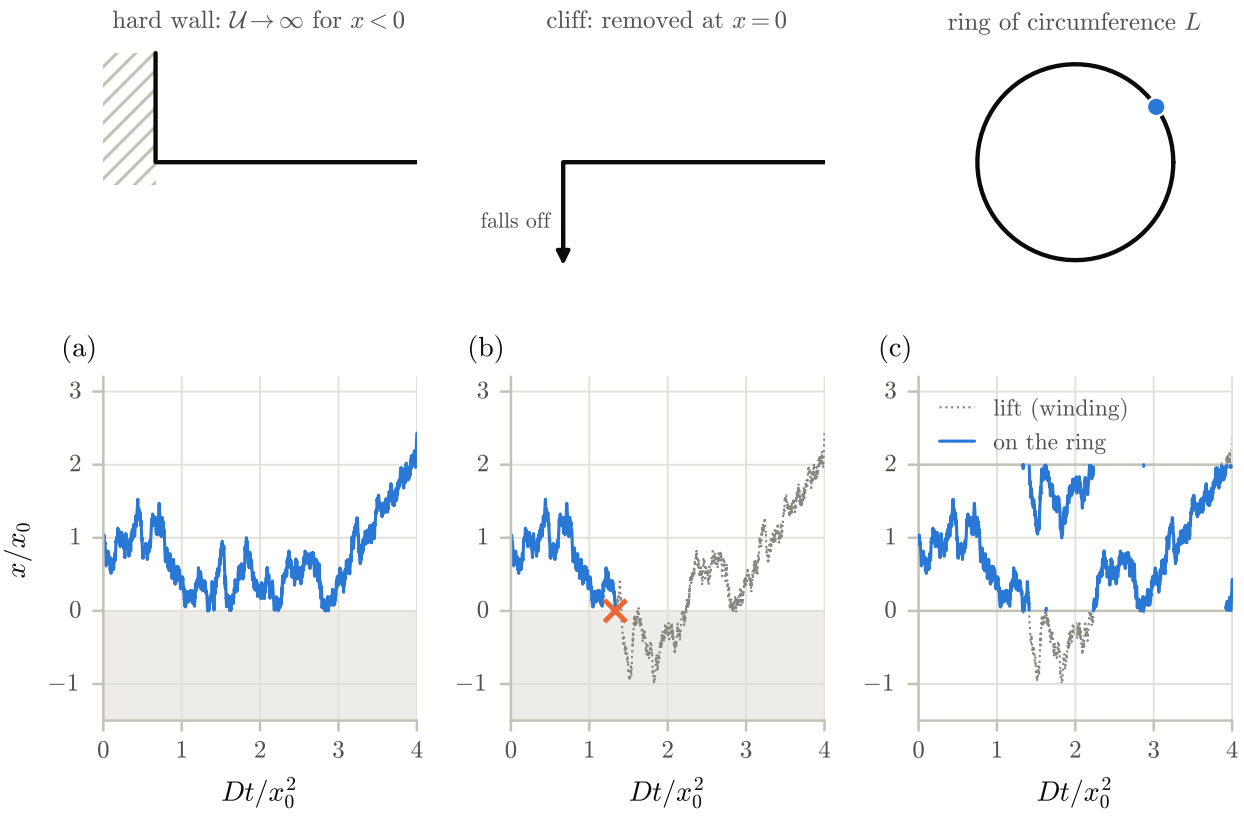


Figure 1: Walls and domains as features of the landscape. Top: the potential. Bottom: one and the same Brownian path from x_0, treated three ways. (a) A hard wall turns the path back; this is $|x|$, the image construction of Section 3. (b) At a cliff the path is removed at its first arrival (cross); the dotted continuation is what it would have done. (c) On a ring of circumference $2x_0$ the path wraps around (solid); the dotted curve is its lift, which counts the windings (Section 1.9).

- An **interface** between two media, or a finite step in $\mathcal{U}$, is a partial wall: it reflects part of the probability and lets the rest through (Section 36).

Table 2 lists the potential and a physical realisation of every example. Each section also has a paragraph *Physical picture*.

1.3 Stationary solution [G §5.2.2]

With zero current at both ends, the stationary equation integrates once to $J = 0$ and once more to Gardiner's potential solution [G (5.2.38)]–[G (5.2.39)],

$$p_s(x) = \frac{\mathcal{N}}{B(x)} \exp \int^x \frac{2A(x')}{B(x')}\,\mathrm{d}x' = \frac{\mathcal{N}\,\psi(x)}{B(x)}, \tag{1.5}$$

with the same ψ that appears in the first-passage formulas [G (5.2.44)], [G (5.2.157)]. For the three drifts of Parts I–III it gives a uniform density ($A = 0$), the barometric $e^{2vx/D}$ ($A = v$), and the Gaussian $e^{-kx^2/D}$ ($A = -kx$). A stationary state exists only if Eq. (1.5) is normalisable [G (5.2.40)]; on the free half-line it is not.

§	Potential $\mathcal{U}(x)$ and walls	Physical realisation
Part I: no force		
2	0	colloid or molecule in a bulk fluid; ink in still water; Perrin's measurement of D
3	0; hard wall	neutrally buoyant bead above a container floor; molecule beside an impermeable membrane
4	0; cliff	walker falling off a plateau; molecule reaching a sticky surface; random-walk neuron firing at threshold
5	0; two hard walls	particle in a closed box or channel; mixing time
6	0; two cliffs	pore open at both ends; gambler's ruin
7	0; hard wall and cliff	escape from a dead-end tube or pore
8	0 on a ring	bead on a circular track; planar rotational diffusion (Debye)
Part II: constant force, $\mathcal{U} = -Fx$, $v = F/\gamma$		
11	$-Fx$	sedimenting colloid; ion in a field; pollutant in a river
12	$-Fx$; floor	sedimentation equilibrium (Perrin, Avogadro's number); barometric formula
13	$-Fx$; cliff	sedimentation onto a sticky floor; drifting neuron (Gerstein–Mandelbrot)
14	$-Fx$; floor and ceiling	sedimentation in a closed column; ultracentrifuge equilibrium
15	$-Fx$; two cliffs	biased gambler's ruin; driven translocation through a nanopore
16	$-Fx$; floor and a rim at height $\lvert F\rvert L$	activated escape over a linear barrier; ion crossing a membrane against a field
17	constant torque on a ring	colloid driven around a ring trap; paddle wheel in a viscous fluid
Part III: harmonic trap, $\mathcal{U} = \frac{1}{2}\mathcal{K}x^2$, $k = \mathcal{K}/\gamma$		
20	$\frac{1}{2}\mathcal{K}x^2$	bead in an optical tweezer (calibration); velocity of a free Brownian particle
21	trap; hard wall at the centre	tweezer focused on a coverslip
22	trap; cliff at the centre	mean-reverting quantity stopped at its mean; leaky integrate-and-fire neuron
23	trap; two hard walls	trapped bead in a channel; box-to-trap crossover
24	trap; two cliffs	trapped bead in a channel with sticky ends
25	trap; cliff at the rim	escape from a finite trap; thermal rupture of a harmonic bond
26	parabolic wells meeting at cusps, on a ring	row of touching optical traps; hopping between traps
27	$-\mathcal{U}_0 \cos(x/R) - Fx$ on a ring	dipole in a field; overdamped pendulum; Josephson junction; corrugated ring trap
Part IV: other solvable potentials		
29	$\mathcal{U}_0 e^{-x/w} + \lvert F\rvert x$ (Morse)	colloid hovering above a charged surface (TIRM)
29	$\lvert F\rvert r - (d-1)k_BT \ln r$	distance from a centre that pulls with a force of constant magnitude
30	$-(d-1)k_BT \ln r$ $(+\frac{1}{2}\mathcal{K}r^2)$, entropic	distance from a target (Smoluchowski rate); amplitude of a noisy field (Rayleigh); CIR
31	$2sk_BT \ln\cosh(x/w)$	saturating restoring force; bead in a rounded V-groove
32	$-2sk_BT \ln\cos(x/w)$, walls by the potential	polar angle of 3D rotational diffusion ($s = \frac{1}{2}$); Wright–Fisher allele frequency
Part V: any potential		
35	$\Delta\mathcal{U}[(x/a)^2 - 1]^2$	bistable optical trap; isomerisation; protein folding; bistable switch
Part VI: interfaces and singular boundaries		
36	0, with friction (or $\mathcal{U}$) stepping at an interface	tracer entering a more viscous medium; partitioning into a membrane; run-and-tumble bacteria
37	$\frac{1}{2}\mathcal{K}r^2 - (d-1)k_BT \ln r$, in $y = r^2$	energy of a particle in a d-dimensional trap; population with demographic noise; CIR interest rate; counterion near a charged rod
38	$-\gamma(x^3/3 + Ix)$, reinjected at $-\infty$	quadratic integrate-and-fire (theta) neuron; phase slips of a pendulum or Josephson junction near critical torque

Table 2: A physical realisation of every example: an overdamped particle in the potential $\mathcal{U} = \gamma U$,

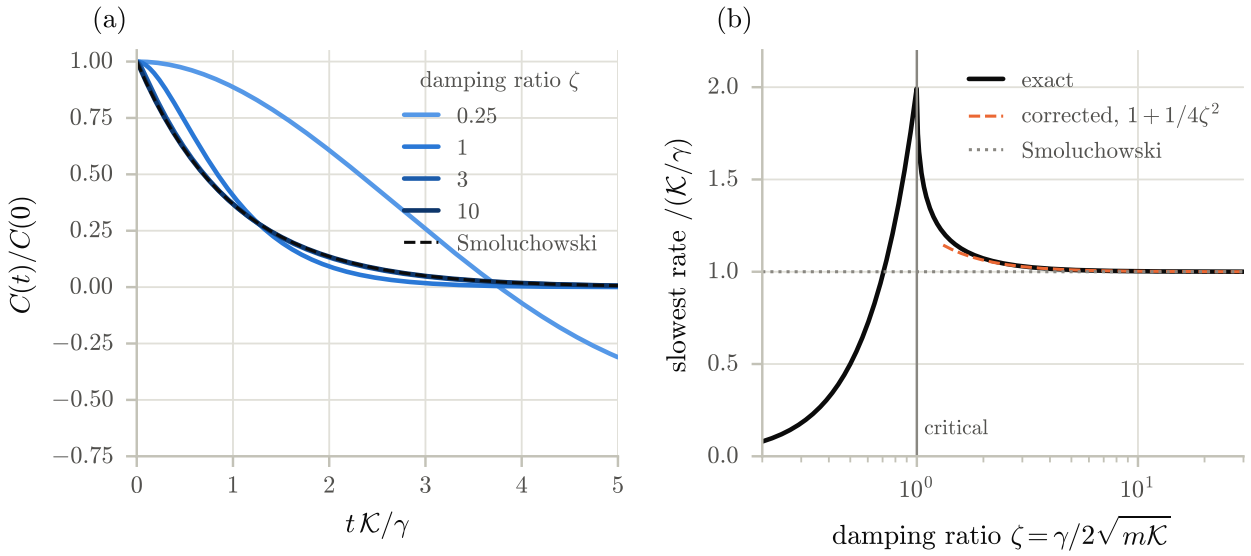


Figure 2: A particle in a harmonic trap with inertia, against the overdamped (Smoluchowski, OU) description. (a) Stationary position correlation for several damping ratios; dashed, the OU prediction $e^{-\mathcal{K}t/\gamma}$. (b) Long-time decay rate in units of the Smoluchowski rate $\mathcal{K}/\gamma$. It is largest, twice the Smoluchowski value, at critical damping. Dashed, Gardiner's corrected Smoluchowski equation [G (6.4.108)].

1.4 Eigenfunction expansion [G §5.2.5]

Gardiner's $P_\lambda(x)$ is an eigenfunction of the forward operator and $Q_\lambda(x)$ one of the backward operator, with the same rate: $\mathcal{L}P_\lambda = -\lambda P_\lambda$ [G (5.2.75)] and $\mathcal{L}^\dagger Q_\lambda = -\lambda Q_\lambda$ [G (5.2.76)] (the operators are Eq. (1.8)). P_λ is a density profile that decays as $e^{-\lambda t}$; Q_λ is a function of the starting point. For a time-independent problem with stationary solution p_s (the reflecting one, even when the walls are absorbing [G (5.2.91)]), they are related by $P_\lambda = p_s\, Q_\lambda$ and normalised by $\int P_\lambda Q_{\lambda'}\,\mathrm{d}x = \int p_s Q_\lambda Q_{\lambda'}\,\mathrm{d}x = \delta_{\lambda\lambda'}$ [G (5.2.79)]–[G (5.2.80)], so that $P_0 = p_s$ and $Q_0 = 1$ [G (5.2.82)]–[G (5.2.83)]. (The relation is [G (5.2.71)] applied to the solutions [G (5.2.73)]–[G (5.2.74)].) Then

$$p(x,t\,|\,x_0,0) = \sum_\lambda P_\lambda(x)\,Q_\lambda(x_0)\,e^{-\lambda t}, \qquad [G\ (5.2.88)] \tag{1.6}$$

$$\langle x(t)x(0)\rangle_s = \sum_\lambda \Big[\int x\,P_\lambda(x)\,\mathrm{d}x\Big]^2 e^{-\lambda t}. \qquad [G\ (5.2.90)] \tag{1.7}$$

With reflecting walls $\lambda = 0$ is in the spectrum and $p \to p_s$; with absorbing walls it is not and $p \to 0$.

Every example's box names its P_λ and Q_λ, normalised so that [G (5.2.88)] holds as it stands. Four situations go beyond Gardiner's [G §5.2.5]:

- **Continuous spectrum.** On an unbounded domain the sum becomes an integral over a continuous label k, and $\delta_{\lambda\lambda'}$ becomes $\delta(k-k')$. Then p_s is often not normalisable, and the weight in $P_\lambda = p_s Q_\lambda$ is the zero-current solution Eq. (1.5), fixed only up to a constant; the boxes say which constant.
- **Complex modes.** The pairing [G (5.2.79)] is $\int P_\lambda Q_{\lambda'}$, with no complex conjugate, so a forward mode e^{iqx} pairs with the backward mode e^{-iqx}.
- **A stationary current.** [G (5.2.71)]–[G (5.2.72)] need $J_s = 0$. With a current (a circle with $\Phi \neq 0$, or reinjection through infinity) P_λ and Q_λ have to be found separately. They are still

biorthogonal [G (5.2.79)], but $P_\lambda \neq p_s Q_\lambda$.

- **Positive solutions off the spectrum.** Some QSDs solve the forward eigen-equation at a λ that is not in the spectrum: their partner $Q_\lambda = P_\lambda/p_s$ grows so fast that $\int P_\lambda Q_\lambda$ diverges (Sections 13, 22 and 37).

1.5 Which operator, on which space

Three things must be fixed before "the spectrum" means anything: the differential expression, the conditions at the walls, and the space of functions it acts on. Change any one of them and the spectrum can change.

Two operators. The forward (Fokker–Planck) operator acts on densities, and the backward operator on functions of the starting point:

$$\mathcal{L}p = -\partial_x(Ap) + \tfrac{1}{2}\partial_x^2(Bp) = -\partial_x J, \qquad \mathcal{L}^\dagger f = A\,\partial_x f + \tfrac{1}{2}B\,\partial_x^2 f. \tag{1.8}$$

$\mathcal{L}^\dagger$ is what probabilists call the *generator* of the process: the average $\langle f(x(t))\rangle$, as a function of the start x_0, obeys $\partial_t\langle f\rangle = \mathcal{L}^\dagger\langle f\rangle$, as in Eq. (1.2). The two operators are adjoint under the pairing $\int f\,p\,\mathrm{d}x$, the average of f over the density p. Integrating by parts, as Gardiner does in [G (5.2.77)],

$$\int f\,\mathcal{L}p\,\mathrm{d}x - \int p\,\mathcal{L}^\dagger f\,\mathrm{d}x = \Big[-f\,J - \tfrac{1}{2}B\,p\,\partial_x f\Big]_{\text{walls}}, \tag{1.9}$$

where J is the current of p.

- **The walls are part of the operator.** The boundary terms vanish when the forward and backward conditions are paired: $J = 0$ with $\partial_x f = 0$ at a reflecting wall [G (5.2.69)], and $p = 0$ with $f = 0$ at an absorbing one [G (5.2.67)]. A circle has no walls, and no boundary terms. So the same differential expression with reflecting or with absorbing walls is a different operator, with a different spectrum (Sections 5 and 6). In operator language, the wall conditions specify the *domain*.
- **What the eigenvalues belong to.** Gardiner's eigen-equations [G (5.2.75)]–[G (5.2.76)] are $\mathcal{L}P_\lambda = -\lambda P_\lambda$ and $\mathcal{L}^\dagger Q_\lambda = -\lambda Q_\lambda$. Every λ in these notes is an eigenvalue of $-\mathcal{L}$ and, with the paired walls, of $-\mathcal{L}^\dagger$. Equation (1.9) with $p = P_\lambda$ and $f = Q_{\lambda'}$ gives $(\lambda' - \lambda)\int P_\lambda Q_{\lambda'}\,\mathrm{d}x = 0$, the biorthogonality [G (5.2.79)].
- **Conservation.** With no absorbing wall, $f = 1$ satisfies the backward condition and $\mathcal{L}^\dagger 1 = 0$, so $Q_0 = 1$ [G (5.2.83)] always exists. Its forward partner $P_0 = p_s$ [G (5.2.82)] exists only if it is normalisable. That is the question in the first item of Section 1.8.

Three spaces. Each answers a different question (Table 3).

Space	Contains	Operator	Used for
L^1	densities, $\int \lvert p\rvert\,\mathrm{d}x < \infty$	$\mathcal{L}$	probability; stationary states and QSDs
$L^2(1/p_s)$	$\int p^2/p_s\,\mathrm{d}x < \infty$	$\mathcal{L}$, self-adjoint if $J_s = 0$	real spectra, gaps, expansions, bounds
L^∞	bounded observables $f(x_0)$	$\mathcal{L}^\dagger$	$S(t\,\vert\,x_0)$, $T(x_0)$, π_a, π_b

Table 3: The spaces on which the operators Eq. (1.8) act.

- L^1 **is where probability lives.** The forward evolution keeps densities positive and never increases $\int |p|$; with no absorption it conserves $\int p$. Stationary states and QSDs are, by definition, positive eigenfunctions in L^1.
- L^∞ **is its dual,** where the backward evolution acts on observables. The survival probability is the backward evolution of $f = 1$ with absorbing walls, and all the first-passage functions of Section 1.7 are backward objects.
- $L^2(1/p_s)$ **is where the operator is self-adjoint.** It is the only one of the three with an inner product, so it is where orthogonality, eigenfunction expansions and variational bounds live. Gardiner's substitution $p = p_s q$ [G (5.2.71)] turns the forward operator into the backward one, $\mathcal{L}(p_s q) = p_s \mathcal{L}^\dagger q$ [G (5.2.72)], whenever the stationary current vanishes. Then Eq. (1.9) becomes

$$\int \frac{g\,\mathcal{L}p}{p_s}\,\mathrm{d}x = \int \frac{p\,\mathcal{L}g}{p_s}\,\mathrm{d}x, \qquad -\int \frac{p\,\mathcal{L}p}{p_s}\,\mathrm{d}x = \tfrac{1}{2}\int B\,p_s\Big[\partial_x \frac{p}{p_s}\Big]^2 \mathrm{d}x \;\geq 0. \tag{1.10}$$

  So $-\mathcal{L}$ is self-adjoint and non-negative: λ is real and ≥ 0, the orthogonality relations [G (5.2.80)]–[G (5.2.81)] hold, and the right-hand side is the numerator of the Rayleigh quotient Eq. (1.15). The weight need not be normalisable. It is the zero-current solution $p_s = \psi/B$ of Eq. (1.5), which exists on any interval or line: the free half-line uses $p_s = 1$, and absorbing problems use the p_s of the same problem with reflecting walls [G (5.2.91)].

On a bounded domain the eigenfunctions are smooth and bounded, so all three spaces give the same discrete spectrum. On an unbounded domain they need not agree. The continua of QSDs in Section 1.8 are eigenvalues in L^1 that are not in the L^2 spectrum, and probability escaping to infinity is visible in L^1 but not in L^2.

Discrete and continuous spectrum. λ is an eigenvalue when its eigenfunction lies in the space. On an unbounded domain there is also continuous spectrum, made of generalised eigenfunctions that are bounded but not normalisable, such as e^{ikx} on the line. The sum [G (5.2.88)] then becomes an integral over k, as in Sections 2 to 4.

The symmetrised form: a Schrödinger operator. The weight can be absorbed into the function. With $B = D$, write $p = \sqrt{p_s}\,\psi$, so that $\int p^2/p_s\,\mathrm{d}x = \int \psi^2\,\mathrm{d}x$. Then $\partial_t p = \mathcal{L}p$ becomes

$$\partial_t \psi = -H\psi, \qquad H = -\tfrac{1}{2}D\,\partial_x^2 + V(x), \qquad V = \frac{A^2}{2D} + \frac{A'}{2}, \tag{1.11}$$

a Schrödinger operator in plain L^2, with D in the role of $\hbar^2/m$. Its spectrum is the $L^2(1/p_s)$ spectrum of $-\mathcal{L}$, and its ground state at $\lambda = 0$ is $\sqrt{p_s}$ when that is normalisable. If ψ_λ are its orthonormal eigenfunctions, Gardiner's are

$$P_\lambda = \sqrt{p_s}\,\psi_\lambda, \qquad Q_\lambda = \psi_\lambda/\sqrt{p_s}, \tag{1.12}$$

which satisfy $P_\lambda = p_s Q_\lambda$ and $\int P_\lambda Q_{\lambda'} = \int \psi_\lambda \psi_{\lambda'} = \delta_{\lambda\lambda'}$ at once; the constant in a non-normalisable p_s cancels from every product $P_\lambda(x)Q_\lambda(x_0)$. The walls become

$$\text{absorbing: } \psi = 0, \qquad \text{reflecting: } \partial_x \psi = \frac{A}{D}\,\psi. \tag{1.13}$$

- **Reading the spectrum off** V**.** A bounded domain, or $V \to \infty$ at infinity, gives a discrete spectrum. Otherwise the continuum starts at the limit of V at infinity: that is where the edges of Section 1.6 come from.

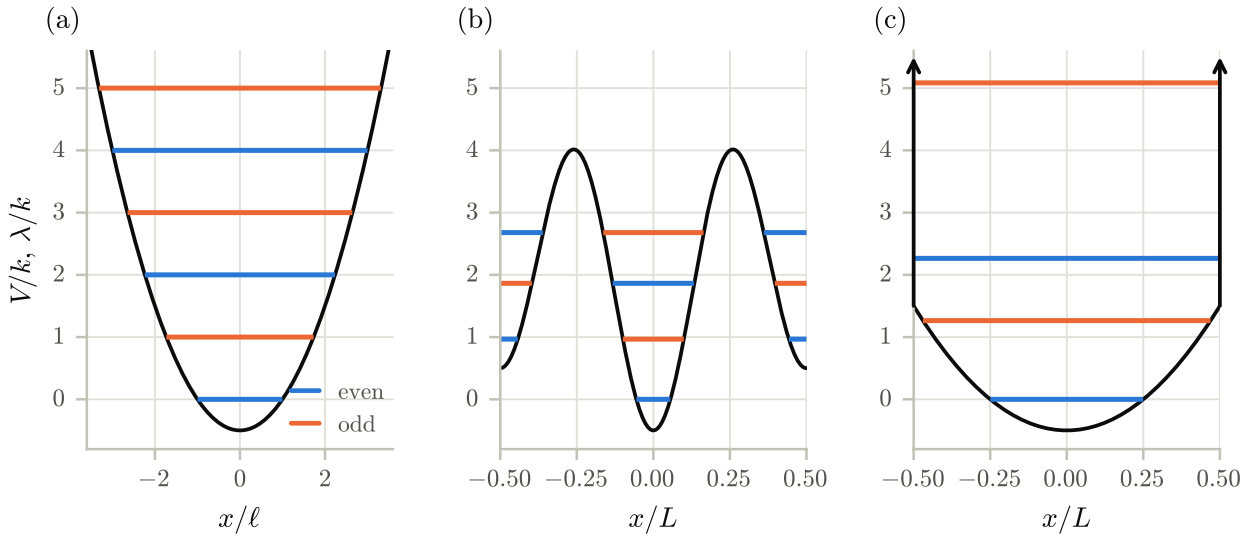


Figure 3: Symmetrised potentials $V = A^2/2D + A'/2$ (black) and their levels, drawn where $V < \lambda$, in units of k. Blue levels are even under $x \to -x$, orange ones odd. (a) OU on the line: a harmonic oscillator with levels nk. A wall at the bottom keeps one parity. (b) The cosine well on a circle, $\kappa = 16$: V has a well at the bottom of U ($x = 0$) and one at its top ($x = \pm L/2$). Each nonzero level has a copy in each well, of opposite parity. (c) The sawtooth on a circle, $L = 4\ell$: one harmonic well and a δ-function spike at the cusp (arrows). The levels are single, in pairs split by exactly k.

- **Part I:** $V = 0$, a free particle, with Dirichlet or Neumann walls.
- **Part II:** $V = v^2/2D = 1/2\tau$, a constant. The gauge Eq. (10.2) is exactly this symmetrisation, since $\sqrt{p_s} \propto e^{\alpha x}$. The constant potential is the shift $1/2\tau$ of every rate, and the Robin condition $u' = \alpha u$ of Section 10 is Eq. (1.13) with $A = v$. On the half-line the continuum starts at $V(\infty) = 1/2\tau$.
- **Part III:** $V = \frac{1}{2}k(\xi^2 - 1)$, a harmonic oscillator whose zero-point energy $k/2$ is cancelled, so $\lambda = nk$ with $\psi_n \propto H_n(\xi)e^{-\xi^2/2}$. A wall at the bottom keeps the even states (reflecting, $\psi' = 0$) or the odd ones (absorbing, $\psi = 0$) (Fig. 3a).
- **Circles.** Near an extremum of U, $A \approx 0$ and $V \approx A'/2 = -U''/2$, and the A^2 term grows away from it. So in a deep enough well both the bottom and the top of U are wells of V.
  - The smooth cosine well (Section 27) has two wells in V, offset by k, and each nonzero level has one copy in each, of opposite parity (Fig. 3b). These are the doublets.
  - The sawtooth (Section 26) has a single harmonic well, plus a repulsive spike $\frac{1}{2}kL\,\delta$ at the cusp, where A' jumps. Its levels are single (Fig. 3c).
- **With a current** (on a circle with nonzero circulation $\Phi = \oint 2A/B$, Section 1.9) there is no zero-current p_s to symmetrise with. In plain L^2 the drift term $-v\partial_x$ is anti-self-adjoint, which is why the eigenvalues of Section 17 have imaginary parts $\pm vq_m$. The operator is still normal, with orthogonal Fourier modes, and biorthogonality [G (5.2.79)] holds with $P_m = e^{iq_m x}/L$ and $Q_m = e^{-iq_m x}$.

1.6 The spectral gap

Most long-time statements in these notes come from one property of this expansion. Order the eigenvalues $\lambda_{\min} < \lambda_{\text{next}} < \cdots$. With reflecting or absorbing walls, or on a circle without a current,

the operator is self-adjoint with respect to the weight $1/p_s$ ([G §5.2.5]; Section 1.5), so the eigenvalues are real and non-negative and the eigenfunctions are complete. (Each section labels modes by their natural quantum number, so the lowest one may be called λ_0 or λ_1. What matters is lowest and next-lowest.)

What the gap is. Keep the two slowest terms of [G (5.2.88)]:

$$p(x,t\,|\,x_0,0) = P_{\min}(x)\,Q_{\min}(x_0)\,e^{-\lambda_{\min}t} + P_{\text{next}}(x)\,Q_{\text{next}}(x_0)\,e^{-\lambda_{\text{next}}t} + \cdots \tag{1.14}$$

- **Reflecting or confining problems** have $\lambda_{\min} = 0$, with $P_{\min} = p_s$ and $Q_{\min} = 1$. Then $p - p_s \approx P_{\text{next}}Q_{\text{next}}(x_0)\,e^{-\lambda_{\text{next}}t}$: the distance from equilibrium decays at λ_{next}. The *spectral gap* is $\lambda_{\text{next}} - \lambda_{\min} = \lambda_{\text{next}}$, and its inverse is the relaxation time. Every expectation value, and every stationary correlation function $C(\tau) = \sum_n c_n^2 e^{-\lambda_n \tau}$ [G (5.2.90)], ends up decaying at the gap.
- **Absorbing problems** have $\lambda_{\min} > 0$. Then $\lambda_{\min}$ is the decay rate of the survival probability, $S \propto e^{-\lambda_{\min}t}$, and the conditioned density $q = p/S$ tends to the QSD $\propto P_{\min}$. The gap $\lambda_{\text{next}} - \lambda_{\min}$ sets how fast q forgets where it started: $q - q_\infty \propto e^{-(\lambda_{\text{next}}-\lambda_{\min})t}$. On an unbounded domain this holds for localised starts; a heavy-tailed start can reach a different QSD (Section 1.8).
- **Selection by symmetry.** If an initial condition or observable has no overlap with the next mode, it relaxes at the next rate it does overlap. Examples: starting at the centre of a symmetric interval (Sections 5 and 6), and $\langle x^2\rangle$ in the Ornstein–Uhlenbeck process (Section 20).

Why the gap has the size it does. With zero-current walls, λ_{next} is the minimum of a Rayleigh quotient over functions with $\int p_s f = 0$:

$$\lambda_{\text{next}} = \min_f \frac{\frac{1}{2}\int B\,p_s\,(f')^2\,\mathrm{d}x}{\int p_s\,f^2\,\mathrm{d}x} \qquad \text{(Poincaré inequality)}. \tag{1.15}$$

A small gap needs a function that is not constant but costs little gradient, weighted by p_s:

- a long domain lets f vary slowly, at cost $\sim D/L^2$, so $\lambda \approx \pi^2 D/2L^2$ (Section 5);
- a barrier lets f be nearly $+1$ on one side and -1 on the other, changing only where p_s is exponentially small, so the rate is exponentially small (Arrhenius, Sections 16 and 25);
- confinement works the other way. A restoring force concentrates p_s and keeps the gap finite even on an infinite domain (the Ornstein–Uhlenbeck gap k, Section 20).

Discrete levels, continua, and no gap.

- Bounded domains, and confining forces, give discrete spectra: the gap is positive and relaxation is exponential.
- An unbounded domain without confinement gives a continuous spectrum down to $\lambda = 0$: there is no gap, and relaxation is algebraic (the $t^{-1/2}$ of Section 4).
- A continuum can start above zero, as with drift toward a wall on the half-line (Section 13). Its edge plays the role of the gap, but the approach carries an algebraic prefactor ($t^{-3/2}$).
- Isolated levels below a continuum (bound states) appear when a force pushes against a reflecting wall: the stationary state of Section 12, or the Arrhenius level of Section 16.
- Symmetry can make levels degenerate. On a circle the nonzero levels of a reflection-symmetric problem come in doublets (Section 8). With a circulating current they become complex-conjugate pairs and the density spirals as it relaxes; the gap is then $\mathrm{Re}\,\lambda$ (Section 17).

Figure 4 shows these cases side by side.

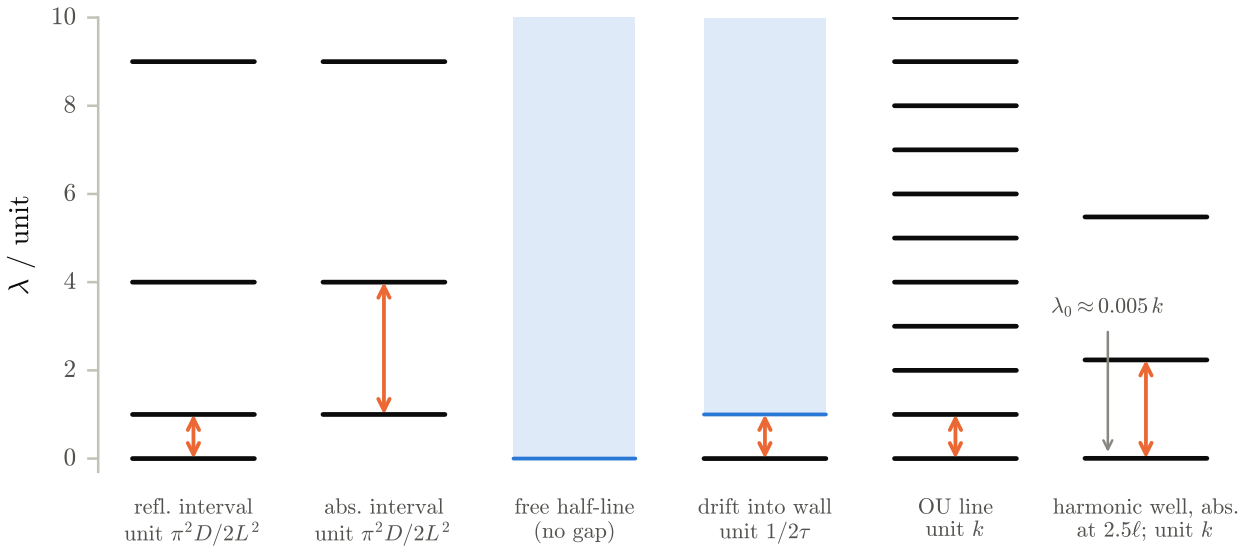


Figure 4: Spectra of six examples, each in its own natural rate. Ticks are discrete eigenvalues, shaded bands continua, and the arrow is the gap that controls the long-time behaviour. For the absorbing interval the lowest level is the decay rate of S, and the arrow is the rate of convergence to the QSD. The last column is the harmonic well with a reflecting wall at its bottom and an absorbing wall at $L = 2.5\ell$ (Section 25); its lowest level, $\approx 0.005\,k$, is the Kramers escape rate.

Caveat. With drift, the operator is self-adjoint only with the weight. In ordinary distance the slowest mode can take a transit time to dominate (Section 10). The gap gives the asymptotic rate, not the length of the transient.

1.7 First passage [G §5.2.7]–[G §5.2.8]

For walls $a < b$, both absorbing, and a start at x_0 (see also Redner [2], ch. 2):

- **Survival** $S(t \mid x_0) = \int_a^b p(x, t \mid x_0, 0)\,\mathrm{d}x = \Pr(T > t)$; this is Gardiner's $G(x_0, t)$ [G (5.2.140)]. It obeys the backward equation with $S(x_0, 0) = 1$, $S(a, t) = S(b, t) = 0$.
- **Exit-time density** $f(t) = -\partial_t S = j_a(t) + j_b(t)$, with outward fluxes $j_a = -J(a, t)$, $j_b = J(b, t)$.
- **Moments** $T_k(x_0) = \langle T^k \rangle = k \int_0^\infty t^{k-1} S\,\mathrm{d}t$ satisfy $A\,T_k' + \frac{1}{2}B\,T_k'' = -k\,T_{k-1}$, $T_0 = 1$, $T_k(a) = T_k(b) = 0$ [G (5.2.154)], [G (5.2.156)]. With $\psi(x) = \exp \int^x 2A/B$, T_1 is given in closed form by [G (5.2.157)]–[G (5.2.158)].
- **Splitting probabilities** $\pi_a + \pi_b = 1$, with $A\pi_a' + \frac{1}{2}B\pi_a'' = 0$, $\pi_a(a) = 1$, $\pi_a(b) = 0$ [G (5.2.186)]–[G (5.2.187)]. The solution is $\pi_b(x) = \int_a^x \mathrm{d}y/\psi(y) \big/ \int_a^b \mathrm{d}y/\psi(y)$, since $\pi' \propto 1/\psi$. **Misprint:** [G (5.2.189)]–[G (5.2.190)] as printed in the 2nd edition have ψ in place of $1/\psi$. That gives the wrong sign of the drift dependence; it only matters once $A \neq 0$.
- **Conditional mean exit times** $T_a(x_0) = \langle T \mid \text{exit at } a \rangle$ from $A(\pi_a T_a)' + \frac{1}{2}B(\pi_a T_a)'' = -\pi_a$, $\pi_a T_a = 0$ at both walls [G (5.2.183)]. Check: $\pi_a T_a + \pi_b T_b = T_1$.
- **Conditioned density** $q = p/S$ obeys $\partial_t q = \mathcal{L}q + \kappa(t)q$ with exit rate $\kappa = -\dot{S}/S = f/S$. Its fixed points are the quasi-stationary distributions (QSDs). On a bounded domain there is exactly one, the positive ground state ϕ_1 of $\mathcal{L}\phi = -\lambda\phi$ with absorbing walls, and $\kappa \to \lambda_1$ [3]. On an unbounded domain there can be none or a continuum (Section 1.8).

1.8 Stationary states and QSDs from the spectrum

A stationary state is a positive, integrable solution of $\mathcal{L}\phi = 0$. A QSD is a positive, integrable solution of $\mathcal{L}\phi = -\lambda\phi$ with $\lambda > 0$ that vanishes at the absorbing walls [3]. Neither definition mentions the spectrum. How much does the spectrum used in these notes reveal? That is the spectrum in $L^2(1/p_s)$, where the operator is self-adjoint (Section 1.5); in Part II it is the spectrum of the gauged problem.

Bounded domains: everything. The spectrum is discrete, and the lowest eigenfunction is the only positive one: every other eigenfunction is orthogonal to the positive ground state of the backward problem (for a conservative problem, $\int P_n \,\mathrm{d}x = 0$), so it must change sign.

- With no absorption the lowest eigenvalue is 0, its eigenfunction is the unique stationary state, and every start converges to it at the gap. This includes the circle with a current (Section 17), whose other eigenvalues are complex while 0 stays isolated.
- With absorption the lowest eigenvalue $\lambda_{\min} > 0$ is the decay rate of S, its eigenfunction is the unique QSD, and every start converges to it at rate $\lambda_{\text{next}} - \lambda_{\min}$.

Stationary states never form a continuum. On a connected domain with $B > 0$ the process can get from anywhere to anywhere, and then it has at most one stationary state. It exists exactly when the stationary solution Eq. (1.5) is normalisable [G (5.2.40)] (Eq. (1.19) on a circle). The continua below are all of QSDs.

Unbounded domains: three things to check.

1. **Eigenvalue, or edge of a continuum.** If the bottom of the spectrum is the edge of a continuum and not an eigenvalue, there is no stationary state or QSD in L^2, and the approach is algebraic. The free line and half-lines are examples (Sections 2 to 4): their would-be eigenfunctions 1 and x are not normalisable. Conversely, 0 can be an eigenvalue at the bottom of a continuum. Then p_s exists but relaxation is slower than exponential. That happens when p_s has power-law tails, which none of the examples here do; in one dimension a gap needs tails of p_s that decay at least exponentially.
2. **L^2 sees only local decay.** The weighted norm does not notice probability running off to infinity.
   - Drift away from a reflecting wall (Section 12, $v > 0$): the spectrum is $[1/2\tau, \infty)$, with no 0, although probability is conserved. $1/2\tau$ is the rate at which the density at a fixed x drops as the bulk drifts away.
   - An absorbing wall with the drift toward it or away from it (Section 13): the spectrum is $[1/2\tau, \infty)$ for both signs, since the shift $v^2/2D$ is even in v. Yet toward the wall $S \sim t^{-3/2}e^{-t/2\tau}$, and away from it $S \to 1 - e^{-2x_0/\ell}$.
3. **Positive solutions below the bottom.** For these one-dimensional problems, positive solutions of $\mathcal{L}\phi = -\lambda\phi$ satisfying the wall condition exist exactly for λ up to the bottom λ_{b} of the L^2 spectrum (in Schrödinger language, the Allegretto–Piepenbrink theorem). Below λ_{b} they are not in L^2, but they can still be integrable, and then they are QSDs (Table 4).

   The last row shows that a discrete spectrum with an isolated bottom does not make the QSD unique on an unbounded domain.

Uniqueness. A theorem of Cattiaux et al. [16] gives the sharp criterion: there is exactly one QSD, and it attracts every initial condition, if and only if $+\infty$ is an *entrance* boundary, meaning the process comes down from infinity in finite mean time. Equivalently, the mean absorption time $T(x_0)$ of Section 1.7 stays bounded as $x_0 \to \infty$.

Absorbing problem	Bottom λ_b of L^2 spectrum	Positive, integrable ϕ	QSDs
interval, Section 6	$\pi^2 D/2L^2$, isolated	only at λ_b	one
free half-line, Section 4	0, edge	none ($\phi = x$)	none
drift toward the wall, Section 13	$1/2\tau$, edge	every $\lambda \in (0, 1/2\tau]$	a continuum
drift away from the wall, Section 13	$1/2\tau$, edge	none (ϕ grows)	none: escape
OU, wall at the bottom, Section 22	k, isolated	every $\lambda \in (0, k]$	a continuum

Table 4: How many QSDs each absorbing problem has.

- Bounded domains pass trivially.
- On the half-line, $T = x_0/|v|$ for drift toward the wall and $T \simeq \tau \ln(2x_0/\ell)$ for OU (Section 22) both grow without bound, hence the continua.
- A drift such as $A = -kx^3$, for which T stays bounded, would have a unique QSD.

Which QSD is reached. With a continuum, the initial condition decides, through its tail (Fig. 5).

- **Point starts**, and anything localised enough, reach the *minimal* QSD, the one at λ_b (the Yaglom limit). Through the gauge (Part II) or the map (Part III) the problem becomes the free absorbing half-line, whose density at large t is $\propto x$ near the wall (Section 4). Transformed back, that x becomes the minimal QSD.
- **Heavy tails** change the outcome. Particles that start far out keep arriving at the wall long after the bulk has been absorbed. If the initial tail delivers survivors more slowly than $e^{-\lambda_b t}$, they dominate, and the conditioned density settles on the QSD with the same tail and a smaller λ. In the free variables, the far-out part of the start contributes a profile that does not spread and decays more slowly than the minimal one (Sections 13 and 22).
- The thresholds are sharp.
  - Drift toward the wall: an initial tail $e^{-\beta x}$ with $\beta < 1/\ell$ selects $\lambda = \beta|v| - \frac{1}{2}D\beta^2$.
  - OU: a tail $x^{-1-\nu}$ with $0 < \nu < 1$ (infinite mean) selects $\lambda = \nu k$.
  - Lighter tails give the minimal QSD.
  - Tails heavier than every exponential (drift) or every power (OU) give $\lambda \to 0$. There the family is not normalisable, and the conditioned density never settles.

1.9 Periodic boundaries: the circle

Gardiner treats periodic boundaries [G §5.2.1d] by taking an interval $[a, b]$, identifying its ends, and requiring

$$\lim_{x\to b^-} p = \lim_{x\to a^+} p \quad [G\ (5.2.17)], \qquad \lim_{x\to b^-} J = \lim_{x\to a^+} J \quad [G\ (5.2.18)].$$

These are borrowed from his conditions at a discontinuity of the coefficients, [G (5.2.15)]–[G (5.2.16)], so A and B are allowed to jump at the seam; that they do not, [G (5.2.19)], is presented as the most frequent case. The stationary solution [G (5.2.41)]–[G (5.2.47)] is then written relative to the seam, through $p_s(a)$ and integrals that start at a.

These notes take the topology at face value instead. The state space is the circle $S^1 = \mathbb{R}/L\mathbb{Z}$, of circumference L and radius $R = L/2\pi$. A circle has no boundary, so there is nothing to impose and no special point.

- A, B and p are functions on the circle, that is, L-periodic functions on $\mathbb{R}$, with the same smoothness at every point. A coordinate needs a cut somewhere, but no result may depend on where the cut is, and every formula below is invariant under moving it.

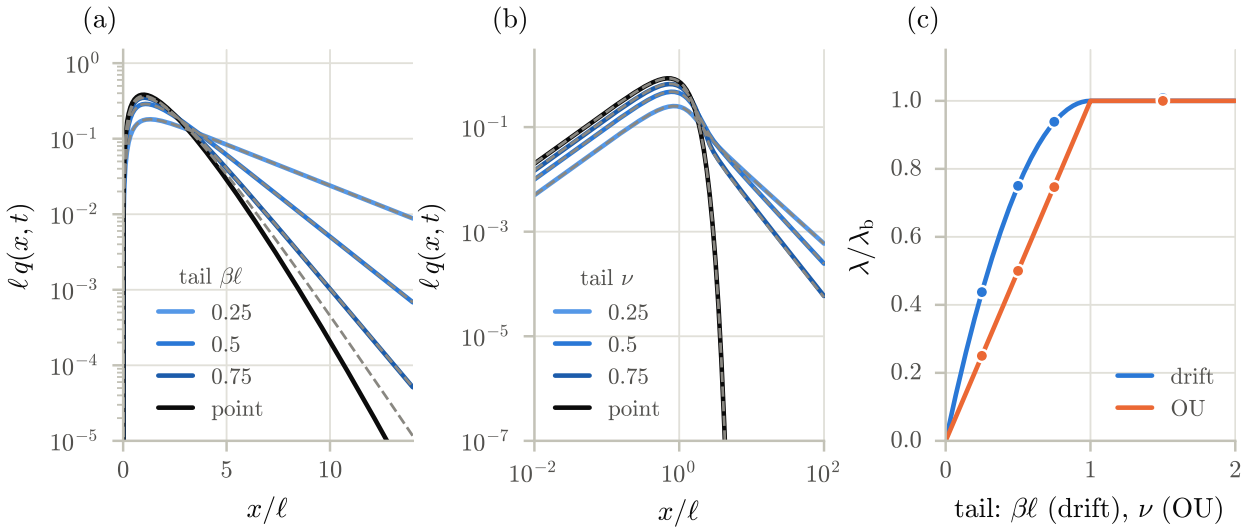


Figure 5: Which QSD a start reaches. (a) Drift toward an absorbing wall: conditioned densities at $t = 60\,\tau$ from exponential starts $\beta e^{-\beta x}$ and from a point start at $x_0 = \ell$. Dashed, the QSDs with the same tails, and the edge QSD $xe^{-x/\ell}/\ell^2$, which the point start is still approaching, algebraically. (b) OU with an absorbing wall at the bottom: conditioned densities at $t = 12\,\tau$ from starts with tails $x^{-1-\nu}$, and from a point start at $x_0 = 2\ell$. Dashed, the QSDs q_ν and the Rayleigh density. (c) Decay rate of S selected by the initial tail, in units of λ_b ($1/2\tau$ for drift, k for OU). Curves are the predictions; dots are measured from the exact evolution.

- [G (5.2.17)]–[G (5.2.18)] are not boundary conditions. Continuity of p and J holds at every point, because $\partial_t p = -\partial_x J$ needs it, and at the "seam" it holds for the same reason as anywhere else. A genuine discontinuity of A or B is a property of the coefficients at that point; [G (5.2.15)]–[G (5.2.16)] then apply there, wherever it is (Section 26).
- Probability is conserved because the circle has no boundary, $\frac{\mathrm{d}}{\mathrm{d}t}\oint p\,\mathrm{d}x = -\oint \partial_x J\,\mathrm{d}x = 0$, not because two currents happen to match. The backward equation needs no condition either: the surface term [G (5.2.65)], which produces [G (5.2.67)] and [G (5.2.69)], is absent.
- Observables must be functions on the circle too. The coordinate x is not one, since it jumps at the cut, so $\langle x\rangle$ and the correlation function [G (5.2.90)] depend on where the cut is. The natural observables are the Fourier modes $e^{iq_m x}$, $q_m = 2\pi m/L$.

The universal cover. The map $\mathbb{R} \to S^1$, $\tilde{x} \mapsto \tilde{x} \bmod L$, unwraps the circle. A path on the circle lifts to a unique path $\tilde{x}(t)$ on $\mathbb{R}$ once its starting point is lifted, and the lift obeys the same Fokker–Planck equation with the coefficients repeated periodically. Summing over where the lift can end,

$$p_{S^1}(x,t\,|\,x_0,0) = \sum_{n\in\mathbb{Z}} p_{\mathbb{R}}(x+nL,t\,|\,x_0,0). \tag{1.16}$$

This is the method of images for the group of translations by L, and every image enters with a plus sign. A mirror imposes a condition, which is why images could change sign on the intervals; a translation imposes nothing, it is just another way of arriving at the same point.

Winding. The lift gives the circle an observable that no interval has: the unwrapped displacement $\Delta\tilde{x}(t) = \tilde{x}(t) - \tilde{x}(0)$, the net distance travelled around the circle. It exists because the circle contains

a loop that cannot be shrunk to a point. Its long-time drift and spread,

$$\langle \Delta\tilde{x} \rangle \simeq v_{\mathrm{w}} t, \qquad \operatorname{Var} \Delta\tilde{x} \simeq D_{\mathrm{eff}}\, t, \tag{1.17}$$

are the transport coefficients of the circle: the winding velocity v_{w} and the effective diffusion constant.

- $v_{\mathrm{w}} = \oint J\,\mathrm{d}x = LJ_s$: the stationary current is the winding rate.
- Let $T_{\mathrm{w}}(x_0)$ be the mean time for the lift to first get a full circumference from its start, either way: the exit time from $(x_0 - L, x_0 + L)$ on the cover, an absorbing-interval problem [G (5.2.158)]. After each such event the process starts afresh at an equivalent point, so the windings form a random walk with steps $\pm L$. If $v_{\mathrm{w}} = 0$, the renewal theorem gives $D_{\mathrm{eff}} = L^2/T_{\mathrm{w}}$; in particular T_{w} is then the same from every starting point.
- Both coefficients also follow from one eigenvalue. $\langle e^{s\Delta\tilde{x}} \rangle \sim e^{\Lambda(s)t}$, where $\Lambda(s)$ is the top eigenvalue of the generator with $\partial_x \to \partial_x - s$, acting on periodic functions; then $v_{\mathrm{w}} = \Lambda'(0)$ and $D_{\mathrm{eff}} = \Lambda''(0)$. The script computes both this way, by perturbation theory in s.

Stationary state and current. Let $\varphi(x) = \int^x 2A/B$ on the cover, so that $\psi = e^{\varphi}$ is Gardiner's [G (5.2.44)], and let

$$\Phi = \oint \frac{2A}{B}\,\mathrm{d}x = \varphi(x + L) - \varphi(x), \tag{1.18}$$

which does not depend on x. Integrate $J = \text{const}$ as in [G (5.2.45)], but once around the loop from x back to itself rather than from a seam:

$$p_s(x) = \frac{\mathcal{N}}{B(x)} \int_x^{x+L} e^{\varphi(x)-\varphi(y)}\,\mathrm{d}y, \qquad J_s = \frac{\mathcal{N}}{2}(1 - e^{-\Phi}), \tag{1.19}$$

with $\mathcal{N}$ fixed by $\oint p_s = 1$. This is [G (5.2.47)], rewritten so that the only point it refers to is x itself; the script checks that the two agree.

- **Detailed balance is topological.** $J_s = 0$ exactly when $\Phi = 0$, that is, when $2A/B$ is the derivative of a function on the circle, a single-valued potential. Then ψ is a function on the circle and Eq. (1.19) reduces to the potential solution Eq. (1.5). If $\Phi \neq 0$ a current with the sign of Φ circulates forever: a non-equilibrium steady state (NESS). Adding a periodic potential to A changes how large J_s is, but never whether it vanishes.
- **Misprint.** [G (5.2.46)] as printed reads $J = [B(b)/\psi(b) - B(a)/\psi(a)]\, p_s(a) / \int_a^b \mathrm{d}x/\psi$. Setting $x = b$ in [G (5.2.45)] and using $p_s(b) = p_s(a)$ gives instead $J = [B(a)/\psi(a) - B(b)/\psi(b)]\, p_s(a)/2\int_a^b \mathrm{d}x/\psi$: the sign is reversed and a factor $\frac{1}{2}$ is missing. For constant drift the printed form gives $J = -2v/L$ instead of v/L (Section 17). [G (5.2.47)] is correct.

Spectrum. If $\Phi = 0$ the operator is self-adjoint with weight $1/p_s$ and [G §5.2.5] applies unchanged, with Q periodic. If $\Phi \neq 0$ no weight makes it self-adjoint, and eigenvalues can come in complex-conjugate pairs: probability circulates while it relaxes. The gap is then the smallest nonzero $\operatorname{Re}\lambda$.

Folding. A reflection of the circle fixes two antipodal points, c and $c + L/2$. If B is even and A odd under it, the problem splits into an even sector and an odd sector. Even densities have $J = 0$ at both fixed points and odd ones have $p = 0$ there, so the even sector is the interval $(c, c + L/2)$ with reflecting walls and the odd sector the same interval with absorbing walls. This is how the circle connects to the intervals.

First passage. A point becomes special only when a question makes it special. To find when the process first reaches a target c, cut the circle there. It becomes the interval $(c, c+L)$ with both ends absorbing, the two ends being the target approached from either side, and the splitting probabilities say which way round the particle arrived. The first full winding, likewise, is the absorbing interval $(x_0 - L, x_0 + L)$ on the cover.

1.10 How the notes are built

Parts I–III each treat one drift in the same seven geometries, ordered so that each section mostly uses earlier ones:

1. the whole line, which gives that part's free propagator;
2. the half-line, reflecting at 0: one image of the same sign as the source;
3. the half-line, absorbing at 0: one image of opposite sign;
4. the interval, both walls reflecting: the image of 2 repeated at both walls, or equivalently the modes of 2 quantised by the second wall;
5. the interval, both walls absorbing: the same with 3;
6. the interval, reflecting at 0 and absorbing at L: unfolded at the mirror into 5 on an interval of length $2L$;
7. the circle, with no walls at all: 1 wrapped up, every image of the same sign. Cut at a target it becomes 5; folded by a reflection it becomes 4 and 5 on half its length.

The three parts then build on each other:

- **Part I** ($A = 0$) builds everything from the Gaussian by images.
- **Part II** ($A = v$) maps each Part I result through one transformation, Eq. (10.2). Absorbing walls carry over unchanged; reflecting walls need one new ingredient, a Robin condition. On the circle the transformation is unavailable, for a topological reason, and a Galilean boost replaces it.
- **Part III** ($A = -kx$) maps walls at the bottom of the potential onto Part I through Gardiner's substitution $y = xe^{kt}$. A wall elsewhere needs Kummer functions, and the results cross over from Part I (small L) to the half-line (large L). OU fits on a circle only with a special point; a smooth periodic well, with and without a tilt, closes the part.
- **Part IV** changes the potential instead of the geometry. The factorisation of the symmetrised operator explains why Parts I–III are solvable and finds the other solvable potentials; the logarithm, $\ln\cosh$ and $-\ln\cos$ are worked out, each joining two earlier cases.
- **Part V** collects what holds for any potential (exact quadratures, bounds, small-noise asymptotics, numerics) and applies it to the double well.
- **Part VI** works the boundary types of Gardiner's [G §5.2.1] that the walls of Parts I–V do not cover: an interface where B jumps, a boundary where B vanishes, and a current through infinity. It ends by sorting every boundary in the notes into Feller's four types (Table 9).
- **Part VII** summarises all the reductions in one table and one diagram, with a procedure for new problems (Section 40).

Where the equation itself comes from, a Brownian particle in the overdamped limit, is in Section 1.2. Each section opens with a box of key results, which names Gardiner's eigenfunctions P_λ and Q_λ (Section 1.4). A paragraph *Physical picture* then names a system the example describes, and a paragraph *From Gardiner* says which equations of the book produce the results. Derivations, a figure, and pointers to the numerical checks follow. A blank template for new sections is at the end of the `.tex` source.

Part I
Free diffusion: $A = 0$, $B = D$

2 Free diffusion on the whole line

Setup and key results

$A = 0$, $B = D$ on $(-\infty, \infty)$, no walls: the Wiener process $\mathrm{d}x = \sqrt{D}\,\mathrm{d}W$ [G §3.8.1].

$$P_k(x) = \frac{e^{ikx}}{2\pi}, \quad Q_k(x) = e^{-ikx}, \quad \lambda = \tfrac{1}{2}Dk^2, \quad k \in \mathbb{R} \ \text{(continuous; } \pm k \text{ degenerate)} \tag{2.1}$$

$$p(x, t \,|\, x_0, 0) = g(x - x_0, t) = \frac{1}{\sqrt{2\pi Dt}}\, e^{-(x-x_0)^2/2Dt}, \qquad \langle e^{ikx(t)} \rangle = e^{ikx_0 - Dk^2 t/2} \tag{2.2}$$

$$\langle x(t) \rangle = x_0, \qquad \operatorname{Var} x(t) = Dt, \qquad \langle x(t)x(s) \rangle - x_0^2 = D \min(t, s) \tag{2.3}$$

$$\tilde{g}(y, s) \equiv \int_0^\infty e^{-st} g(y, t)\,\mathrm{d}t = \frac{e^{-k|y|}}{Dk}, \qquad k = \sqrt{2s/D} \tag{2.4}$$

First passage to a level a is Section 4 with $x_0 \to |x_0 - a|$.

Physical picture. No force and no walls: a colloidal bead or a molecule diffusing in a fluid far from any surface, Brown's pollen grains, or ink spreading in still water. Here $\mathcal{U} = 0$, and $D/2 = k_B T/\gamma$ is the Einstein diffusion coefficient that Perrin measured from exactly this spreading [6]. It is also the continuum limit of a random walker: steps of $\pm\delta$ every Δt give $D = \delta^2/\Delta t$ [G §3.8.2].

From Gardiner. [G §3.8.1] solves this case with $B = 1$ by the characteristic function $\phi(s, t) = \langle e^{isx} \rangle$. It obeys $\partial_t \phi = -\frac{1}{2}s^2\phi$ [G (3.8.4)], so $\phi = \exp(isx_0 - \frac{1}{2}s^2 t)$ [G (3.8.6)], and inverting gives the Gaussian [G (3.8.7)] with mean and variance [G (3.8.8)]–[G (3.8.9)]. With $B = D$ the only change is $s^2 \to Ds^2$. The two-time function follows from independent increments, and the resolvent from a Laplace transform; neither is in the book.

2.1 The building block

Every propagator in Part I is a signed sum of copies of g, one per image (Fig. 14a,b):

- a reflecting wall adds an image of the same sign, an absorbing wall one of opposite sign (Sections 3 and 4);
- two walls give an image lattice of period $2L$ (Sections 5 and 6); mixed walls alternate the sign, which doubles the period to $4L$ (Section 7);
- on a circle every translate by L enters with a plus sign, since there is nothing to reflect (Section 8).

In Fourier space $\partial_t \hat{p} = -\frac{1}{2}Dk^2\hat{p}$. In Gardiner's notation the modes are $P_k = e^{ikx}/2\pi$ and $Q_k = e^{-ikx}$, with $\int P_k Q_{k'}\,\mathrm{d}x = \delta(k - k')$, and [G (5.2.88)] with $\sum \to \int \mathrm{d}k$ is the Fourier integral of g. The weight is the uniform zero-current solution: $P_k = p_s \overline{Q_k}$ with $p_s = 1/2\pi$, or, with the real pair $Q_k = \cos kx$, $\sin kx$ ($k > 0$), $P_k = Q_k/\pi$. The walls select which combinations of $e^{\pm ikx}$ are allowed (cos, sin) and, on an interval, which k. The spectrum here is continuous down to $\lambda = 0$: no gap (Section 1.6).

2.2 Recurrence and correlations

- As $s \to 0$, $\tilde{g}(0,s) = 1/\sqrt{2Ds}$ diverges, so the expected time spent near the starting point, $\int_0^\infty g(0,t)\,\mathrm{d}t$, is infinite. One-dimensional Brownian motion is recurrent: it returns to every level infinitely often. That is why the half-line walls of Sections 3 and 4 are always reached.
- The increments are independent, so $\langle x(t)x(s)\rangle = x_0^2 + D\min(t,s)$. There is no stationary state and so no stationary correlation function.

3 Free diffusion on the half-line, reflecting at 0

Setup and key results

$A = 0$, $B = D$ on $(0,\infty)$, $\partial_x p = 0$ at $x = 0$. The process is reflected Brownian motion, $x(t) = |x_0 + \sqrt{D}\,W(t)|$. Lengths in the figures are in units of x_0, times in x_0^2/D.

$$P_k(x) = \frac{2}{\pi}\cos kx, \quad Q_k(x) = \cos kx, \quad \lambda = \tfrac{1}{2}Dk^2, \quad k \geq 0 \quad \text{(no stationary state)} \tag{3.1}$$

$$p(x,t\,|\,x_0,0) = g(x-x_0,t) + g(x+x_0,t) = \frac{2}{\pi}\int_0^\infty \cos kx\,\cos kx_0\,e^{-Dk^2t/2}\,\mathrm{d}k \tag{3.2}$$

$$\langle x(t)\rangle = x_0\,\mathrm{erf}\,\frac{x_0}{\sqrt{2Dt}} + \sqrt{\frac{2Dt}{\pi}}\,e^{-x_0^2/2Dt} \xrightarrow{t\to\infty} \sqrt{\frac{2Dt}{\pi}} \tag{3.3}$$

$$\langle x^2(t)\rangle = x_0^2 + Dt \quad \text{(exact)}, \qquad \sigma_x^2 \to \Big(1 - \frac{2}{\pi}\Big)Dt \tag{3.4}$$

$$p(x,t\,|\,x_0,0) \simeq \sqrt{\frac{2}{\pi Dt}}\,e^{-x^2/2Dt} \quad (Dt \gg x_0^2) \quad \text{(half-Gaussian)} \tag{3.5}$$

Physical picture. A hard wall (Fig. 1a): a neutrally buoyant bead above the floor of its container, or a molecule beside a membrane it cannot cross. $\mathcal{U} = 0$ for $x > 0$ and $\mathcal{U} \to \infty$ for $x < 0$. Nothing holds the particle near the wall, so it wanders away and never settles.

From Gardiner. The reflecting condition [G (5.2.13)], $J = 0$, reduces to $\partial_x p = 0$ when $A = 0$ and B is constant. The book does not use images. It reaches the same propagator through the cosine eigenfunctions of [G §5.2.6b]; the integral form of Eq. (3.2) is the $L \to \infty$ limit of [G (5.2.107)], which reappears in Section 5.

3.1 Images and spectrum

A reflecting wall is a mirror. Add to the free Gaussian of Section 2 a copy centred at the mirror point $-x_0$. The sum is even in x, so $\partial_x p = 0$ at the wall, and it solves the free equation, so it is the answer Eq. (3.2). In Fourier language the wall keeps only the even combination $\cos kx$ of the line's modes $e^{\pm ikx}$. It satisfies both reflecting conditions, $J = 0$ forward and $Q' = 0$ backward [G (5.2.69)], so P_k and Q_k are both cosines; $\int_0^\infty \cos kx\cos k'x\,\mathrm{d}x = \frac{\pi}{2}\delta(k-k')$ fixes the factor $2/\pi$, which plays the role of the constant, non-normalisable p_s. The spectrum is still continuous down to $\lambda = 0$; a second wall will quantise k and open a gap (Section 5).

With no gap there is no exponential relaxation (Section 1.6). The would-be stationary state, a uniform density, cannot be normalised. Instead p spreads self-similarly: at fixed x, $p \sim (2/\pi Dt)^{1/2}$ decays as $t^{-1/2}$, and the memory of x_0 fades as $p = 2g(x,t)\,[1 + O(xx_0/Dt)]$.

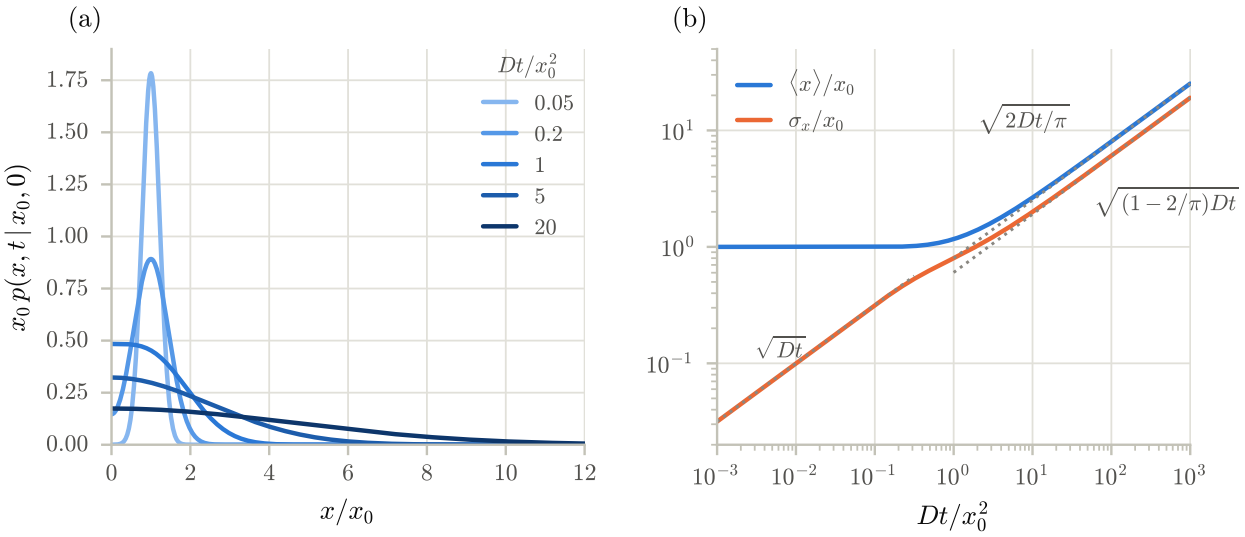


Figure 6: Half-line, reflecting at 0. Lengths are in units of x_0. (a) Density snapshots. The density spreads without limit and develops zero slope at the wall. (b) Mean and standard deviation, with their short- and long-time asymptotes (dotted).

3.2 Moments

Since $x = |y|$ with $y = x_0 + \sqrt{D}\,W$ Gaussian, the moments are those of a folded normal.

- $\langle x^2\rangle = \langle y^2\rangle = x_0^2 + Dt$ exactly. The wall does not change the second moment, because the current through $x = 0$ vanishes and x^2 is even.
- $\langle x\rangle$ grows from x_0 to $\sqrt{2Dt/\pi}$; the wall pushes the mean outward.
- $\sigma_x \approx \sqrt{Dt}$ while the wall is not yet felt, and $\sigma_x \to \sqrt{(1-2/\pi)Dt} \approx 0.60\sqrt{Dt}$ once it is (Fig. 6b).

Passage to a level. The time to first reach $x = L > x_0$ is the exit time of Section 7: $\langle T\rangle = (L^2 - x_0^2)/D$, finite because of the wall. Compare Section 4, where the mean first-passage time to 0 is infinite.

4 Free diffusion on the half-line, absorbing at 0

Setup and key results

$A = 0$, $B = D$ on $(0, \infty)$, $p = 0$ at $x = 0$. Brownian motion killed on first reaching 0; equivalently, the first passage of free Brownian motion to a level a distance x_0 away.

$$P_k(x) = \frac{2}{\pi} \sin kx, \quad Q_k(x) = \sin kx, \quad \lambda = \tfrac{1}{2}Dk^2, \quad k > 0 \quad \text{(continuous; no QSD)} \tag{4.1}$$

$$p(x, t \,|\, x_0, 0) = g(x - x_0, t) - g(x + x_0, t) = \frac{2}{\pi} \int_0^\infty \sin kx \, \sin kx_0 \, e^{-Dk^2 t/2} \, \mathrm{d}k \tag{4.2}$$

$$S(t \,|\, x_0) = \operatorname{erf} \frac{x_0}{\sqrt{2Dt}} \xrightarrow{t \to \infty} x_0 \sqrt{\frac{2}{\pi Dt}} \tag{4.3}$$

$$f(t \,|\, x_0) = \frac{x_0}{\sqrt{2\pi D t^3}} \, e^{-x_0^2/2Dt}, \qquad \tilde{f}(s) = e^{-x_0\sqrt{2s/D}} \quad \text{(Lévy–Smirnov)} \tag{4.4}$$

$$\Pr(T < \infty) = 1, \qquad \langle T \rangle = \infty, \qquad t_{\text{mode}} = \frac{x_0^2}{3D}, \qquad t_{\text{median}} \approx 2.20 \, \frac{x_0^2}{D} \tag{4.5}$$

$$q(x, t) \to \frac{x}{Dt} \, e^{-x^2/2Dt} \quad \text{(Rayleigh)}, \qquad \kappa(t) \to \frac{1}{2t} \tag{4.6}$$

Physical picture. A cliff (Fig. 1b): a random walker on a plateau whose edge is at $x = 0$; whoever reaches the edge falls off and is gone. Physically, a perfectly sticky or reactive surface, a detector, or an exit. It is also the simplest random-walk neuron: the membrane potential wanders and the cell fires when it first reaches threshold [7]. Without drift the time to fire is the Lévy–Smirnov first-passage time, with an infinite mean.

From Gardiner. The wall is [G (5.2.14)]. The survival probability is Gardiner's $G(x_0, t)$ [G (5.2.140)], which obeys the backward equation [G (5.2.144)] with $G(x_0, 0) = 1$ and $G(0, t) = 0$ [G (5.2.145)]–[G (5.2.146)]; $\operatorname{erf}(x_0/\sqrt{2Dt})$ is that solution. The mean exit time for "b reflecting, a absorbing" [G (5.2.161)] with $\psi = 1$ gives $T = \frac{2}{D} \int_0^{x_0} (b - y) \, \mathrm{d}y$, which diverges as the reflecting wall b moves to infinity: that is the $\langle T \rangle = \infty$ in the box.

4.1 Images and first passage

The absorbing wall needs $p = 0$ at $x = 0$, so the copy of g at $-x_0$ now enters with a minus sign: the image of Section 3 with the opposite sign. Integrating over $x > 0$ gives S, and $f = -\partial_t S$ is the Lévy–Smirnov density Eq. (4.4). The spectrum keeps only $\sin kx$, which vanishes at the wall as both P and Q must [G (5.2.91)]; the weight $2/\pi$ is the same as with the reflecting wall.

Because the line is recurrent (Section 2), every path is eventually absorbed. But $S \sim t^{-1/2}$ goes to zero too slowly for $\int S \, \mathrm{d}t = \langle T \rangle$ to converge. The typical time is finite: the density f peaks at $x_0^2/3D$, and half of all paths are absorbed by $2.20 \, x_0^2/D$, where $\operatorname{erf}(x_0/\sqrt{2Dt}) = \frac{1}{2}$.

The same function answers a question about free motion on the line (reflection principle). The probability that a free path from x_0 has hit a level a by time t is $\Pr(\min_{s \le t} x(s) \le a) = \operatorname{erfc}((x_0 - a)/\sqrt{2Dt}) = 1 - S$.

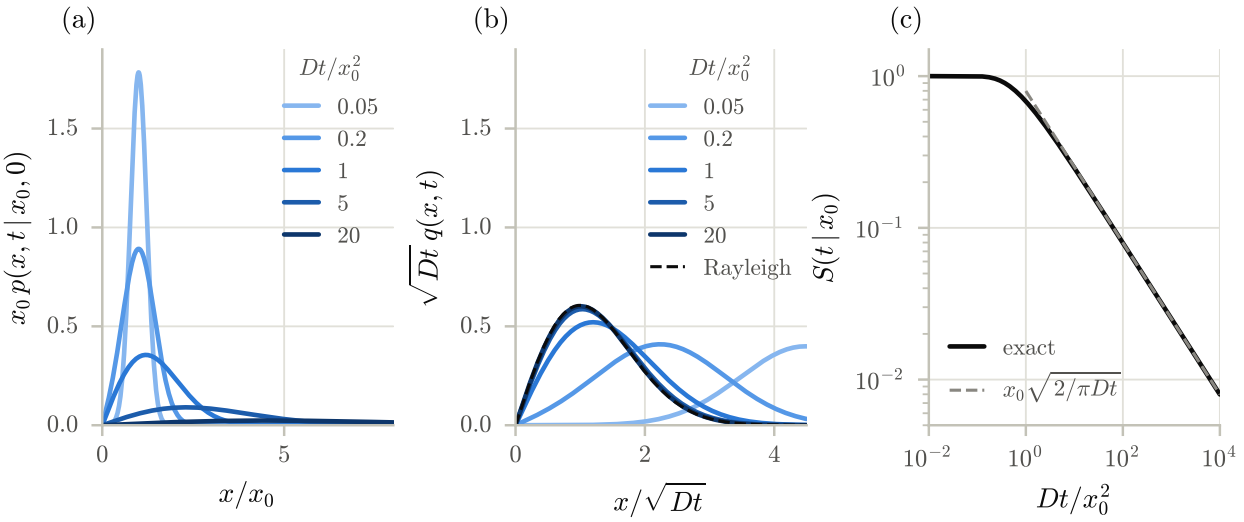


Figure 7: Half-line, absorbing at 0; lengths in units of x_0. (a) Density snapshots: mass leaks out at the wall and the remainder spreads. (b) The conditioned density, rescaled by the diffusion length $\sqrt{Dt}$, collapses onto the Rayleigh profile $u\,e^{-u^2/2}$ for $Dt \gg x_0^2$; it does not settle in x. (c) Survival decays algebraically, $S \sim x_0\sqrt{2/\pi Dt}$.

4.2 Conditioned on survival: no QSD

For $Dt \gg x_0^2$, $g(x-x_0) - g(x+x_0) \simeq 2g(x,t)\,xx_0/Dt$. Dividing by S gives

$$q(x,t) \simeq \frac{x}{Dt}\,e^{-x^2/2Dt}, \tag{4.7}$$

a Rayleigh density with scale $\sqrt{Dt}$ and mean $\sqrt{\pi Dt/2}$. It keeps spreading, so there is no quasi-stationary state (Fig. 7b). This is the result in `qsd-context.md` with $D' = D/2$. The exit rate $\kappa = f/S \to 1/2t$ goes to zero.

- Why no QSD: the spectrum starts at $\lambda = 0$ with no gap, and no $\sin kx$ is positive. The $k \to 0$ limit of the eigenfunctions, $P_k/k \to 2x/\pi$, is positive but not normalisable. The Rayleigh profile is that x times a spreading Gaussian envelope.
- With a drift toward the wall a continuum of QSDs appears; that comes in Part II.
- Conditioning on never being absorbed is the Doob transform with $h = x$. It adds the drift D/x, which gives the three-dimensional Bessel process. The same transform reappears in Section 6, for paths conditioned to leave through the far wall.
- A confining force freezes the Rayleigh profile into a QSD, the one a point start reaches: see the Ornstein–Uhlenbeck version, which also has heavier-tailed QSDs, Section 22.

5 Free diffusion, reflecting walls

Setup and key results

$A = 0$, $B = D$ on $(0, L)$, $\partial_x p = 0$ at $x = 0, L$. SDE $\mathrm{d}x = \sqrt{D}\,\mathrm{d}W$, reflected at the walls. Gardiner [G §5.2.6b] does the case $L = D = 1$.

$$\lambda_n = \frac{n^2\pi^2 D}{2L^2}, \quad n = 0, 1, 2, \ldots \quad Q_0 = 1, \quad Q_n(x) = \sqrt{2}\cos\frac{n\pi x}{L}, \quad P_n = \frac{Q_n}{L} \tag{5.1}$$

$$\begin{aligned} p(x, t \,|\, x_0, 0) &= \frac{1}{L} + \frac{2}{L}\sum_{n\geq 1} \cos\frac{n\pi x}{L}\cos\frac{n\pi x_0}{L}\, e^{-\lambda_n t} \\ &= \sum_{m\in\mathbb{Z}} \left[g(x - x_0 + 2mL, t) + g(x + x_0 + 2mL, t)\right] \end{aligned} \tag{5.2}$$

$$p_s = 1/L, \qquad \text{relaxation time } 1/\lambda_1 = \frac{2L^2}{\pi^2 D} \approx 0.203\,\frac{L^2}{D} \tag{5.3}$$

$$C(\tau) \equiv \langle x(\tau)x(0)\rangle_s - \langle x\rangle_s^2 = \frac{8L^2}{\pi^4}\sum_{n \text{ odd}} \frac{e^{-\lambda_n|\tau|}}{n^4}, \qquad C(0) = \frac{L^2}{12} \tag{5.4}$$

with free kernel $g(y, t) = (2\pi D t)^{-1/2} e^{-y^2/2Dt}$.

Physical picture. A particle in a closed box or channel of length L, with hard walls at both ends: a bead in a sealed microfluidic chamber, or a protein in a cell compartment. $\mathcal{U} = 0$ inside. The relaxation time $2L^2/\pi^2 D$ is the mixing time of the box.

From Gardiner. [G §5.2.6b] does exactly this case for $L = D = 1$. It gives the cosine series [G (5.2.102)]–[G (5.2.107)], the stationary autocorrelation [G (5.2.109)]–[G (5.2.110)], and the identity [G (5.2.114)]. He does not name P_n and Q_n there, but [G (5.2.107)] is [G (5.2.88)] with $P_n = Q_n = \sqrt{2}\cos n\pi x$, since $p_s = 1$. To restore L and D, replace $n\pi x$ by $n\pi x/L$ and $\lambda_n = \frac{1}{2}n^2\pi^2$ [G (5.2.106)] by $n^2\pi^2 D/2L^2$. The stationary density $1/L$ is Eq. (1.5) with $A = 0$. The moments, τ_c and the image form are additions.

5.1 Spectrum and propagator

Start from the half-line of Section 3, whose modes are $\cos kx$ for every $k \geq 0$. The second wall at L needs $Q'(L) = 0$ as well, which allows only $k_n = n\pi/L$: Eq. (5.1). The continuum becomes a ladder, and a gap $\lambda_1 = \pi^2 D/2L^2$ opens (Section 1.6). Now $p_s = 1/L$ is normalisable, and Gardiner's normalisation $\int p_s Q_n^2 = 1$ gives $Q_0 = 1$, $Q_n = \sqrt{2}\cos(n\pi x/L)$ and $P_n = p_s Q_n = Q_n/L$, which is the mode sum in Eq. (5.2).

The image form says the same thing in position space. Mirror the source at 0, as in Section 3; that pair must also be mirrored at L, which gives another pair, which must be mirrored at 0, and so on. The result is a lattice of image pairs with period $2L$. Mode sum and image sum are the same function by Poisson summation. The image sum converges fast for $Dt \ll L^2$ (only the source and the nearest images matter), the mode sum for $Dt \gtrsim L^2$. They agree to machine precision (Appendix A).

5.2 Relaxation and moments

Every mode but $n = 0$ decays, so $p \to 1/L$. The slowest rate is λ_1, unless the start is symmetric: for $x_0 = L/2$ all odd modes vanish, since $\cos(n\pi/2) = 0$, and the relaxation goes at $\lambda_2 = 4\lambda_1$.

Using $\int_0^L x \cos(n\pi x/L)\,\mathrm{d}x = [(-1)^n - 1]L^2/n^2\pi^2$ and $\int_0^L x^2 \cos(n\pi x/L)\,\mathrm{d}x = 2(-1)^n L^3/n^2\pi^2$,

$$\langle x(t)\rangle = \frac{L}{2} - \frac{4L}{\pi^2} \sum_{n \text{ odd}} \frac{1}{n^2} \cos\frac{n\pi x_0}{L}\, e^{-\lambda_n t}, \tag{5.5}$$

$$\langle x^2(t)\rangle = \frac{L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n\geq 1} \frac{(-1)^n}{n^2} \cos\frac{n\pi x_0}{L}\, e^{-\lambda_n t}. \tag{5.6}$$

At $t = 0$ these reduce to x_0 and x_0^2, by the Fourier series of θ and θ^2 on $(0, \pi)$. For $Dt \ll \min(x_0, L - x_0)^2$ the walls are not felt and $\sigma_x^2 \approx Dt$; for $Dt \gg L^2$, $\langle x\rangle \to L/2$ and $\sigma_x^2 \to L^2/12$ (Fig. 8b).

5.3 Stationary autocorrelation

Only odd modes have $\int x\, P_n \neq 0$, which gives $C(\tau)$ in the box [G (5.2.110)]. The identity $\sum_{n \text{ odd}} n^{-4} = \pi^4/96$ [G (5.2.114)] gives $C(0) = L^2/12$, the variance of the uniform distribution.

- The slowest mode carries a fraction $96/\pi^4 \approx 0.986$ of the variance, so $C(\tau)$ is a single exponential to within 1.5 % (Fig. 8c).
- The integrated correlation time is exact: with $\sum_{n \text{ odd}} n^{-6} = \pi^6/960$,

$$\tau_c = \int_0^\infty \frac{C(\tau)}{C(0)}\,\mathrm{d}\tau = \frac{L^2}{5D} \qquad \text{vs.} \qquad \frac{1}{\lambda_1} = \frac{2L^2}{\pi^2 D} \approx \frac{L^2}{4.93\,D}. \tag{5.7}$$

- The initial slope is $C'(0^+) = -D/2$, since $C(0) - C(\tau) = \frac{1}{2}\langle[x(\tau) - x(0)]^2\rangle \approx \frac{1}{2}D\tau$. The one-mode approximation gives $-4D/\pi^2 \approx -0.41\,D$ instead; the missing part sits in the fast modes.

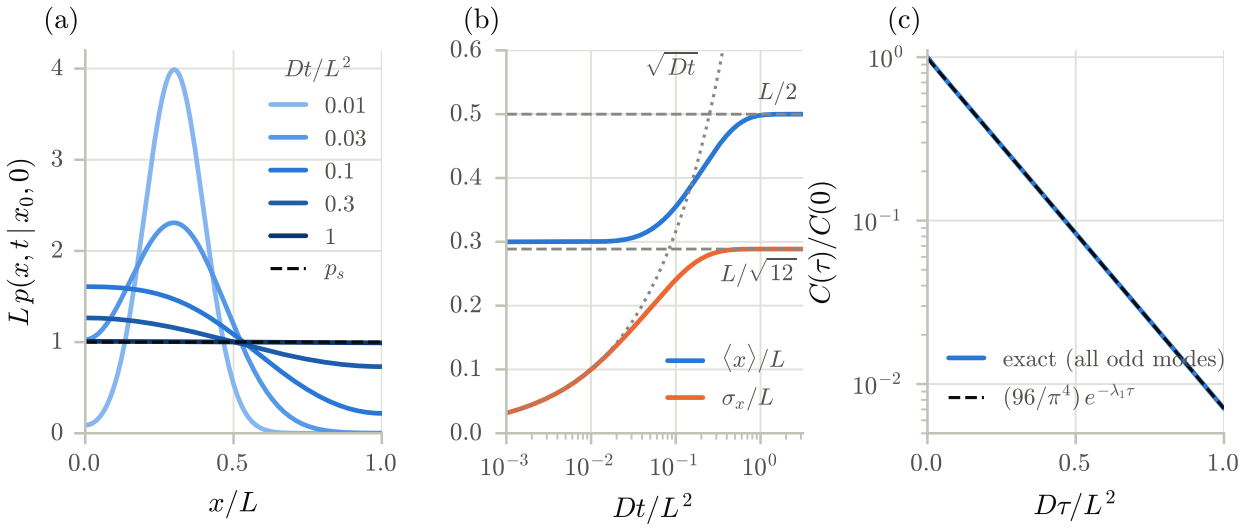


Figure 8: Reflecting walls, $x_0 = 0.3L$. (a) Density snapshots relaxing to $p_s = 1/L$. (b) Mean and standard deviation. At short times $\sigma_x \approx \sqrt{Dt}$ (dotted); both saturate on the time scale $1/\lambda_1 \approx 0.2\,L^2/D$. Here x_0 happens to lie close to $L/\sqrt{12}$, so $\langle x\rangle$ starts near the lower dashed line. (c) Normalised stationary autocorrelation. The exact series and the single slowest mode are indistinguishable on this scale.

6 Free diffusion, absorbing walls

Setup and key results

$A = 0$, $B = D$ on $(0, L)$, $p = 0$ at $x = 0, L$. SDE $\mathrm{d}x = \sqrt{D}\,\mathrm{d}W$, killed at the walls. Gardiner [G §5.2.6a] does the propagator for $L = D = 1$; first passage is [G §5.2.7]–[G §5.2.8].

$$\lambda_n = \frac{n^2\pi^2 D}{2L^2}, \quad n = 1, 2, \ldots \quad Q_n(x) = \sqrt{2}\sin\frac{n\pi x}{L}, \quad P_n = \frac{Q_n}{L} \tag{6.1}$$

$$\begin{aligned} p(x,t\,|\,x_0,0) &= \frac{2}{L}\sum_{n\geq 1}\sin\frac{n\pi x}{L}\sin\frac{n\pi x_0}{L}\,e^{-\lambda_n t} \\ &= \sum_{m\in\mathbb{Z}}\left[g(x - x_0 + 2mL, t) - g(x + x_0 + 2mL, t)\right] \end{aligned} \tag{6.2}$$

$$S(t\,|\,x_0) = \frac{4}{\pi}\sum_{n\ \text{odd}}\frac{1}{n}\sin\frac{n\pi x_0}{L}\,e^{-\lambda_n t} \xrightarrow{t\to\infty} \frac{4}{\pi}\sin\frac{\pi x_0}{L}\,e^{-\lambda_1 t} \tag{6.3}$$

$$\pi_0 = 1 - \frac{x_0}{L}, \qquad \pi_L = \frac{x_0}{L} \tag{6.4}$$

$$T(x_0) = \frac{x_0(L - x_0)}{D}, \qquad \operatorname{Var} T = \frac{x_0(L - x_0)\left[x_0^2 + (L - x_0)^2\right]}{3D^2} \tag{6.5}$$

$$T_0(x_0) = \frac{x_0(2L - x_0)}{3D}, \qquad T_L(x_0) = \frac{L^2 - x_0^2}{3D} \tag{6.6}$$

$$q_\infty(x) = \frac{\pi}{2L}\sin\frac{\pi x}{L}, \qquad \kappa(t) \to \lambda_1 = \frac{\pi^2 D}{2L^2} \approx 4.93\,\frac{D}{L^2} \tag{6.7}$$

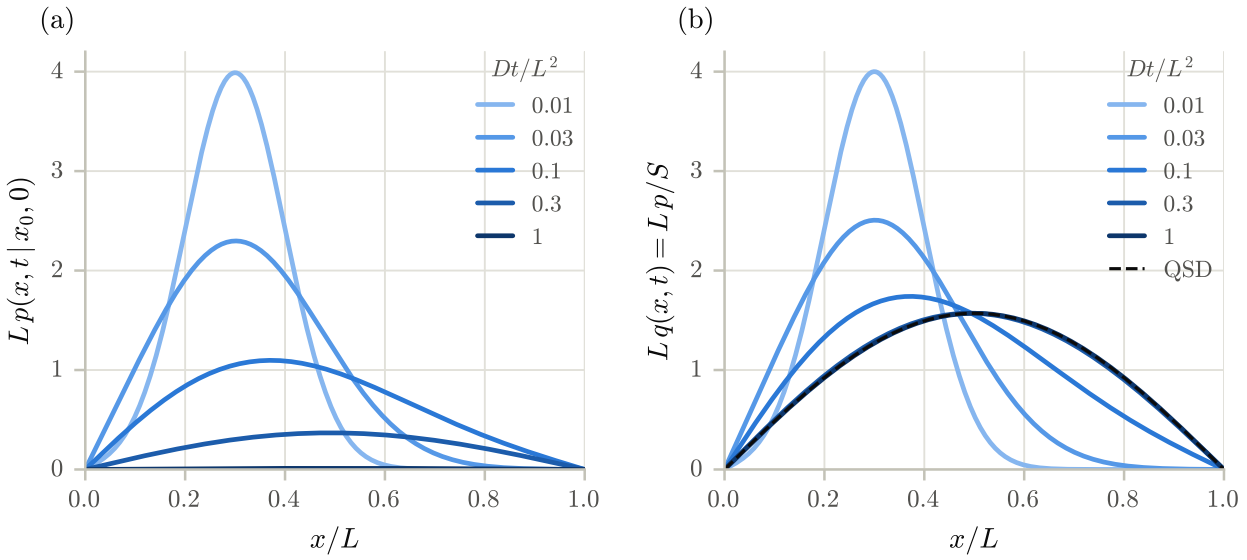


Figure 9: Absorbing walls, $x_0 = 0.3L$. (a) $p(x,t\,|\,x_0,0)$ loses mass through both walls. (b) The density conditioned on survival, $q = p/S$, converges to the QSD $\frac{\pi}{2L}\sin\frac{\pi x}{L}$ (dashed) by $Dt/L^2 \approx 0.3$.

Physical picture. A plateau between two cliffs, or a pore open at both ends from which a molecule escapes through either mouth. The classic discrete version is the gambler's ruin: a fortune of x_0 against $L - x_0$, changed by fair bets until one player is broke, with $\pi_L = x_0/L$.

From Gardiner. [G §5.2.6a] gives the sine series [G (5.2.92)]–[G (5.2.100)] for $L = D = 1$, with the absorbing-wall convention for P_λ, Q_λ of [G (5.2.91)]: [G (5.2.100)] is [G (5.2.88)] with $P_n = Q_n = \sqrt{2}\sin n\pi x$. With $\psi = 1$ the first-passage machinery of [G §5.2.7]–[G §5.2.8] integrates in elementary functions:

- T from [G (5.2.154)], or directly from [G (5.2.158)];
- T_2 from the hierarchy [G (5.2.156)] with $n = 2$;
- π_0, π_L from [G (5.2.186)]–[G (5.2.190)];
- T_0, T_L from [G (5.2.183)] with [G (5.2.185)].

The QSD is not in the book; it is the $n = 1$ term of [G (5.2.100)], normalised.

6.1 Spectrum and propagator

This is Section 4 with a second absorbing wall, exactly as Section 5 was Section 3 with a second mirror. The half-line's $\sin kx$ must also vanish at L, so $k_n = n\pi/L$. The nonzero spectrum is identical to the reflecting interval, but $n = 0$ is missing, so nothing survives. Gardiner's convention takes $p_s = 1/L$ from the reflecting problem [G (5.2.91)], so the normalisation is the same as there: $Q_n = \sqrt{2}\sin(n\pi x/L)$, $P_n = Q_n/L$, both zero at the walls. In the image sum each image is the negative mirror image of the one before, which makes p vanish at both walls.

Letting $L \to \infty$ recovers Section 4, and the interval results show how the half-line's peculiarities arise:

- $\pi_L = x_0/L \to 0$: the path reaches 0 with probability one (recurrence);
- $T = x_0(L - x_0)/D \to \infty$: it takes infinitely long on average;
- $\lambda_1 = \pi^2 D/2L^2 \to 0$: the gap closes and the decay of S becomes algebraic.

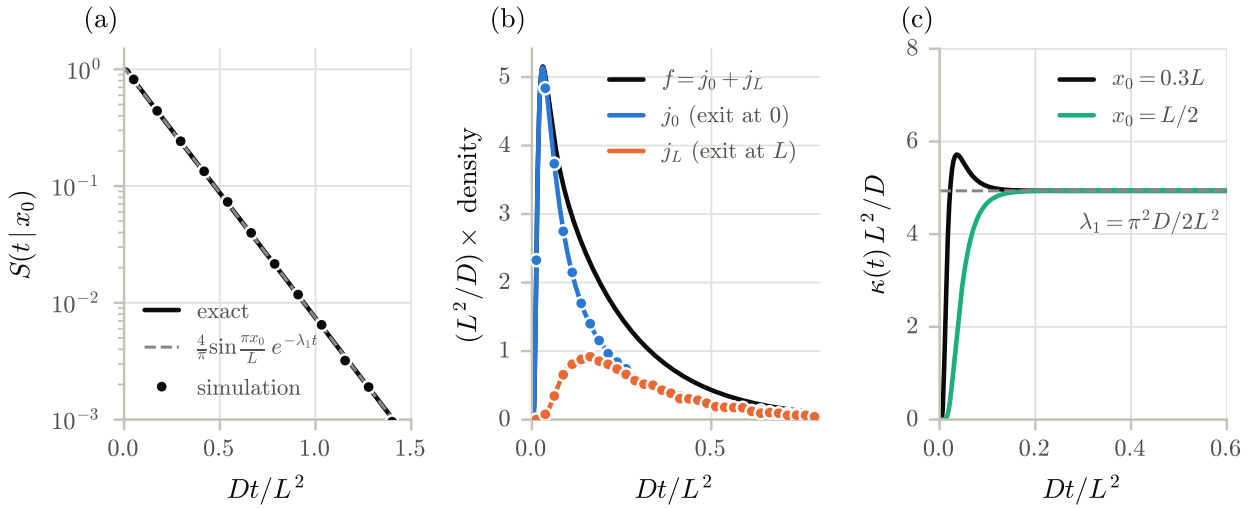


Figure 10: Absorbing walls, exit statistics. (a) Survival from $x_0 = 0.3L$ against the one-mode asymptote, with simulation points. (b) Exit-time density f and its split into exits through each wall, Eq. (6.9); dots are histograms of 5×10^4 simulated paths. (c) Exit rate $\kappa = f/S$ from two starting points. Both settle at λ_1; from $x_0 = 0.3L$ it overshoots first.

6.2 Survival, exit-time density, exit through each wall

Integrating the mode sum gives Eq. (6.3). Only odd n contribute, and $S(0) = 1$ because $\sum_{n\,\text{odd}} \sin(n\theta)/n = \pi/4$ on $(0,\pi)$. The exit-time density and the outward fluxes through each wall, $j_0 = \frac{1}{2}D\,\partial_x p|_{x=0}$ and $j_L = -\frac{1}{2}D\,\partial_x p|_{x=L}$, are

$$f(t\,|\,x_0) = -\partial_t S = \frac{2\pi D}{L^2}\sum_{n\ \text{odd}} n \sin\frac{n\pi x_0}{L}\, e^{-\lambda_n t}, \tag{6.8}$$

$$j_0(t\,|\,x_0) = \frac{\pi D}{L^2}\sum_{n\ge 1} n \sin\frac{n\pi x_0}{L}\, e^{-\lambda_n t} = \sum_{m\in\mathbb{Z}} \frac{x_0 + 2mL}{\sqrt{2\pi D t^3}} \exp\Big[-\frac{(x_0+2mL)^2}{2Dt}\Big], \tag{6.9}$$

and $j_L(t\,|\,x_0) = j_0(t\,|\,L - x_0)$. The image form is a sum of half-line Lévy–Smirnov densities Eq. (4.4), one per image; the $m = 0$ term is Section 4 itself. It dominates while $Dt \ll x_0^2$. In that regime the walls act independently: $1 - S \approx \operatorname{erfc}(x_0/\sqrt{2Dt}) + \operatorname{erfc}((L-x_0)/\sqrt{2Dt})$.

Laplace transforms. With $k = \sqrt{2s/D}$, solving $\frac{1}{2}D\tilde{f}'' = s\tilde{f}$ in x_0 with the appropriate wall values gives

$$\tilde{j}_0(s) = \frac{\sinh k(L-x_0)}{\sinh kL}, \qquad \tilde{j}_L(s) = \frac{\sinh kx_0}{\sinh kL}, \qquad \tilde{f}(s) = \frac{\cosh k(x_0 - L/2)}{\cosh(kL/2)}. \tag{6.10}$$

Expanding in s gives all the moments below, e.g. $\tilde{f} = 1 - s\,x_0(L-x_0)/D + O(s^2)$.

6.3 Splitting probabilities and exit times

With $A = 0$, $\psi = 1$ in [G (5.2.157)], and the backward equations of Section 1.7 integrate directly.

- $\pi_L = x_0/L$ [G (5.2.190)]: the gambler's-ruin result.
- $T(x_0) = x_0(L-x_0)/D$, which follows from $\frac{1}{2}DT'' = -1$. Its maximum is $L^2/4D$ at the centre; averaged over a uniform start it is $L^2/6D$.

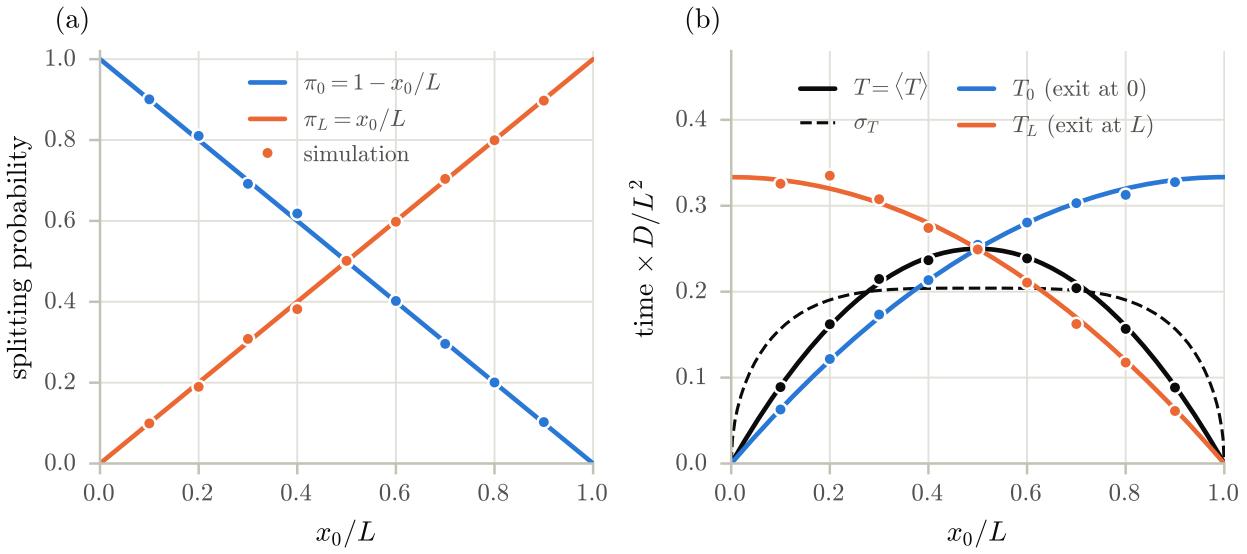


Figure 11: Absorbing walls, dependence on the starting point. (a) Splitting probabilities. (b) Mean exit time T, its standard deviation σ_T, and the conditional means T_0, T_L. Dots are simulation means over 4000 paths per starting point. At the centre $\sigma_T = 0.204\, L^2/D$, close to but not equal to $1/\lambda_1 = 0.203\, L^2/D$.

- The second moment is $T_2 = x_0(L - x_0)(L^2 + Lx_0 - x_0^2)/3D^2$, which gives $\operatorname{Var} T$ in the box. At the centre $\sigma_T = L^2/\sqrt{24}\, D$, so the coefficient of variation is $\sqrt{2/3} \approx 0.82$: the exit time is not sharply defined.
- Conditional exit times. Paths that start near 0 but leave through L take $T_L \to L^2/3D$. Conditioning on exit at L is the Doob transform with $h = \pi_L = x/L$; it adds the drift $B\,\partial_x \ln h = D/x$, which makes the conditioned process a three-dimensional Bessel process.

6.4 Long times: quasi-stationary distribution

For $t \gg 1/(\lambda_2 - \lambda_1)$ only $n = 1$ survives: $p \simeq \frac{2}{L} \sin\frac{\pi x_0}{L} \sin\frac{\pi x}{L} e^{-\lambda_1 t}$. Therefore

- $q = p/S \to q_\infty = \frac{\pi}{2L} \sin\frac{\pi x}{L}$, independent of x_0 (Fig. 9b).
- $\kappa(t) \to \lambda_1$. The approach goes at $\lambda_2 - \lambda_1 = 3\lambda_1$, or at $\lambda_3 - \lambda_1 = 8\lambda_1$ for $x_0 = L/2$, where the even modes vanish by symmetry (Fig. 10c).
- Starting in q_∞, $S = e^{-\lambda_1 t}$ exactly and T is exponential with mean $1/\lambda_1 = 2L^2/\pi^2 D$. Consistency check: $\int_0^L q_\infty(x_0)\, T(x_0)\, \mathrm{d}x_0 = 1/\lambda_1$, using $\int_0^L x(L - x) \sin(\pi x/L)\, \mathrm{d}x = 4L^3/\pi^3$.
- The QSD is the ground state P_1, normalised as a density: $q_\infty = P_1 / \int P_1$.
- Conditioning on survival *forever* (the Q-process) gives a different stationary density, $P_1 Q_1 = \frac{2}{L} \sin^2\frac{\pi x}{L}$, already normalised by [G (5.2.79)]. The process is $\mathrm{d}x = \frac{\pi D}{L} \cot\frac{\pi x}{L}\, \mathrm{d}t + \sqrt{D}\, \mathrm{d}W$.

Overshoot in κ. For $x_0 = 0.3L$, $\kappa(t)$ rises above λ_1 before settling. Early on, q sits nearer the wall at 0 than q_∞ does, so it leaks faster than the QSD. From $x_0 = L/2$ the density is far from both walls and κ rises monotonically.

6.5 Reflecting vs. absorbing

- The spectra are the same, $\{n^2\pi^2 D/2L^2\}$, except that only the reflecting case includes $n = 0$.

- The reflecting decay rate λ_1 toward p_s equals the absorbing decay rate of S. The absorbing q relaxes faster, at $3\lambda_1$.
- The eigenfunctions are cos vs. sin, and the image signs are $+$ vs. $-$.

7 Free diffusion, reflecting at 0, absorbing at L

Setup and key results

$A = 0$, $B = D$ on $(0, L)$, $\partial_x p = 0$ at $x = 0$, $p = 0$ at $x = L$. SDE $\mathrm{d}x = \sqrt{D}\,\mathrm{d}W$, reflected at 0 and killed at L. Every exit is through L. Gardiner gives T in [G (5.2.160)]. With $k_n = (2n+1)\pi/2L$:

$$\lambda_n = \frac{(2n+1)^2\pi^2 D}{8L^2}, \quad n = 0, 1, 2, \ldots \qquad Q_n(x) = \sqrt{2}\cos k_n x, \quad P_n = \frac{Q_n}{L} \tag{7.1}$$

$$\begin{aligned} p(x, t \,|\, x_0, 0) &= \frac{2}{L}\sum_{n\geq 0} \cos k_n x \, \cos k_n x_0 \, e^{-\lambda_n t} \\ &= \sum_{m\in\mathbb{Z}} (-1)^m \left[g(x - x_0 + 2mL, t) + g(x + x_0 + 2mL, t)\right] \end{aligned} \tag{7.2}$$

$$S(t \,|\, x_0) = \frac{4}{\pi}\sum_{n\geq 0} \frac{(-1)^n}{2n+1} \cos k_n x_0 \, e^{-\lambda_n t} \xrightarrow{t\to\infty} \frac{4}{\pi}\cos\frac{\pi x_0}{2L}\, e^{-\lambda_0 t} \tag{7.3}$$

$$f(t \,|\, x_0) = \sum_{m\in\mathbb{Z}} (-1)^m \frac{(2m+1)L - x_0}{\sqrt{2\pi D t^3}} \exp\Big[-\frac{((2m+1)L - x_0)^2}{2Dt}\Big], \qquad \tilde{f}(s) = \frac{\cosh kx_0}{\cosh kL} \tag{7.4}$$

$$T(x_0) = \frac{L^2 - x_0^2}{D}, \qquad \operatorname{Var} T = \frac{2(L^4 - x_0^4)}{3D^2} \tag{7.5}$$

$$q_\infty(x) = \frac{\pi}{2L}\cos\frac{\pi x}{2L}, \qquad \kappa(t) \to \lambda_0 = \frac{\pi^2 D}{8L^2} \approx 1.23\,\frac{D}{L^2} \tag{7.6}$$

Physical picture. A dead-end tube: a hard wall at one end and an open end, a cliff, at the other. Examples are a molecule leaving a closed-ended pore, or a protein leaving a dendritic spine through its neck. The mean time to get out from the closed end, L^2/D, is four times that from the middle of a tube of the same length open at both ends.

From Gardiner. This is the case "a reflecting, b absorbing" of [G §5.2.7b]. Its backward boundary conditions [G (5.2.159)] lead to the closed form [G (5.2.160)], which with $\psi = 1$ is $T = \frac{2}{D}\int_{x_0}^{L} y\,\mathrm{d}y$. The Q_n obey the reflecting backward condition [G (5.2.69)] at 0 and the absorbing one [G (5.2.67)] at L, so they are cosines with $\cos k_n L = 0$; $P_n = p_s Q_n$ with $p_s = 1/L$ from the reflecting interval [G (5.2.91)]. Everything else follows by unfolding into Section 6.

7.1 Unfolding into an interval of length $2L$

A reflecting wall is a mirror, as in Section 3. Extend p evenly to $(-L, L)$: the extension solves the free equation with $p = 0$ at $\pm L$, started from $\frac{1}{2}[\delta(x - x_0) + \delta(x + x_0)]$. So this problem is the absorbing interval of Section 6 with $L \to 2L$, restricted to initial data symmetric about the centre.

- The spectrum of the length-$2L$ interval is $n'^2\pi^2 D/8L^2$. The symmetric modes are those with odd $n' = 2n + 1$; they are the $\cos k_n x$ above. The antisymmetric (even n') modes are never excited.

- The ground state is 4× slower than with both walls absorbing: $\lambda_0 = \pi^2 D/8L^2$ vs. $\pi^2 D/2L^2$. The gap is $\lambda_1 - \lambda_0 = 8\lambda_0$.
- In the image sum Eq. (7.2) each image pair repeats with period $2L$ and alternating sign, so the full pattern has period $4L$.
- Every result for the starting point $x_0 = 0$ here equals the result for the centre of an absorbing interval of length $2L$. For example $T(0) = L^2/D = (2L)^2/4D$.

The other orientation, absorbing at 0 and reflecting at L, follows from $x \to L - x$.

7.2 Survival and exit time

With $\int_0^L \cos k_n x \, \mathrm{d}x = (-1)^n/k_n$, the mode sum gives S; $S(0) = 1$ because $\sum_{n\geq 0}(-1)^n \cos((2n+1)\theta)/(2n+1) = \pi/4$ for $|\theta| < \pi/2$. The image form of f in Eq. (7.4) is the absorbing-interval flux Eq. (6.9) on $(-L, L)$, summed over both walls. Its $m = 0$ term is the Lévy–Smirnov density for the distance $L - x_0$.

The backward equation $\frac{1}{2}D\tilde{f}'' = s\tilde{f}$ with $\tilde{f}'(0) = 0$ and $\tilde{f}(L) = 1$ gives $\tilde{f} = \cosh kx_0 / \cosh kL$, with $k = \sqrt{2s/D}$. Its expansion gives the moments.

- $T = (L^2 - x_0^2)/D$. This is [G (5.2.160)] with $\psi = 1$: $T = 2\int_{x_0}^L \mathrm{d}y \int_0^y \mathrm{d}z/D$. The largest value is L^2/D, at the reflecting wall.
- $T_2 = (5L^4 - 6L^2x_0^2 + x_0^4)/3D^2$, so $\operatorname{Var} T = 2(L^4 - x_0^4)/3D^2$. At $x_0 = 0$ the coefficient of variation is $\sqrt{2/3}$, the same as at the centre of the absorbing interval, as the unfolding requires.
- Starting from the QSD, T is exponential with mean $1/\lambda_0 = 8L^2/\pi^2 D \approx 0.81\, L^2/D$. Check: $\int_0^L q_\infty(x_0)\, T(x_0)\, \mathrm{d}x_0 = 1/\lambda_0$.

7.3 Long times

The ground state dominates once $t \gg 1/8\lambda_0$:

- $q \to q_\infty = \frac{\pi}{2L} \cos \frac{\pi x}{2L}$. The QSD is flat at the reflecting wall and vanishes linearly at the absorbing one.
- $\kappa \to \lambda_0$.
- The Q-process has stationary density $P_0 Q_0 = \frac{2}{L} \cos^2 \frac{\pi x}{2L}$, from the SDE $\mathrm{d}x = -\frac{\pi D}{2L} \tan \frac{\pi x}{2L} \, \mathrm{d}t + \sqrt{D}\, \mathrm{d}W$.

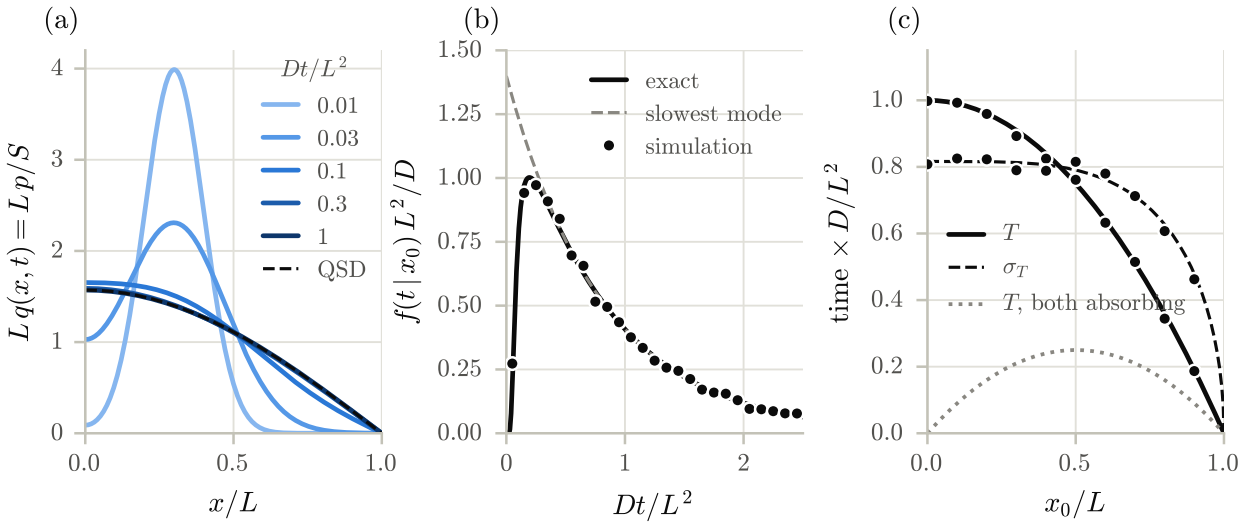


Figure 12: Reflecting at 0, absorbing at L. (a) From $x_0 = 0.3L$, $q = p/S$ converges to the QSD $\frac{\pi}{2L}\cos\frac{\pi x}{2L}$ (dashed). (b) The exit-time density Eq. (7.4) from $x_0 = 0.3L$, against the slowest mode alone and a histogram of 2×10^4 simulated paths. (c) Mean and standard deviation of the exit time; dots are means over 4000 paths. The dotted curve is T with both walls absorbing, for comparison.

8 Free diffusion on the circle

Setup and key results

$A = 0$, $B = D$ on the circle $S^1 = \mathbb{R}/L\mathbb{Z}$ (Section 1.9): no walls and no special point. With $R = L/2\pi$ and angle $\theta = x/R$ this is the rotational diffusion of a planar rotor. With $q_m = 2\pi m/L$:

$$p(x,t\,|\,x_0,0) = \sum_{n\in\mathbb{Z}} g(x - x_0 + nL, t) = \frac{1}{L} + \frac{2}{L}\sum_{m\geq 1}\cos q_m(x-x_0)\,e^{-\lambda_m t} \tag{8.1}$$

$$\lambda_m = \tfrac{1}{2}Dq_m^2 = \frac{2\pi^2 m^2 D}{L^2}, \quad P_m = \frac{e^{iq_m x}}{L}, \quad Q_m = e^{-iq_m x}, \quad m \in \mathbb{Z} \tag{8.2}$$

$$p_s = \frac{1}{L}, \qquad J_s = 0, \qquad \text{relaxation time } \frac{1}{\lambda_1} = \frac{L^2}{2\pi^2 D} \approx 0.051\,\frac{L^2}{D} \tag{8.3}$$

$$\langle e^{iq_m[x(t)-x(0)]}\rangle = e^{-\lambda_m|t|}, \qquad \text{e.g. } \langle\cos\Delta\theta(t)\rangle = e^{-Dt/2R^2} \tag{8.4}$$

$$\Delta\tilde{x} \text{ Gaussian, mean 0, variance } Dt \text{ exactly}: \qquad v_{\mathrm{w}} = 0, \quad D_{\mathrm{eff}} = D \tag{8.5}$$

$$T(d) = \frac{d(L-d)}{D}, \qquad \Pr(\text{arrive moving in the } +x \text{ direction}) = \frac{d}{L}, \qquad T_{\mathrm{w}} = \frac{L^2}{D} \tag{8.6}$$

Here $T(d)$ is the mean time to reach a target from a distance d above it (measured in the $+x$ direction), and T_{w} the mean time to the first full winding.

Physical picture. A bead on a circular track with no force along it (Fig. 1c): a colloid in a ring-shaped optical trap of uniform intensity, or the orientation of a rod rotating in a plane. For the rotor, x/R is the angle, $D/2R^2$ the rotational diffusion coefficient, and $\langle\cos\Delta\theta\rangle = e^{-Dt/2R^2}$ Debye's orientational relaxation in two dimensions.

From Gardiner. The book's periodic results are [G (5.2.17)]–[G (5.2.19)] and the stationary solution [G (5.2.41)]–[G (5.2.47)]. With $A = 0$, $\psi = 1$, so [G (5.2.46)] gives $J = 0$ and [G (5.2.47)] a uniform p_s. The propagator comes from the Wiener process of [G §3.8.1] in two ways:

- summing the Gaussian [G (3.8.7)] over the lifts Eq. (1.16) gives the image form;
- for a variable that lives on a circle, only the values $s = q_m$ of the characteristic function [G (3.8.6)] are defined, since e^{isx} is a function on the circle only for those s. They are exactly the Fourier coefficients of p, and $\phi(q_m, t) = e^{iq_m x_0 - Dq_m^2 t/2}$ gives the mode sum.

[G §5.2.5] is written for reflecting and absorbing walls; with periodic Q its boundary terms are simply absent. With $p_s = 1/L$, the real form of each doublet is $Q_m = \sqrt{2}\cos q_m x$, $\sqrt{2}\sin q_m x$ with $P_m = p_s Q_m = Q_m/L$, exactly as on the reflecting interval; the complex pair in the box is the same doublet.

8.1 From the line

Equation (1.16) applied to the Gaussian of Section 2 gives the image form of Eq. (8.1): the free Gaussian, wrapped around the circle (Fig. 13b). Poisson summation turns it into the mode sum. As on the intervals, the image sum converges fast for $Dt \ll L^2$ and the mode sum for $Dt \gtrsim L^2$, and they agree to 10^{-15}. Compare the reflecting interval Eq. (5.2), where the images come in mirror pairs with period $2L$. Here there is one image per lattice point, with period L and no mirrors.

8.2 Folding: the circle is two intervals

$A = 0$ is odd under every reflection, so the folding of Section 1.9 works about any point. Fold about the start x_0, which fixes x_0 and its antipode $x_0 + L/2$:

- The even sector is the reflecting interval of length $L/2$ (Section 5), with levels $n^2\pi^2 D/2(L/2)^2 = 2\pi^2 n^2 D/L^2$, $n \geq 0$. The odd sector is the absorbing interval of length $L/2$ (Section 6), with the same nonzero levels. Together they give each λ_m twice: the doublet $\cos q_m(x - x_0)$, $\sin q_m(x - x_0)$ (Fig. 13c).
- A point start is even about itself, so only the even sector is excited. The propagator on the circle is exactly half the reflecting propagator on $(0, L/2)$ started at a wall: $p(x, t \,|\, x_0, 0) = \frac{1}{2} p_{\mathrm{R}}(|x - x_0|, t \,|\, 0, 0)$.
- The gap $2\pi^2 D/L^2$ is four times that of the reflecting interval of length L: on the circle the farthest point is only $L/2$ away.

8.3 Observables and relaxation

- $\langle x(t) \rangle$ has no meaning here: x jumps at the cut, so the answer depends on where the cut is. The Fourier modes are good observables, and each is a single eigenfunction: $\langle e^{iq_m x(t)} \rangle = e^{iq_m x_0 - \lambda_m t}$ exactly, with no admixture of other rates. For the rotor, $\lambda_1 = D/2R^2$ and this is Debye's $\langle \cos \Delta\theta \rangle = e^{-Dt/2R^2}$.
- The density relaxes to $1/L$ at λ_1, in a time $\approx 0.05\, L^2/D$ (Fig. 13a).

8.4 Winding and first passage

- The lift is exactly free motion on the line: $\Delta\tilde{x}$ is Gaussian with variance Dt at all times, so $v_{\mathrm{w}} = 0$ and $D_{\mathrm{eff}} = D$. The circle changes where the particle is, not how far it has gone.
- The first full winding is the absorbing interval $(x_0 - L, x_0 + L)$ on the cover, started at its centre: $T_{\mathrm{w}} = (2L)^2/4D = L^2/D$ by Section 6. Then $L^2/T_{\mathrm{w}} = D = D_{\mathrm{eff}}$, as the renewal argument of Section 1.9 requires.

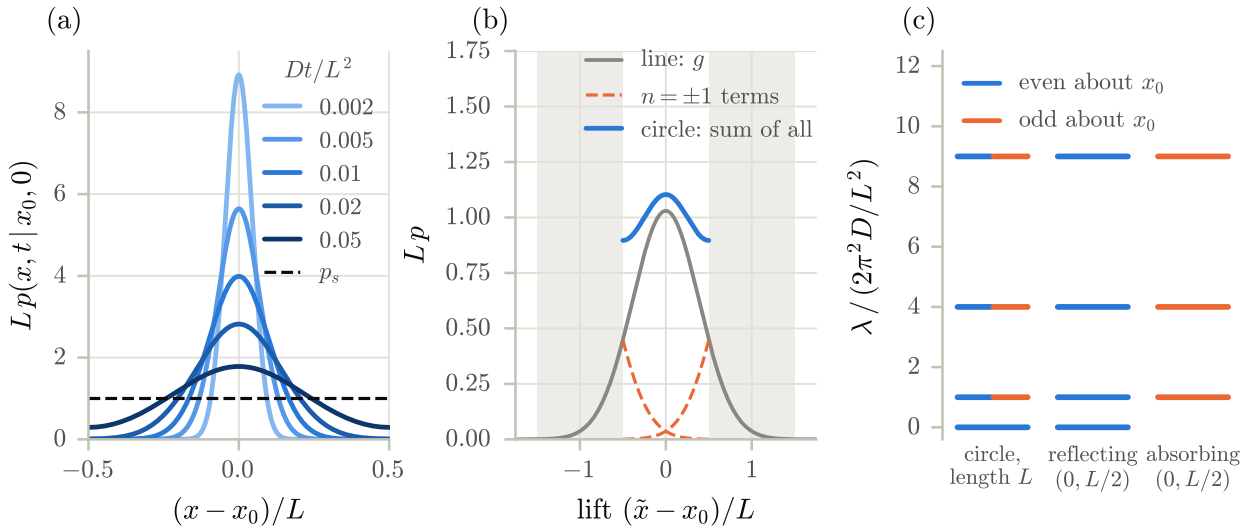


Figure 13: Free diffusion on the circle, centred on the starting point (the only distinguished point). (a) Density snapshots relaxing to $1/L$ by $Dt/L^2 \approx 0.05$. (b) The universal cover at $Dt/L^2 = 0.15$: the free Gaussian on the line (grey), its translates by $\pm L$ folded into one period (dashed), and their sum, the density on the circle (blue). Shading marks the neighbouring copies of the circle. (c) Spectrum in units of $2\pi^2 D/L^2$. Each nonzero level of the circle is a doublet: one level from the reflecting interval of length $L/2$ (even about x_0) and one from the absorbing one (odd).

- First passage to a target: cut there, and the circle becomes the absorbing interval of length L of Section 6, with $x_0 \to d$. The mean time is longest, $L^2/4D$, from the antipode. The particle arrives by the shorter way round with probability $\max(d, L-d)/L$.

9 Summary of free diffusion

Walls	Slowest rate	Long-time law	Mean exit time $T(x_0)$
reflecting 0, L	$\pi^2 D/2L^2$	$p \to 1/L$	—
absorbing 0, L	$\pi^2 D/2L^2$	$S \propto e^{-\pi^2 Dt/2L^2}$	$x_0(L-x_0)/D$
refl. 0, abs. L	$\pi^2 D/8L^2$	$S \propto e^{-\pi^2 Dt/8L^2}$	$(L^2-x_0^2)/D$
reflecting 0 only	0 (no gap)	$p \simeq \sqrt{2/\pi Dt}\, e^{-x^2/2Dt}$	—
absorbing 0 only	0 (no gap)	$S \simeq x_0\sqrt{2/\pi Dt}$	∞
none	0 (no gap)	$p = g(x-x_0,t)$	—
circle	$2\pi^2 D/L^2$ (doublet)	$p \to 1/L$	to a target: $d(L-d)/D$

Table 5: Free diffusion, $\partial_t p = \frac{1}{2} D\, \partial_x^2 p$, in every geometry.

- A bounded domain has a discrete spectrum with a gap, so relaxation and decay are exponential. An unbounded domain has a continuous spectrum reaching $\lambda = 0$, so everything is algebraic in t.
- With absorption on a bounded domain the conditioned density goes to a QSD, the positive ground state. Without a bound it spreads (Rayleigh) and there is no QSD.

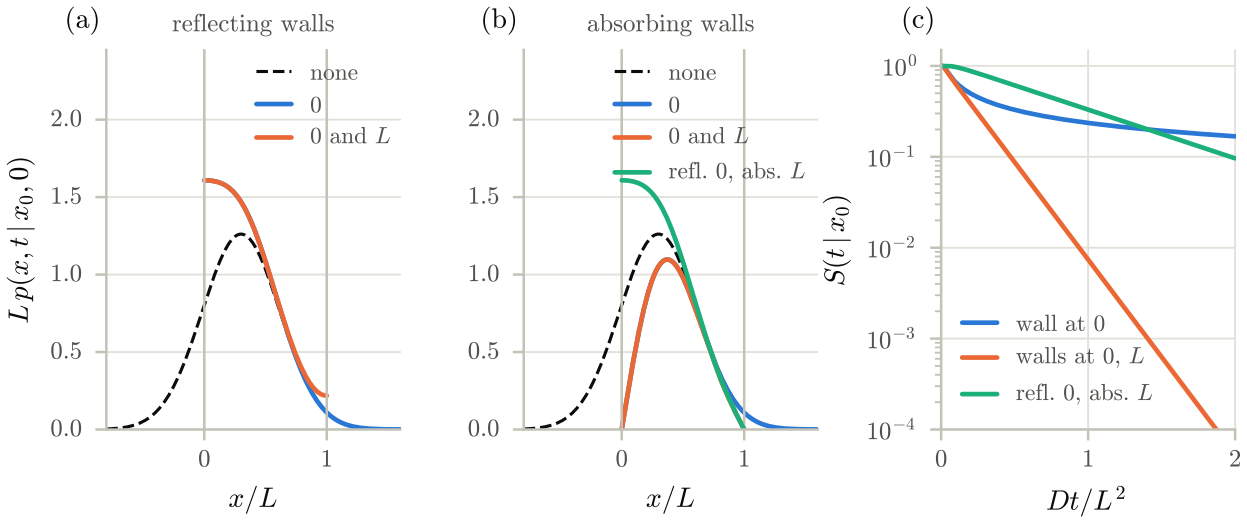


Figure 14: Free diffusion in every geometry, all from $x_0 = 0.3L$. (a,b) Densities at $Dt/L^2 = 0.1$; grey vertical lines mark $x = 0$ and $x = L$, and the dashed curve is the free Gaussian. A wall "at 0" alone is the half-line. (c) Survival for the three absorbing geometries: $e^{-\pi^2 Dt/2L^2}$, $e^{-\pi^2 Dt/8L^2}$, and $t^{-1/2}$ for the half-line, which overtakes the mixed interval near $Dt/L^2 \approx 1.4$.

- The slowest rate depends on the walls, not just the size. A reflecting wall is a mirror and doubles the effective length, so the rate drops by 4 (Fig. 14c).
- The two reflecting cases differ only near the far wall at L (Fig. 14a). The absorbing ones differ everywhere, because each absorbing wall pins p to 0 (Fig. 14b).
- The circle has no walls. Folded about the start it is a reflecting interval of half its length, so it relaxes four times faster than the reflecting interval of the same length. Its winding is exactly free motion on the line.

Part II
Constant drift: $A = v$, $B = D$

10 Constant drift: reduction to free diffusion

The equation is $\partial_t p = -v\,\partial_x p + \frac{1}{2}D\,\partial_x^2 p$, from $\mathrm{d}x = v\,\mathrm{d}t + \sqrt{D}\,\mathrm{d}W$; $v > 0$ drifts toward $+x$. It is the overdamped motion in the linear potential $U = -vx$ ($A = -U'$). In every drift figure blue means $v < 0$ (toward $x = 0$), red $v > 0$, and grey $v = 0$.

Physical picture. A constant force F, so $\mathcal{U} = -Fx$ and $v = F/\gamma$ (Section 1.2): gravity minus buoyancy on a colloid (sedimentation), an electric field on a charged particle (electrophoresis), or a uniform flow carrying a solute (advection). The drift length $\ell = D/|v| = 2k_BT/|F|$ is the distance over which the force does work $2k_BT$, so the Péclet number L/ℓ compares the work done by the force across the system with the thermal energy.

From Gardiner. The stationary solution Eq. (1.5) [G (5.2.39)] with $A = v$, $B = D$ gives $p_s \propto e^{2vx/D}$ at once. Everything else in Part II comes from Part I through the transformation below,

which is not in the book. It is the one-dimensional case of removing a constant drift from the heat equation, equivalently the Girsanov factor.

Drift scales. Unlike free diffusion, the dynamics now has its own length and time,

$$\ell \equiv \frac{D}{|v|}, \qquad \tau \equiv \frac{\ell^2}{D} = \frac{\ell}{|v|} = \frac{D}{v^2}\,: \tag{10.1}$$

over a length ℓ drift and diffusion are comparable, and τ is the time to cross it either way.

- Every other length enters only through a Péclet number, $L/\ell = |v|L/D$ for the interval or $x_0/\ell = |v|x_0/D$ for the start. These describe the geometry and the initial condition, not the dynamics.
- Results at fixed $v \neq 0$ below are written in ℓ and τ. Figures that sweep v through 0 keep L or x_0 as the unit, because $\ell \to \infty$ as $v \to 0$.
- Factor of two: because $B = D$ means a diffusion constant $D/2$, the barometric profile decays over $\ell/2$. Defining ℓ as $D/|v|$ rather than $D/2|v|$ keeps the gauge constant below simple: $\alpha = \pm 1/\ell$.

Gauge transformation. With $\alpha \equiv v/D = \mathrm{sgn}(v)/\ell$,

$$p(x,t) = \exp\Big[\alpha(x - x_0) - \frac{t}{2\tau}\Big]\, u(x,t), \qquad \partial_t u = \tfrac{1}{2} D\, \partial_x^2 u. \tag{10.2}$$

Substituting, the terms in $\partial_x u$ cancel because $\alpha = v/D$, and the terms in u cancel because the exponent decays at $1/2\tau = v^2/2D$. The prefactor equals 1 at $x = x_0$, $t = 0$, so u also starts as $\delta(x - x_0)$. This is the symmetrisation of Section 1.5: $e^{\alpha x} \propto \sqrt{p_s}$, the potential Eq. (1.11) is the constant $V = 1/2\tau$, and u solves the free problem once that constant is taken out.

Walls. Since $J = vp - \frac{1}{2} D\, \partial_x p = \frac{1}{2} D e^{(\cdots)}[\alpha u - \partial_x u]$:

- absorbing, $p = 0 \iff u = 0$, so absorbing problems are free problems times the prefactor;
- reflecting, $J = 0 \iff \partial_x u = \alpha u = \pm u/\ell$, a Robin condition, so reflecting problems change character.

Consequences.

- The spectrum is shifted: $\lambda = 1/2\tau + \frac{1}{2} D k^2$, where k comes from the u-problem. Drift alone sets a decay rate $1/2\tau = v^2/2D$, independent of its sign.
- Gardiner's eigenfunctions are $P_n \propto e^{\alpha x} u_n$ (forward) and $Q_n \propto e^{-\alpha x} u_n$ (backward), with u_n orthogonal in plain L^2. Since $\sqrt{p_s} \propto e^{\alpha x}$, this is Eq. (1.12) with $\psi_n = u_n/\sqrt{N_n}$, and his normalisation is

  $$P_n = \sqrt{\frac{p_s}{N_n}}\, u_n, \qquad Q_n = \frac{u_n}{\sqrt{N_n p_s}}, \qquad N_n = \int u_n^2 \, \mathrm{d}x. \tag{10.3}$$

  Here p_s is the zero-current solution $\propto e^{2\alpha x}$: normalised where it can be (on an interval, and on the half-line for $v < 0$), any multiple of $e^{2\alpha x}$ otherwise, and that of the reflecting problem when a wall absorbs [G (5.2.91)]. Hence

  $$p(x, t \,|\, x_0, 0) = e^{\alpha(x - x_0)} \sum_n \frac{u_n(x)\, u_n(x_0)}{N_n}\, e^{-\lambda_n t}. \tag{10.4}$$

  The operator is not self-adjoint in L^2. The prefactor $e^{\alpha(x-x_0)}$, as large as $e^{L/\ell}$, means the slowest mode can take a transit time $L/|v| = (L/\ell)\,\tau$ to dominate.
- Through an absorbing wall at $x = w$ the exit flux is $j_w(t) = e^{\alpha(w - x_0) - t/2\tau}\, j_w^{\mathrm{free}}(t)$.
- For the first-passage formulas of Section 1.7, $\psi = e^{2\alpha x}$.

11 Constant drift on the whole line

Setup and key results

$A = v$, $B = D$ on $(-\infty, \infty)$: Brownian motion with drift [G §3.8.3].

$$p(x, t \mid x_0, 0) = g(x - x_0 - vt, t), \qquad \langle x \rangle = x_0 + vt, \qquad \operatorname{Var} x = Dt \tag{11.1}$$

$$P_k = \frac{e^{(\alpha+ik)x}}{2\pi}, \quad Q_k = e^{-(\alpha+ik)x}, \quad \lambda = \frac{1}{2\tau} + \tfrac{1}{2}Dk^2 \quad \text{(in the gauge)} \tag{11.2}$$

$$\langle e^{ikx(t)} \rangle = e^{ik(x_0+vt) - Dk^2 t/2} \tag{11.3}$$

$$f_a(t) = \frac{a - x_0}{\sqrt{2\pi D t^3}}\, e^{-(a-x_0-vt)^2/2Dt} \qquad \text{(first passage to } a > x_0\text{)} \tag{11.4}$$

$$\Pr(\text{reach } a) = \begin{cases} 1 & v \geq 0 \\ e^{-2(a-x_0)/\ell} & v < 0 \end{cases} \tag{11.5}$$

The spreading $\sqrt{Dt}$ equals the displacement $|v|t$ at $t = \tau$, where both are ℓ: diffusion dominates for $t \ll \tau$, drift for $t \gg \tau$.

Physical picture. A colloid sedimenting far above the floor of its container, an ion drifting in a field, or a pollutant carried down a river: the cloud moves at v and spreads as $\sqrt{Dt}$.

From Gardiner. The book meets this Gaussian in [G §3.8.3], as the small-step approximation of the Poisson process, a one-sided random walk: [G (3.8.58)] is the density with mean vt and variance Dt, and [G (3.8.59)] its Fokker–Planck equation. (Gardiner stresses that this is an expansion, not a limit.) Here it is Section 2 pushed through Eq. (10.2).

- A Galilean shift $x \to x - vt$ removes the drift on the line, but not in the presence of a wall, which does not move. That is why every bounded example in this part depends on v.
- The gauge Eq. (10.2) is the statement that $g(x - x_0 - vt) = e^{\alpha(x-x_0) - t/2\tau} g(x - x_0)$. In path language it is the Girsanov factor, which depends only on the endpoints and t. That is why absorbing problems, which constrain only endpoints, transform so simply.
- **Two sets of eigenfunctions.** The gauge gives the P_k, Q_k of the box, with real $\lambda = 1/2\tau + \frac{1}{2}Dk^2$: the spectrum in $L^2(1/p_s)$, $p_s \propto e^{2\alpha x}$. But the plain Fourier modes $P_k = e^{ikx}/2\pi$, $Q_k = e^{-ikx}$ are eigenfunctions too, bounded, with complex $\lambda = \frac{1}{2}Dk^2 + ivk$; the characteristic function in the box is their statement. Both sums [G (5.2.88)] give $g(x - x_0 - vt)$. Which is "the" spectrum depends on the space (Section 1.5); on the circle only the Fourier modes survive (Section 17).
- First passage to a level will be Section 13 with $x_0 \to a - x_0$ and $v \to -v$. Drift toward the level makes $\langle T \rangle = (a - x_0)/v$ finite; drift away makes reaching it uncertain.

12 Constant drift on the half-line, reflecting at 0

Setup and key results

$A = v$, $B = D$ on $(0,\infty)$, $J = 0$ at $x = 0$ (Smoluchowski's sedimentation problem). With $\alpha = v/D = \pm 1/\ell$ and $\tau = D/v^2$:

$$p(x,t\,|\,x_0,0) = g(x-x_0-vt,t) + e^{-2\alpha x_0} g(x+x_0-vt,t) - \alpha\, e^{2\alpha x}\,\mathrm{erfc}\,\frac{x+x_0+vt}{\sqrt{2Dt}} \tag{12.1}$$

$$P_k = \sqrt{\frac{p_s}{N_k}}\,u_k, \quad Q_k = \frac{u_k}{\sqrt{N_k p_s}}, \quad \lambda = \frac{1}{2\tau} + \tfrac{1}{2}Dk^2, \quad k>0 \tag{12.2}$$

$$u_k = \cos kx + \frac{\alpha}{k}\sin kx, \qquad N_k = \frac{\pi}{2}\Big(1+\frac{\alpha^2}{k^2}\Big), \qquad p_s \propto e^{2\alpha x} \tag{12.3}$$

$$v<0: \quad p_s(x) = \frac{2}{\ell}\,e^{-2x/\ell}, \quad \langle x\rangle_s = \frac{\ell}{2}; \quad P_0 = p_s,\ Q_0 = 1; \quad \text{spectrum } \{0\}\cup\Big[\frac{1}{2\tau},\infty\Big) \tag{12.4}$$

$$v>0: \quad p \simeq g(x-x_0-vt,t), \quad \langle x\rangle \simeq vt = \ell\,\frac{t}{\tau} \quad \text{(no stationary state)} \tag{12.5}$$

In drift units. With $\xi = x/\ell$, $s = t/\tau$, $\xi_0 = x_0/\ell$ and $G_s(y) = e^{-y^2/2s}/\sqrt{2\pi s}$, the case $v<0$ reads

$$\ell\, p = G_s(\xi-\xi_0+s) + e^{2\xi_0} G_s(\xi+\xi_0+s) + e^{-2\xi}\,\mathrm{erfc}\,\frac{\xi+\xi_0-s}{\sqrt{2s}}. \tag{12.6}$$

There is one parameter, ξ_0, and at long times $\ell p \to 2e^{-2\xi}$ whatever it is.

Physical picture. For $v<0$: colloids settling onto the floor of a container. The barometric profile $e^{-2x/\ell} = e^{-|F|x/k_BT}$ is the sedimentation equilibrium from which Perrin measured Avogadro's number [6]; for gas molecules in gravity it is the barometric formula of the atmosphere. For $v>0$ the force points away from the floor, as for buoyant droplets released at the bottom of a tall column: they rise and never return.

From Gardiner. The wall is the zero-current condition [G (5.2.13)]. The stationary solution Eq. (1.5) with $\psi = e^{2vx/D}$ gives p_s, which on $(0,\infty)$ is normalisable only for $v<0$ (compare the infinite-range discussion of [G §5.2.2c]). The time dependence needs the Robin image below, which is not in the book.

12.1 The Robin image

Pushed through Eq. (10.2), the mirror of Section 3 becomes the Robin condition $u' = \alpha u$, and a single image no longer suffices. The combination $w = u' - \alpha u$ obeys the free equation with $w = 0$ at the wall, so it is a Dirichlet problem, solved by one image pair. Then integrating $u' - \alpha u = w$ inward from $x = \infty$ gives

$$u = g(x-x_0) + g(x+x_0) - 2\alpha\int_x^\infty e^{-\alpha(\xi-x)} g(\xi+x_0,t)\,\mathrm{d}\xi, \tag{12.7}$$

a point image plus a line of images beyond $-x_0$ with weight $e^{\alpha\xi}$. Completing the square in the integral gives the erfc term, and Eq. (10.2) turns the point image into $e^{-2\alpha x_0} g(x+x_0-vt)$. The script checks normalisation, $J(0) = 0$, and the PDE residual of Eq. (12.1).

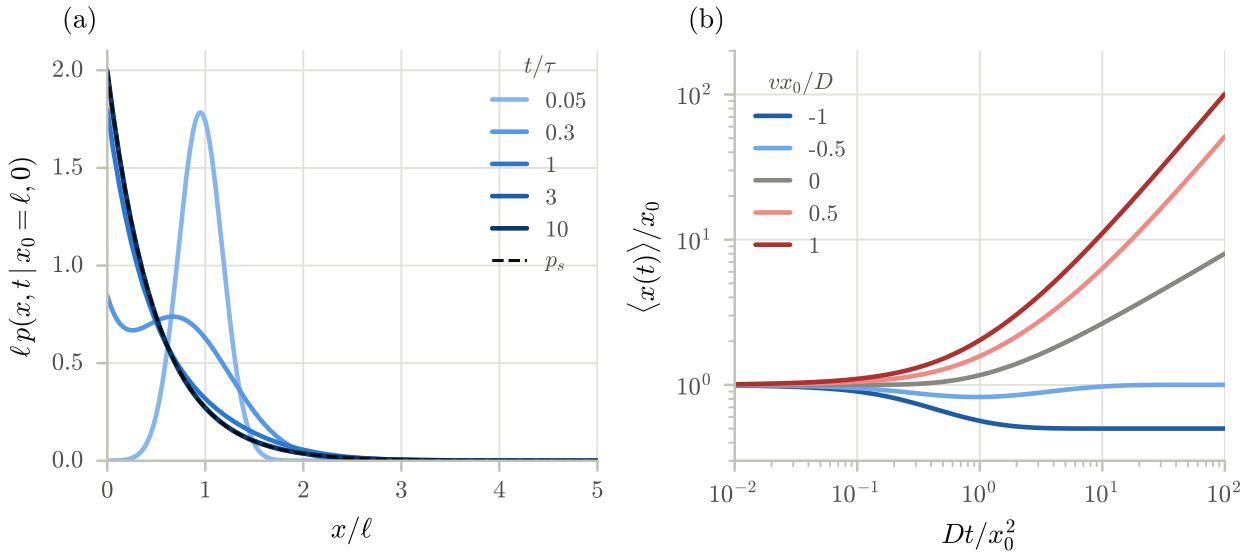


Figure 15: Constant drift on the half-line, reflecting at 0. (a) Relaxation to the barometric profile for $v<0$, in drift units $\ell = D/|v|$, $\tau = D/v^2$, from $x_0 = \ell$. (b) Mean position for several drifts, in units of x_0 because the sweep passes through $v = 0$: saturation at $\ell/2$ for $v<0$, $\sqrt{t}$ for $v=0$, vt for $v>0$.

The same in eigenfunctions. The Robin modes of Section 14, with k now continuous, are the u_k of the box; $\int_0^\infty u_k u_{k'}\,\mathrm{d}x = N_k\,\delta(k-k')$ fixes N_k, and Eq. (10.3) gives P_k and Q_k, with $p_s \propto e^{2\alpha x}$ (for $v>0$ any multiple). For $v<0$ the Robin condition also has the bound state $u = e^{\alpha x}$, which decays and gives $P_0 = p_s$, $Q_0 = 1$; for $v>0$ it grows and is not an eigenfunction. Then $\int_0^\infty P_k(x)Q_k(x_0)e^{-\lambda t}\,\mathrm{d}k$, plus $P_0(x)Q_0(x_0) = p_s(x)$ if $v<0$, is Eq. (12.1) (checked to 10^{-15}).

12.2 Long times

- $v<0$ (drift into the wall): the erfc tends to 2 and the Gaussians leave, so $p \to (2/\ell)e^{-2x/\ell}$: sedimentation equilibrium (Fig. 15a). The spectrum is the isolated $\lambda = 0$ plus a continuum from $1/2\tau$, so the gap is $1/2\tau$ although the domain is unbounded.
- $v>0$: the packet moves off at speed v, and the wall only matters while it is within about $\sqrt{Dt}$ of it.
- $v=0$ is Section 3: $\langle x\rangle \sim \sqrt{2Dt/\pi}$. The three regimes are visible in Fig. 15b. At $vx_0/D = -0.5$ the start sits at $\langle x\rangle_s = \ell/2$, and the mean dips before returning.

13 Constant drift on the half-line, absorbing at 0

Setup and key results

$A = v$, $B = D$ on $(0, \infty)$, $p = 0$ at $x = 0$. With $\alpha = v/D = \pm 1/\ell$ and $\tau = D/v^2$:

$$p(x, t \,|\, x_0, 0) = g(x - x_0 - vt, t) - e^{-2\alpha x_0} g(x + x_0 - vt, t) \tag{13.1}$$

$$P_k = \sqrt{\tfrac{2p_s}{\pi}}\, \sin kx, \quad Q_k = \sqrt{\tfrac{2}{\pi p_s}}\, \sin kx, \quad \lambda = \frac{1}{2\tau} + \tfrac{1}{2} D k^2 \quad (p_s \propto e^{2\alpha x}) \tag{13.2}$$

$$S(t \,|\, x_0) = \tfrac{1}{2}\Big[\operatorname{erfc}\Big(-\frac{x_0 + vt}{\sqrt{2Dt}}\Big) - e^{-2\alpha x_0} \operatorname{erfc} \frac{x_0 - vt}{\sqrt{2Dt}}\Big] \tag{13.3}$$

$$f(t \,|\, x_0) = \frac{x_0}{\sqrt{2\pi D t^3}}\, e^{-(x_0 + vt)^2/2Dt}, \qquad \tilde{f}(s) = e^{-\alpha x_0 - x_0\sqrt{\alpha^2 + 2s/D}} \quad \text{(inverse Gaussian)} \tag{13.4}$$

$$\Pr(T < \infty) = \begin{cases} 1 & v \le 0 \\ e^{-2x_0/\ell} & v > 0 \end{cases}, \qquad v < 0 : \ \langle T \rangle = \frac{x_0}{\ell}\, \tau, \ \ \operatorname{Var} T = \frac{x_0}{\ell}\, \tau^2 \tag{13.5}$$

$$v < 0 : \quad \phi_\lambda \propto e^{r_+ x} - e^{r_- x}, \quad r_\pm = -\frac{1 \mp \sqrt{1 - 2\lambda\tau}}{\ell}, \quad 0 < \lambda \le \frac{1}{2\tau} \quad \text{(a continuum of QSDs)} \tag{13.6}$$

$$q(x, t) \to \frac{x}{\ell^2}\, e^{-x/\ell} \ \text{(point start)}, \qquad S \simeq \sqrt{\frac{2}{\pi}}\, \frac{x_0}{\ell}\, e^{x_0/\ell} \Big(\frac{t}{\tau}\Big)^{-3/2} e^{-t/2\tau}, \qquad \kappa \simeq \frac{1}{2\tau} + \frac{3}{2t} \tag{13.7}$$

$$q(x, t) \to \phi_\lambda \text{ with } r_+ = -\beta, \quad \lambda = \beta|v| - \tfrac{1}{2} D\beta^2 \quad \text{if } p(x, 0) \sim e^{-\beta x}, \ \beta < 1/\ell \tag{13.8}$$

This is the problem of `qsd-context.md` §3–4 (there with drift $-\mu$ and $D' = D/2$) and of the preset `halfline` in `sde-explorer`.

Physical picture. A walker on a slope that ends in a cliff. For $v < 0$ the slope runs toward the edge: colloids sedimenting onto a sticky floor, or the Gerstein–Mandelbrot neuron, whose membrane potential drifts toward threshold [7]; its interspike interval is then the inverse-Gaussian first-passage time of this section. For $v > 0$ the slope runs away from the edge, and a walker that starts a few ℓ from it almost surely never falls.

From Gardiner. With the far wall at infinity, everything in [G §5.2.7]–[G §5.2.8] applies with $\psi = e^{2vx/D}$.

- The probability of ever reaching 0 is π_0 from the corrected [G (5.2.189)] with $b \to \infty$: $\pi_0 = \int_{x_0}^{\infty} \mathrm{d}y/\psi / \int_0^{\infty} \mathrm{d}y/\psi$. This is $e^{-2x_0/\ell}$ for $v > 0$, and 1 for $v \le 0$, where both integrals diverge.
- For $v < 0$, $\langle T \rangle = x_0/|v|$ solves [G (5.2.154)] with $T(0) = 0$ and T bounded at infinity.

13.1 Propagator and first passage

Absorbing, so the gauge maps the image pair of Section 4 straight over. The exit-time density is the Lévy–Smirnov density times $e^{-\alpha x_0 - t/2\tau}$: the inverse Gaussian.

- $v < 0$: absorption is certain, with $\langle T \rangle = x_0/|v|$ (ballistic) and a tail cut off at $t \sim \tau$ (Fig. 16b). The relative spread is $\sigma_T/\langle T \rangle = \sqrt{\ell/x_0}$, so the exit time is sharp only for a start many drift lengths away.

- $v > 0$: the path escapes to ∞ with probability $1 - e^{-2x_0/\ell}$, and f integrates to $e^{-2x_0/\ell}$. A start a few drift lengths out almost always escapes.
- $v = 0$: the $t^{-3/2}$ tail and $\langle T \rangle = \infty$ of Section 4.

13.2 A continuum of QSDs

For $v < 0$, $\frac{1}{2}D\phi'' - v\phi' + \lambda\phi = 0$ with $\phi(0) = 0$ has roots $r_\pm$ that are both negative for $0 < \lambda \le 1/2\tau$, so ϕ_λ is positive and integrable: every such λ gives a QSD (Fig. 16c). For $\lambda > 1/2\tau$ the roots are complex and ϕ oscillates. In the symmetrised variable, $\phi = e^{\alpha x}\psi$, the L^2 spectrum is the continuum $[1/2\tau, \infty)$. The QSDs below the edge have ψ growing like $e^{(r_+ - \alpha)x}$: not in L^2, but ϕ is in L^1. In Gardiner's terms ϕ_λ is a forward eigenfunction P_λ whose partner $Q_\lambda = \phi_\lambda/p_s$ grows like $e^{(r_+ + 2/\ell)x}$, so $\int P_\lambda Q_\lambda$ [G (5.2.79)] diverges and λ has no place in [G (5.2.88)]. A second wall at L removes all but one; compare Section 15, where $\lambda_1 = 1/2\tau + \pi^2 D/2L^2 \to 1/2\tau$ as $L \to \infty$.

13.3 Which QSD is reached

The tail of the initial density decides (Section 1.8, Fig. 5a).

- **Point start.** The gauge turns the problem into the free absorbing half-line, whose conditioned density spreads as the Rayleigh profile $(x/Dt)e^{-x^2/2Dt}$ (Section 4). Near the wall that is $\propto x$, and the gauge factor $e^{-x/\ell}$ freezes it into the edge QSD $xe^{-x/\ell}/\ell^2$: a Gamma(2) density with mean 2ℓ, independent of x_0. The approach is algebraic, because the edge of a continuum has no gap: $\kappa \simeq 1/2\tau + 3/2t$ (Fig. 16a). The drift length sets the QSD completely; x_0 only sets how long the transient lasts.
- **Exponential start**, $p(x, 0) = \beta e^{-\beta x}$. In the gauge the initial condition is $w_0 = e^{x/\ell}p(x, 0) = \beta e^{\gamma x}$ with $\gamma = 1/\ell - \beta$.
  - If $\gamma > 0$, w_0 grows, and the free evolution of a growing exponential does not spread: $u \simeq 2\beta e^{\gamma^2 Dt/2} \sinh \gamma x$ once $\gamma^2 Dt \gg 1$. Back through the gauge, $p \propto e^{-(1/2\tau - D\gamma^2/2)t}(e^{-\beta x} - e^{-(2/\ell - \beta)x})$. That is exactly the QSD ϕ_λ with $r_+ = -\beta$, and $\lambda = 1/2\tau - \frac{1}{2}D\gamma^2 = \beta|v| - \frac{1}{2}D\beta^2$. It is approached exponentially fast.
  - If $\gamma \le 0$, w_0 does not grow, and the start behaves like a point start: it reaches the edge QSD, algebraically.

  Both cases are closed-form, since the free evolution of $e^{\gamma y}$ is a pair of Gaussian integrals; the script checks them.
- **General starts.** Martínez, Picco and San Martín [17] showed that an initial density with a definite exponential tail rate β goes to the QSD this rule picks. That is the one with the same tail if $\beta < 1/\ell$, and the edge otherwise, including tails lighter than every exponential. A tail heavier than every exponential gives no limit at all: the survivors are particles still drifting in from far out, and the conditioned density follows them outward.
- **Why the threshold is $1/\ell$.** A particle starting near y arrives after about $y/|v|$, so a tail $e^{-\beta y}$ supplies survivors at the rate $\beta|v|$; diffusion lowers this by $\frac{1}{2}D\beta^2$. The far supply wins when it is slower than the bulk's $1/2\tau$, which is exactly when $\beta < 1/\ell$ (Fig. 5c).

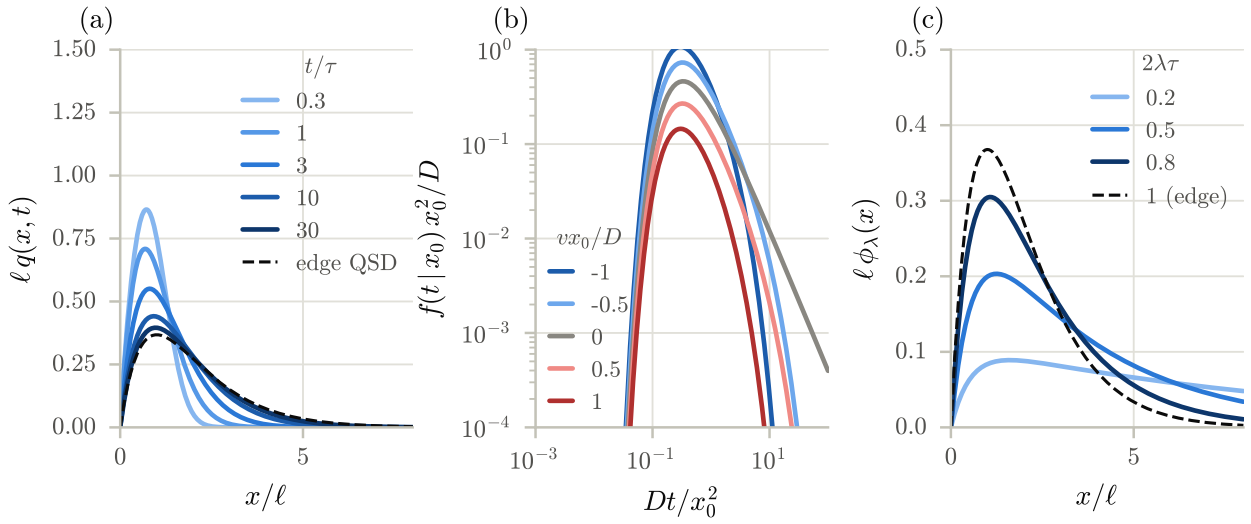


Figure 16: Constant drift on the half-line, absorbing at 0. (a) Conditioned density for $v < 0$ in drift units $\ell = D/|v|$, $\tau = D/v^2$, from $x_0 = \ell$, creeping toward the edge QSD $xe^{-x/\ell}/\ell^2$. (b) Exit-time densities, in units of x_0 because the sweep passes through $v = 0$. For $v > 0$ the area is $e^{-2vx_0/D} < 1$. (c) The QSD family, in drift units: smaller λ means a heavier tail.

14 Constant drift, reflecting walls

Setup and key results

$A = v$, $B = D$ on $(0, L)$, $J = 0$ at both walls. With $\alpha = v/D = \pm 1/\ell$ and $\tau = D/v^2$ (Eq. (10.1)):

$$p_s(x) = \frac{2\alpha\, e^{2\alpha x}}{e^{2\alpha L} - 1}, \qquad \langle x\rangle_s = \frac{L}{1 - e^{-2\alpha L}} - \frac{1}{2\alpha} \tag{14.1}$$

$$\lambda_0 = 0, \qquad \lambda_n = \frac{1}{2\tau} + \frac{n^2\pi^2 D}{2L^2}, \quad n \geq 1 \tag{14.2}$$

$$\phi_n(x) = \cos k_n x + \frac{\alpha}{k_n}\sin k_n x, \qquad k_n = \frac{n\pi}{L}, \qquad N_n = \int_0^L \phi_n^2\, \mathrm{d}x = \frac{L}{2}\Big(1 + \frac{\alpha^2}{k_n^2}\Big) \tag{14.3}$$

$$P_0 = p_s, \quad Q_0 = 1; \qquad P_n = \sqrt{\frac{p_s}{N_n}}\,\phi_n, \quad Q_n = \frac{\phi_n}{\sqrt{N_n p_s}}, \quad n \geq 1 \tag{14.4}$$

$$p(x, t\,|\,x_0, 0) = p_s(x) + \frac{2}{L}\, e^{\alpha(x - x_0)} \sum_{n\geq 1} \frac{\phi_n(x)\,\phi_n(x_0)}{1 + \alpha^2/k_n^2}\, e^{-\lambda_n t} \tag{14.5}$$

Physical picture. Sedimentation in a closed column of height L: the barometric profile of Section 12, cut off by a ceiling. An analytical ultracentrifuge reaches the same kind of sedimentation equilibrium, with the centrifugal force in place of gravity, roughly constant across a thin cell.

From Gardiner. p_s is Eq. (1.5). The eigenfunctions are Gardiner's $P_\lambda = p_s Q_\lambda$ ([G (5.2.71)], [G (5.2.73)]–[G (5.2.74)]), with the reflecting backward condition [G (5.2.69)], $Q' = 0$, at both walls. In the gauge $Q \propto e^{-\alpha x}u$, so $Q' = 0$ is $u' = \alpha u$; forward, $J(P_n) = 0$ is the same condition. The normalisation [G (5.2.80)] is Eq. (10.3), and the expansion [G (5.2.84)]–[G (5.2.88)] then gives Eq. (14.5).

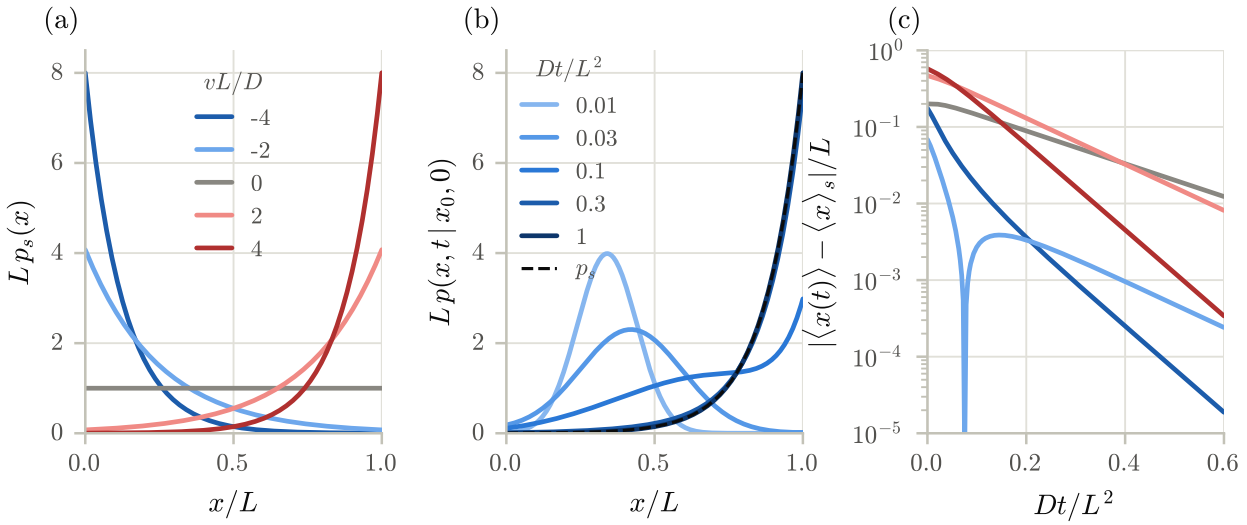


Figure 17: Constant drift, reflecting walls. (a) Stationary densities for several vL/D. (b) Relaxation from $x_0 = 0.3L$ at $vL/D = 4$: the peak is carried to the right wall while it spreads. (c) Distance of the mean from its stationary value, from $x_0 = 0.3L$; colours as in (a).

14.1 Spectrum

This is the Robin wall of Section 12, now at both ends, just as Section 5 put the mirror of Section 3 at both ends. The solution regular at $x = 0$ is $\phi = \cos kx + (\alpha/k)\sin kx$; the condition at L reduces to $(k + \alpha^2/k)\sin kL = 0$, so $k_n = n\pi/L$ exactly as without drift. The extra solution with imaginary $k = \pm i\alpha$ is $u = e^{\alpha x}$, which gives $p_s \propto e^{2\alpha x}$ with $\lambda = 0$. Using $\int_0^L \phi_n^2 = \frac{L}{2}(1 + \alpha^2/k_n^2)$ in Eq. (10.4) gives Eq. (14.5); at $v = 0$ it is Eq. (5.2).

14.2 Stationary state and relaxation

- $p_s \propto e^{-2U/D}$ with $U = -vx$: a barometric (sedimentation) profile, piled against the wall the drift points to, with decay length $\ell/2$ (Fig. 17a).
- The gap $\lambda_1 = 1/2\tau + \pi^2 D/2L^2$ grows with $|v|$ in either direction. For $L \gg \pi\ell$ the relaxation time is 2τ, shorter than the transit time $(L/\ell)\,\tau$ by a factor $2\ell/L$. Until the transit is over, the large prefactor $e^{\alpha(x-x_0)}$ keeps many modes in play (Fig. 17b).
- In Fig. 17c the deviation of the mean decays at $\lambda_1(v)$, with the same slope for $\pm v$. For $vL/D = -2$ it changes sign once: from $x_0 = 0.3L$ the mean undershoots $\langle x\rangle_s = 0.23L$ before settling.

15 Constant drift, absorbing walls

Setup and key results

$A = v$, $B = D$ on $(0, L)$, $p = 0$ at both walls. With $\alpha = v/D = \pm 1/\ell$, $\tau = D/v^2$, and $k_n = n\pi/L$:

$$\lambda_n = \frac{1}{2\tau} + \frac{n^2\pi^2 D}{2L^2}, \quad n \geq 1, \qquad P_n = \sqrt{\frac{2p_s}{L}}\,\sin k_n x, \quad Q_n = \sqrt{\frac{2}{Lp_s}}\,\sin k_n x \tag{15.1}$$

$$p(x, t \,|\, x_0, 0) = e^{\alpha(x-x_0) - t/2\tau} \times [\text{free absorbing propagator Eq. (6.2)}] \tag{15.2}$$

$$S(t \,|\, x_0) = \frac{2}{L}\, e^{-\alpha x_0 - t/2\tau} \sum_{n\geq 1} \frac{k_n[1 - (-1)^n e^{\alpha L}]}{\alpha^2 + k_n^2}\, \sin k_n x_0\, e^{-Dk_n^2 t/2} \tag{15.3}$$

$$j_0 = e^{-\alpha x_0 - t/2\tau}\, j_0^{\text{free}}, \qquad j_L = e^{\alpha(L-x_0) - t/2\tau}\, j_L^{\text{free}} \quad \text{(free fluxes from Eq. (6.9))} \tag{15.4}$$

$$\pi_L(x_0) = \frac{1 - e^{-2\alpha x_0}}{1 - e^{-2\alpha L}}, \qquad T(x_0) = \frac{L\,\pi_L(x_0) - x_0}{v} \tag{15.5}$$

$$T_L(x_0) = \frac{L\coth \alpha L - x_0 \coth \alpha x_0}{v}, \qquad T_0(x_0) = \frac{L\coth \alpha L - (L - x_0)\coth \alpha(L - x_0)}{v} \tag{15.6}$$

$$q_\infty(x) = \frac{\alpha^2 + k_1^2}{k_1(1 + e^{\alpha L})}\, e^{\alpha x} \sin k_1 x, \qquad \kappa \to \lambda_1 = \frac{1}{2\tau} + \frac{\pi^2 D}{2L^2} \tag{15.7}$$

Here $p_s = 2\alpha e^{2\alpha x}/(e^{2\alpha L} - 1)$ is that of the reflecting interval [G (5.2.91)] (Section 14), and $q_\infty = P_1 / \int P_1$.

Physical picture. A slope between two cliffs: the biased gambler's ruin, or a charged molecule driven by a voltage through a pore and leaving through either end, as in the simplest models of DNA translocation through a nanopore. π_L is the probability of being pushed through rather than falling back.

From Gardiner. With $\psi = e^{2vx/D}$:

- T from [G (5.2.158)], or by solving [G (5.2.154)];
- π_0, π_L from the corrected [G (5.2.189)]–[G (5.2.190)];
- T_0, T_L from [G (5.2.183)].

A compact general form is $T = \pi_L(x_0)F(L) - F(x_0)$ with $F(x) = \frac{2}{D}\int_0^x \frac{\mathrm{d}y}{\psi(y)} \int_0^y \psi(z)\,\mathrm{d}z$. That is the particular solution of [G (5.2.154)] plus a multiple of π_L, chosen so that $T(L) = 0$. Part III uses the same form.

15.1 Propagator, survival, fluxes

Absorbing walls keep $u = 0$, so u is exactly the free absorbing propagator and everything follows from Section 6; letting $L \to \infty$ gives back Section 13. The $u_n = \sin k_n x$ have $N_n = L/2$, and Eq. (10.3) gives the P_n, Q_n of the box: Part I's $\sqrt{2}\sin k_n x$, tilted by $e^{\pm\alpha x}$. Survival uses $\int_0^L e^{\alpha x}\sin k_n x\,\mathrm{d}x = k_n[1 - (-1)^n e^{\alpha L}]/(\alpha^2 + k_n^2)$. Each exit flux is the free one times $e^{\alpha(w - x_0)}$, which favours the downstream wall, times the common decay $e^{-t/2\tau}$.

15.2 Splitting probabilities and exit times

With $\psi = e^{2\alpha x}$, $\pi_L' \propto e^{-2\alpha x}$ (Section 1.7).

- π_L is the biased gambler's ruin. For $v > 0$ and $L \gg \ell$, $\pi_L \approx 1 - e^{-2x_0/\ell}$: exit is downstream unless the start is within about $\ell/2$ of the upstream wall (Fig. 18b).
- $T = (L\pi_L - x_0)/v$ solves $vT' + \frac{1}{2}DT'' = -1$. For strong drift $T \approx (L - x_0)/v$, the transit time (Fig. 18c). The maximum of T moves upstream.
- **Conditional exit times are even in v.** Conditioning on exit at L is a Doob transform with $h = \pi_L$. It replaces the drift by $v + D\,\partial_x \ln \pi_L = v \coth \alpha x$, which is even in v. So the paths that leave through L have the same statistics whether the drift helps or opposes them (Fig. 18d). Girsanov says the same: the likelihood ratio between drifts v and $-v$ for a path exiting at L is $e^{2\alpha(L-x_0)}$, independent of when it exits. As $v \to 0$ these reduce to $T_L = (L^2 - x_0^2)/3D$ and $T_0 = x_0(2L - x_0)/3D$.

15.3 Long times

The QSD is the sine tilted by $e^{\alpha x}$ (this is the preset `walls` in `sde-explorer`). For $L \gg \ell$ it piles up against the downstream wall in a layer of width about ℓ, and $\lambda_1 \approx 1/2\tau$. Starting from the QSD, T is exponential with mean $1/\lambda_1$; the check $\int q_\infty T \, \mathrm{d}x_0 = 1/\lambda_1$ holds for every v.

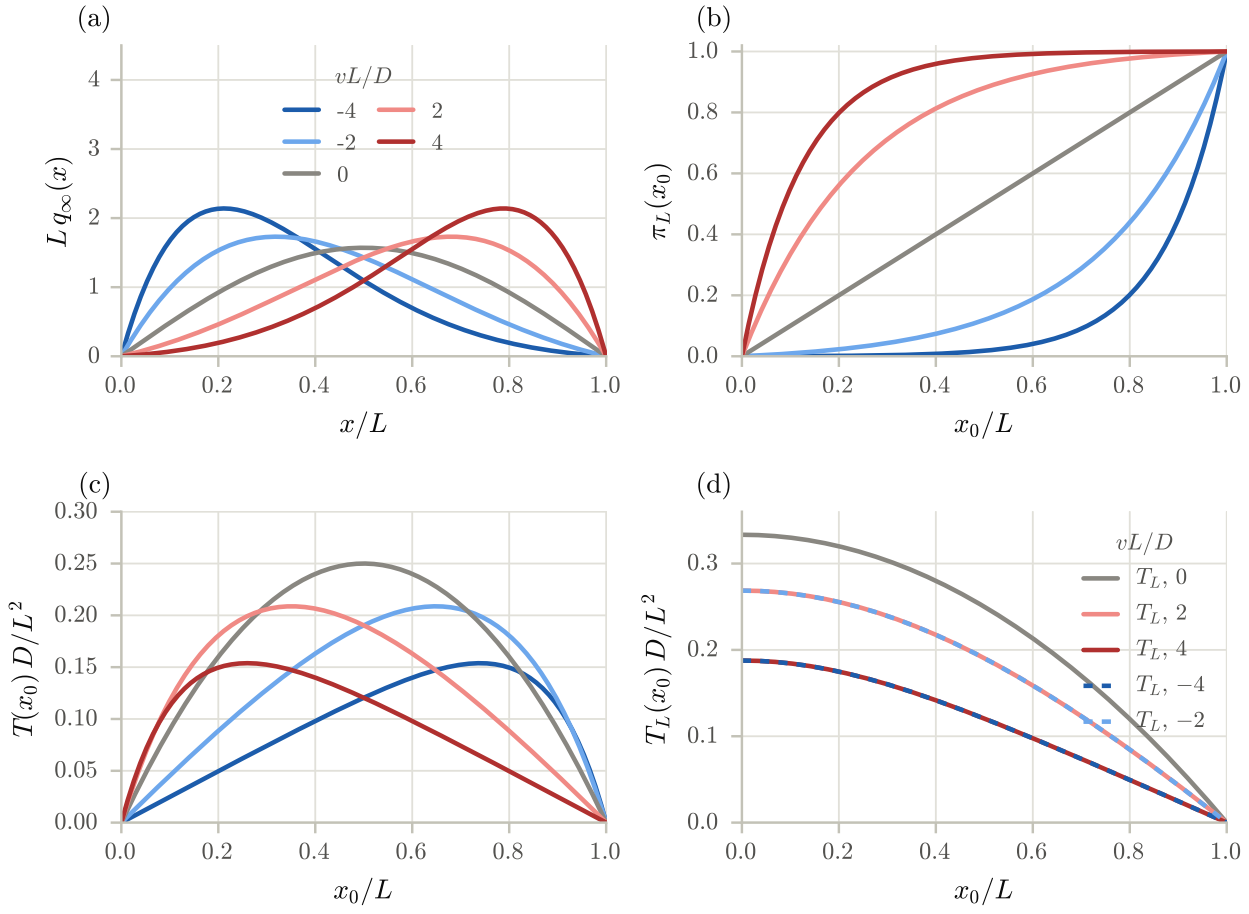


Figure 18: Constant drift, absorbing walls; colours as in (a) throughout. (a) Quasi-stationary densities. (b) Probability of leaving through L. (c) Mean exit time. (d) Mean exit time given exit through L. Dashed curves ($v < 0$) lie exactly on the solid ones with the opposite drift.

16 Constant drift, reflecting at 0, absorbing at L

Setup and key results

$A = v$, $B = D$ on $(0, L)$, $J = 0$ at $x = 0$, $p = 0$ at $x = L$. With $\alpha = v/D = \pm 1/\ell$ and $\tau = D/v^2$:

$$u_n = \cos k_n x + \frac{\alpha}{k_n} \sin k_n x, \qquad k_n \cos k_n L + \alpha \sin k_n L = 0, \qquad \lambda_n = \frac{1}{2\tau} + \tfrac{1}{2} D k_n^2 \tag{16.1}$$

$$u_0 = \cosh \kappa x + \frac{\alpha}{\kappa} \sinh \kappa x, \qquad \kappa \cosh \kappa L + \alpha \sinh \kappa L = 0 \qquad \text{if } v < 0,\ L > \ell,$$

$$\lambda_0 = \frac{1}{2\tau} - \tfrac{1}{2} D \kappa^2 \quad \text{(bound state)} \tag{16.2}$$

$$P_n = \sqrt{\frac{p_s}{N_n}}\, u_n, \quad Q_n = \frac{u_n}{\sqrt{N_n p_s}}, \quad N_n = \int_0^L u_n^2 \, \mathrm{d}x, \quad p_s \text{ of Section 14} \tag{16.3}$$

$$p(x, t \,|\, x_0, 0) = \sum_n P_n(x)\, Q_n(x_0)\, e^{-\lambda_n t} = e^{\alpha(x - x_0)} \sum_n \frac{u_n(x)\, u_n(x_0)}{N_n}\, e^{-\lambda_n t} \tag{16.4}$$

$$T(x_0) = \frac{1}{v}\Big[(L - x_0) + \frac{e^{-2\alpha L} - e^{-2\alpha x_0}}{2\alpha}\Big] \tag{16.5}$$

$$\lambda_0 \simeq \frac{1}{T} \simeq \frac{2}{\tau}\, e^{-2L/\ell} \qquad (v < 0,\ L \gg \ell:\ \text{Arrhenius}) \tag{16.6}$$

Physical picture. For $v < 0$, a particle held against a floor by gravity in a container whose rim, at height L, is a cliff: to escape it must climb FL, an activated escape over a linear barrier. For $v > 0$ the force carries it over the rim. The same geometry describes an ion crossing a membrane against a field into a sink.

From Gardiner. [G (5.2.160)] with $\psi = e^{2vx/D}$ integrates in closed form to the boxed T. The eigenvalue condition comes from [G (5.2.69)] at 0 and [G (5.2.67)] at L, imposed on Q_n; $P_n = p_s Q_n$ with the reflecting p_s [G (5.2.91)]. [G §5.2.7c] ([G (5.2.162)]–[G (5.2.174)]) is Gardiner's escape problem. Note that there he writes $\partial_t p = \partial_x(U'p) + D\,\partial_x^2 p$, so his D is our $D/2$: his Arrhenius factor $e^{\Delta U/D}$ [G (5.2.174)] is our $e^{2\Delta U/D}$.

16.1 Spectrum

Now the Robin condition of Section 12 at 0 meets a Dirichlet condition at L, so the k_n solve the transcendental Eq. (16.1), $\tan k_n L = -k_n/\alpha$. The roots move as the drift changes:

- $v > 0$ (toward the absorbing wall): $k_n L \in ((n - \frac{1}{2})\pi, n\pi)$, $n \geq 1$; the ground state sits just above $1/2\tau$.
- $v = 0$: $k_n L = (n + \frac{1}{2})\pi$, Section 7.
- $v < 0$, $L < \ell$: the lowest root falls toward $k = 0$, reached at $L = \ell$ with $u_0 = 1 - x/L$.
- $v < 0$, $L > \ell$: the lowest root moves onto the imaginary axis, $k = i\kappa$, giving the bound state Eq. (16.2) with $\lambda_0 < 1/2\tau$. Its rate falls exponentially with L/ℓ (Fig. 19b). The bound state exists once the interval is longer than the drift length.

The eigenfunctions u_n are orthogonal in L^2 (Robin plus Dirichlet is self-adjoint), so Eq. (10.4) applies; the script finds the roots numerically.

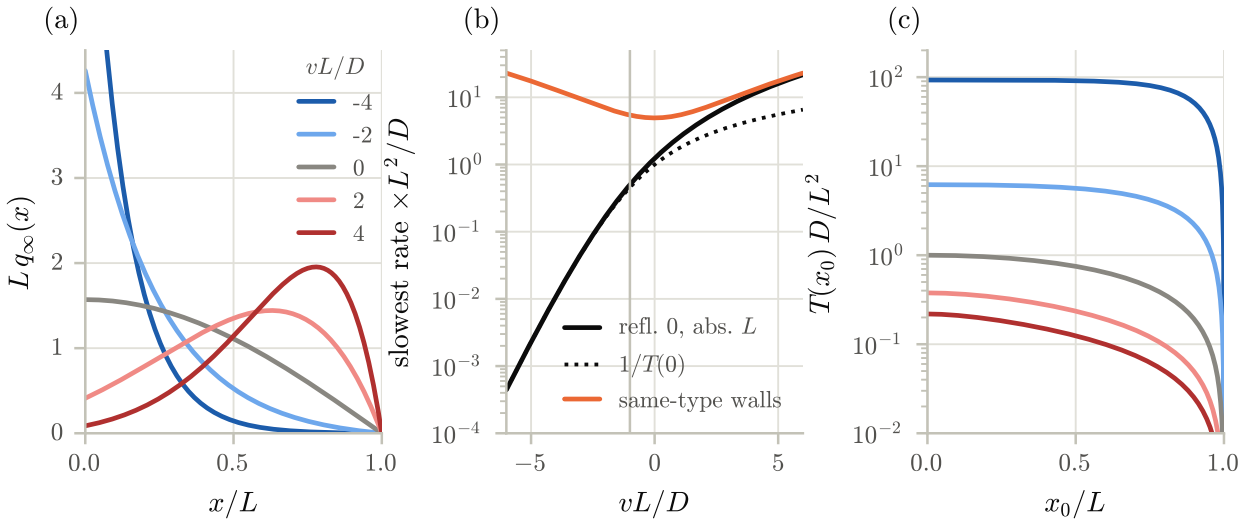


Figure 19: Constant drift, reflecting at 0, absorbing at L. (a) Quasi-stationary densities. (b) Slowest decay rate against $vL/D = \pm L/\ell$; the grey line marks $L = \ell$ with $v < 0$, where the bound state appears. For comparison, the orange curve is the gap $1/2\tau + \pi^2 D/2L^2$, shared by the two same-type intervals; the dotted curve is $1/T(0)$. (c) Mean exit time (log scale); colours as in (a).

16.2 Exit time and escape over a barrier

[G (5.2.160)] with $\psi = e^{2\alpha x}$ gives $T = \frac{2}{D}\int_{x_0}^{L} \mathrm{d}y\, e^{-2\alpha y}\int_0^y e^{2\alpha z}\mathrm{d}z$, which integrates to the boxed form. At $v = 0$ it is $(L^2 - x_0^2)/D$.

- $v > 0$: $T \approx (L - x_0)/v$, the transit time.
- $v < 0$: the particle sits in the well $U = |v|x$ against the reflecting wall and must climb $\Delta U = |v|L$ to escape. Since $p_s \propto e^{-2U/D}$, $\frac{1}{2}D$ plays the role of k_BT, so $2\Delta U/D = 2L/\ell$ and $T \simeq \frac{1}{2}\tau\, e^{2L/\ell}$. This is independent of x_0 except within about ℓ of L (Fig. 19c). It is Gardiner's escape problem [G §5.2.7c] for a linear potential, and here it is exact.
- In the same regime $\kappa \to 1/\ell$, $\lambda_0 \to 1/T$, and the QSD becomes the barometric profile of Section 14, cut off near L (Fig. 19a). At $L = 4\ell$: $\lambda_0 = 0.010786$, $1/T(0) = 0.010767$, and $(2/\tau)e^{-2L/\ell} = 0.010735$ (all in D/L^2).

17 Constant drift on the circle

Setup and key results

$A = v$, $B = D$ on the circle of circumference L: an overdamped rotor driven by a constant torque. With $\ell = D/|v|$, $\tau = D/v^2$, $q_m = 2\pi m/L$, and p_{free} the propagator Eq. (8.1):

$$p(x,t\,|\,x_0,0) = p_{\text{free}}(x - vt, t\,|\,x_0,0) = \frac{1}{L} + \frac{2}{L}\sum_{m\geq 1} \cos q_m(x - x_0 - vt)\, e^{-Dq_m^2 t/2} \tag{17.1}$$

$$\lambda_m = \tfrac{1}{2}Dq_m^2 + ivq_m, \quad m \in \mathbb{Z} \text{ (complex pairs)}, \qquad \text{gap } \operatorname{Re}\lambda_{\pm 1} = \frac{2\pi^2 D}{L^2} \text{ for every } v \tag{17.2}$$

$$P_m = \frac{e^{iq_m x}}{L}, \quad Q_m = e^{-iq_m x} \quad (\text{as for } v = 0), \qquad P_m \neq p_s Q_m \tag{17.3}$$

$$p_s = \frac{1}{L}, \qquad J_s = \frac{v}{L} \neq 0, \qquad \Phi = \frac{2vL}{D} = \pm\frac{2L}{\ell} \quad (\text{NESS}) \tag{17.4}$$

$$v_{\text{w}} = v, \qquad D_{\text{eff}} = D, \qquad \text{entropy production } \frac{2v^2}{D} = \frac{2}{\tau} \tag{17.5}$$

$$T_{\text{w}} = \frac{L}{|v|}\tanh\frac{L}{\ell}; \qquad \text{to a target: } \pi_L,\ T \text{ of Section 15 with } x_0 \to d \tag{17.6}$$

Physical picture. An overdamped rotor driven by a constant torque: a colloid pushed around a ring-shaped optical trap by a constant force [10], or a paddle wheel turned steadily in a viscous fluid. The steady state is uniform but carries a current; the work done by the torque is dissipated as heat, at the entropy-production rate $2/\tau$ (in units of k_B).

From Gardiner. [G (5.2.44)]–[G (5.2.47)] with $\psi = e^{2vx/D}$ and $B = D$: [G (5.2.47)] gives $p_s = p_s(a)$, uniform, and the corrected [G (5.2.46)] gives $J = v/L$; as printed it gives $-2v/L$ (Section 1.9). The propagator is the drifting Gaussian [G (3.8.58)] summed over the lifts Eq. (1.16), or its characteristic function at $s = q_m$. T_{w} is the $T = (L'\pi_{L'} - x_0)/v$ of Section 15 on the cover, with $L' = 2L$ and $x_0 = L$.

17.1 The gauge fails; a boost works

The reduction of Part II to Part I, Eq. (10.2), multiplies by $e^{\alpha(x-x_0)}$. That is not a function on the circle: going once around multiplies it by $e^{\alpha L}$. In the language of Section 1.9, $2A/B = 2\alpha$ is not the derivative of a function on the circle; its period is $\Phi = 2\alpha L \neq 0$. One can still gauge, but then u lives on a *twisted* circle, $u(x + L) = e^{-\alpha L}u(x)$, not on the circle of Part I (the script checks this).

What works instead is the Galilean boost $x \to x - vt$ that Section 11 mentioned for the line. There it was useless once a wall appeared, because the wall does not move. A rotation is a symmetry of the circle and there is no wall to break it, so $p(x,t\,|\,x_0,0) = p_{\text{free}}(x - vt, t\,|\,x_0,0)$ exactly: in the rotating frame the drift is gone (Fig. 20a). The circle is the only bounded domain in these notes on which a change of frame removes the drift.

17.2 Spectrum: complex, with the gap unchanged

- Translations still commute with the generator, so the Fourier modes are still eigenfunctions. Their eigenvalues $\frac{1}{2}Dq_m^2 + ivq_m$ are complex: mode m decays and rotates at m times the revolution

frequency $2\pi v/L$ (Fig. 20b). The doublet of Section 8 becomes a complex-conjugate pair, the rotating waves $e^{\pm iq_m(x-vt)}$. Since $A = v$ is odd under no reflection, the circle no longer folds into intervals.

- The real parts do not depend on v. Relaxation goes at $2\pi^2 D/L^2$ whatever the drift, because the drift only rotates the density. On an interval drift raised every rate by $1/2\tau$ (Section 10), because the walls make the density pile up against them.
- The generator is not self-adjoint, but it is normal: the Fourier modes are orthonormal in plain L^2. So the transient of Section 10, in which the slowest mode can take a transit time to dominate, does not occur.
- Gardiner's P_m and Q_m are the Fourier modes of the free circle, unchanged; only λ_m moves. [G (5.2.79)] holds, but $P_m = p_s Q_m$ does not, since [G (5.2.71)]–[G (5.2.72)] need $J_s = 0$. What holds instead is $P_m = p_s \overline{Q_m}$: forward and backward modes are complex conjugates, as for a normal operator. The backward modes are rotating waves too: $\langle e^{-iq_m x(t)} \rangle = Q_m(x_0)\, e^{-\lambda_m t}$, rotating at v.

17.3 A steady state with a current

The stationary density is uniform, exactly as in Section 8, yet this is not equilibrium: $J_s = v/L$ flows forever, because $\Phi = 2L/\ell \neq 0$. Nothing in p_s reveals the current. It shows up in the winding, $v_{\mathrm{w}} = LJ_s = v$ Eq. (1.17), and in the dissipation: the drift does work against friction at rate vv_{w}, and dividing by the effective temperature $D/2$ gives the entropy production rate $2v^2/D = 2/\tau$. The same drift has no steady state on the line, and on an interval it gives the barometric p_s with $J = 0$ (Section 14). Only the circle lets it flow.

17.4 Winding and first passage

- The lift is the drifting Gaussian of Section 11, so $v_{\mathrm{w}} = v$ and $D_{\mathrm{eff}} = D$.
- The first full winding is, on the cover, the absorbing interval of length $2L$ of Section 15, started in the middle. It gives $T_{\mathrm{w}} = (L/|v|)\tanh(L/\ell)$, and a winding in the direction of the drift with probability $(1 + e^{-2L/\ell})^{-1}$. The renewal argument then gives $v_{\mathrm{w}} = L\tanh(vL/D)/T_{\mathrm{w}} = v$, as it must. For $L \gg \ell$, $T_{\mathrm{w}} \to L/|v|$, the revolution time; for $L \ll \ell$, $T_{\mathrm{w}} \to L^2/D$ (Section 8).
- First passage to a target: cut there to get Section 15 with $x_0 \to d$. For strong drift $v > 0$ the target is reached from below, after travelling $L - d$ in a time $T \approx (L-d)/v$, unless the start is within about $\ell/2$ above it (Fig. 20c).

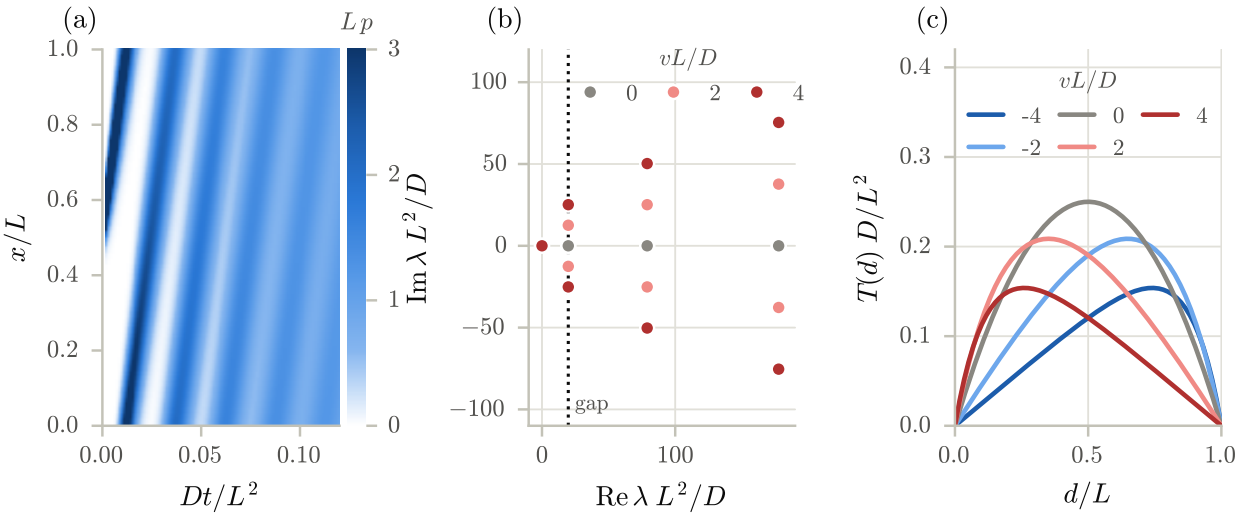


Figure 20: Constant drift on the circle. (a) Density against time for $vL/D = 40$, from $x_0 = L/2$: the peak winds around while it spreads, a helix on the cylinder $S^1 \times \mathbb{R}_t$. (b) Eigenvalues λ_m, $|m| \le 3$, in the complex plane. Drift moves them vertically but not horizontally, so the gap (dotted) does not depend on v. Reversing v gives the same set. (c) Mean time to reach a target from a distance d above it, measured in the $+x$ direction.

18 Summary of constant drift

Walls	Slowest rate	Long times
reflecting 0, L	$\frac{1}{2\tau} + \frac{\pi^2 D}{2L^2}$ (gap)	$p_s \propto e^{2vx/D}$
absorbing 0, L	$\frac{1}{2\tau} + \frac{\pi^2 D}{2L^2}$	QSD $\propto e^{vx/D} \sin \frac{\pi x}{L}$
refl. 0, abs. L	root of Eq. (16.1), or bound state if $v < 0$, $L > \ell$	$\lambda_0 \simeq \frac{2}{\tau} e^{-2L/\ell}$ for $v < 0$, $L \gg \ell$
reflecting 0 only	$1/2\tau$ (gap) if $v < 0$; none if $v \ge 0$	$p_s = \frac{2}{\ell} e^{-2x/\ell}$ if $v < 0$
absorbing 0 only	edge $1/2\tau$ if $v < 0$	QSDs $\lambda \le \frac{1}{2\tau}$; point start: $\frac{x}{\ell^2} e^{-x/\ell}$
none	—	$g(x - x_0 - vt)$
circle	$\frac{2\pi^2 D}{L^2}$ (gap of $\mathrm{Re}\,\lambda$), complex pairs	$p_s = 1/L$ with $J_s = v/L$ (NESS)

Table 6: Constant drift $A = v$, $B = D$, in every geometry, with $\ell = D/|v|$ and $\tau = D/v^2$.

- The dynamics sets the scales $\ell = D/|v|$ and $\tau = D/v^2$. Lengths L and x_0 only enter as L/ℓ and x_0/ℓ. Long-time objects (p_s, the half-line QSD, the gap $1/2\tau$) depend on ℓ and τ alone; x_0 shapes only transients and first passage.
- Absorbing walls commute with the gauge, so drift only multiplies free results by $e^{\alpha(x-x_0)-t/2\tau}$. Reflecting walls turn into Robin conditions, $u' = \pm u/\ell$, which can create a bound state (the barometric p_s, or the Arrhenius ground state of the mixed interval).
- Drift adds $1/2\tau$ to every rate of the continuum; with two same-type walls this is the whole effect on the spectrum, symmetric in v.
- With one absorbing and one reflecting wall the sign of v matters enormously. Drift toward the absorbing wall gives transit, $T \approx (L/\ell)\,\tau$; drift away gives activated escape, $T \simeq \frac{1}{2}\tau\, e^{2L/\ell}$.

- On the half-line, drift toward the wall turns the free algebraic behaviour into a gap: a stationary state with a reflecting wall, a continuum of QSDs with an absorbing one. Drift away gives escape to infinity.
- Exit statistics conditioned on the exit wall do not depend on the sign of v (Doob/Girsanov).
- On the circle the gauge fails, because its factor is not periodic, and a boost removes the drift instead. The gap does not depend on v, and the uniform p_s carries a current v/L.

Part III
Linear drift: Ornstein–Uhlenbeck, $A = -kx$, $B = D$

19 Ornstein–Uhlenbeck: scales and the map to free diffusion

The equation is $\partial_t p = \partial_x(kxp) + \frac{1}{2}D\,\partial_x^2 p$ [G (3.8.61)], from $\mathrm{d}x = -kx\,\mathrm{d}t + \sqrt{D}\,\mathrm{d}W$ [G (4.4.22)]: overdamped motion in the well $U = \frac{1}{2}kx^2$. All walls in this part are placed relative to the bottom of the well at $x = 0$.

Physical picture. A harmonic trap, $\mathcal{U} = \frac{1}{2}\mathcal{K}x^2$ with $k = \mathcal{K}/\gamma$: a bead in an optical tweezer, an AFM cantilever in liquid, or any overdamped spring. The same equation also describes the velocity of a free Brownian particle, which is where Ornstein and Uhlenbeck started: there x is the velocity and $k = \gamma/m$ [G (6.4.2)]–[G (6.4.3)].

From Gardiner. The book solves this process three ways, and each is used below.
- [G §3.8.4] uses the characteristic function: the first-order equation [G (3.8.63)] is solved by characteristics to give [G (3.8.71)], a Gaussian.
- [G §4.4.4] solves the SDE directly with the substitution $y = xe^{kt}$ [G (4.4.23)], which removes the drift, $\mathrm{d}y = \sqrt{D}e^{kt}\mathrm{d}W$ [G (4.4.25)]. This gives the mean, the variance and the correlation function [G (4.4.26)]–[G (4.4.30)].
- [G §5.2.6c] gives the eigenfunctions: Hermite polynomials [G (5.2.115)]–[G (5.2.122)], with $\lambda = nk$ [G (5.2.119)] and p_s [G (5.2.120)].

Scales. As in Part II, the dynamics fixes a length and a time,

$$\ell \equiv \sqrt{D/k}, \qquad \tau \equiv 1/k, \qquad \ell^2 = D\tau, \qquad \xi \equiv x/\ell. \tag{19.1}$$

The stationary variance is $D/2k = \ell^2/2$, so $p_s = e^{-\xi^2}/(\ell\sqrt{\pi})$. A wall at distance L from the bottom sits at height $\Delta U = \frac{1}{2}kL^2$, and $\Delta U/(D/2) = (L/\ell)^2$: ξ^2 measures energy in units of the "temperature" $D/2$.

The map. Gardiner's substitution turns OU into free diffusion with $B = D$, run on the clock $\theta(t) = \int_0^t e^{2ks}\,\mathrm{d}s = (e^{2kt} - 1)/2k$. A wall at $x = 0$ stays at $y = 0$, so for any walls at the bottom of the well

$$p_{\mathrm{OU}}(x, t \,|\, x_0, 0) = e^{kt}\, p_{\mathrm{free}}(e^{kt}x, \theta(t) \,|\, x_0, 0), \tag{19.2}$$

with p_{free} taken from Part I with the same walls. For $t \ll \tau$, $\theta \approx t$ and OU looks free. For $t \gg \tau$ the free spreading $\sqrt{D\theta} \propto e^{kt}$ is exactly cancelled by the factor e^{-kt}: this is how a confining force turns

algebraic spreading into a stationary state. A wall at $x = a \neq 0$ moves under the map ($y = ae^{kt}$), so the intervals of Sections 23 to 25 need eigenfunctions instead.

Eigenfunctions. In the variable ξ, the backward eigen-equation is Hermite's equation, as in Gardiner's [G (5.2.116)] and [G (5.2.117)]:

$$Q'' - 2\xi Q' + 2\nu Q = 0, \qquad \lambda = \nu k, \tag{19.3}$$

whose even and odd solutions for any ν are Kummer functions,

$$Q_{\text{even}} = M\left(-\tfrac{\nu}{2}, \tfrac{1}{2}, \xi^2\right), \qquad Q_{\text{odd}} = \xi\, M\left(\tfrac{1-\nu}{2}, \tfrac{3}{2}, \xi^2\right). \tag{19.4}$$

On the line, normalisability forces $\nu = n$ and the Hermite polynomials [G (5.2.118)]. With walls, the wall conditions fix ν. The eigenfunctions are orthogonal with weight $p_s \propto e^{-\xi^2}$, and Gardiner's forward eigenfunctions are $P = p_s Q$. On every interval of this part p_s is the truncated half-Gaussian of the reflecting interval (Section 23), also when a wall absorbs [G (5.2.91)], and Q_n is normalised by $\int p_s Q_n^2 \, \mathrm{d}x = 1$ [G (5.2.80)]. In the symmetrised form of Section 1.5 this is the harmonic oscillator, $V = \frac{1}{2}k(\xi^2 - 1)$ (Fig. 3a).

A ladder. Differentiate the backward eigen-equation $A\,Q' + \frac{1}{2}DQ'' = -\lambda Q$: with $A' = -k$ constant, Q' obeys the same equation with $\lambda \to \lambda - k$. On the line this is the Hermite ladder $H_n' = 2nH_{n-1}$. With walls, a reflecting eigenfunction has $Q' = 0$ at the walls, so its derivative is an absorbing eigenfunction: *the reflecting spectrum is the absorbing spectrum shifted up by k*, plus $\lambda = 0$. For constant A (Parts I and II) the shift is zero, which is why the reflecting and absorbing intervals there share their nonzero eigenvalues.

20 Ornstein–Uhlenbeck on the whole line

Setup and key results

$A = -kx$, $B = D$ on $(-\infty, \infty)$; $\ell = \sqrt{D/k}$, $\tau = 1/k$, $\xi = x/\ell$.

$$p(x, t \,|\, x_0, 0) = \text{Gaussian}, \quad \langle x \rangle = x_0 e^{-t/\tau}, \quad \operatorname{Var} x = \tfrac{1}{2}\ell^2(1 - e^{-2t/\tau}) \tag{20.1}$$

$$= \frac{e^{-\xi^2}}{\ell\sqrt{\pi}} \sum_{n \geq 0} \frac{H_n(\xi) H_n(\xi_0)}{2^n n!}\, e^{-nt/\tau} \quad \text{(Mehler)} \tag{20.2}$$

$$\lambda_n = nk, \quad Q_n = \frac{H_n(\xi)}{\sqrt{2^n n!}}, \quad P_n = p_s Q_n, \quad p_s = \frac{e^{-x^2/\ell^2}}{\ell\sqrt{\pi}}, \quad \text{gap } k \tag{20.3}$$

$$C(t) = \langle x(t)x(0) \rangle_s = \tfrac{1}{2}\ell^2 e^{-|t|/\tau} \quad \text{(exactly one exponential)} \tag{20.4}$$

Physical picture. A bead in an optical tweezer. Its stationary variance $\ell^2/2 = k_B T/\mathcal{K}$ (equipartition) and its relaxation rate $k = \mathcal{K}/\gamma$ are the two standard ways of calibrating the trap stiffness; the rate appears as the corner frequency $\mathcal{K}/2\pi\gamma$ of the position power spectrum.

From Gardiner. The Gaussian is [G (3.8.71)] inverted, or [G (4.4.26)] with the moments [G (4.4.27)]–[G (4.4.29)]. Q_n is his [G (5.2.118)], normalised as in [G (5.2.79)]–[G (5.2.81)], and p_s is [G (5.2.120)] (our ξ is his $x\sqrt{k/D}$). The expansion Eq. (20.2) is [G (5.2.121)] with the coefficients [G (5.2.122)] evaluated for a sharp start, and $C(t)$ is the stationary limit of [G (4.4.30)]: only P_1 has $\int x P_n \neq 0$ [G (5.2.125)].

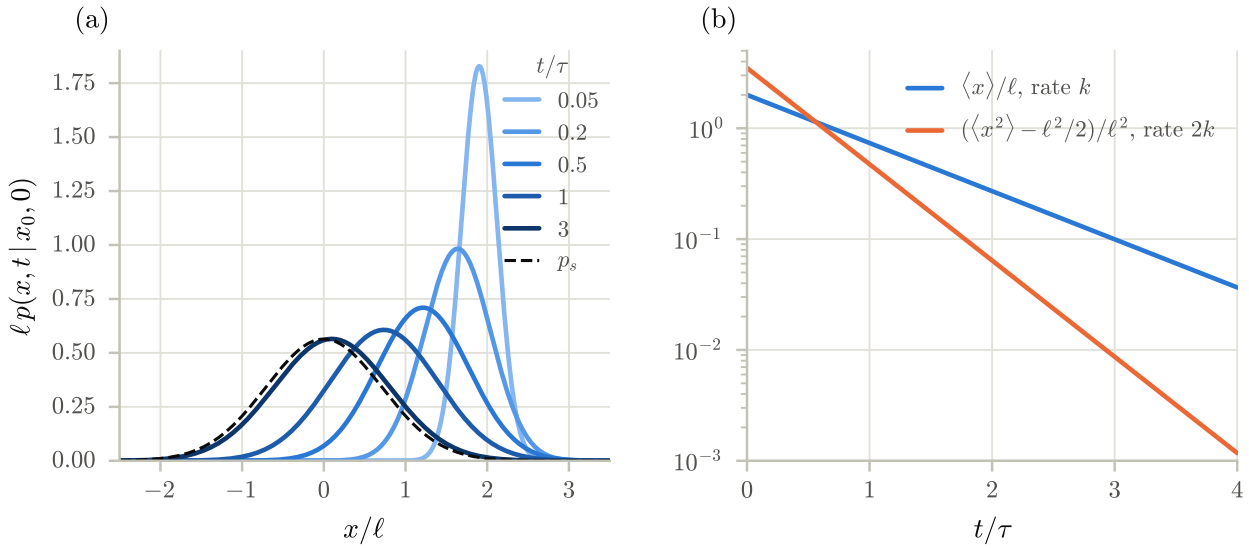


Figure 21: Ornstein–Uhlenbeck on the line, from $x_0 = 2\ell$; units $\ell = \sqrt{D/k}$, $\tau = 1/k$. (a) The Gaussian slides to the origin while widening to p_s. (b) $\langle x\rangle$ relaxes at the gap k; the second moment, with no overlap on mode 1, at $2k$.

20.1 From the free line

Put the Gaussian of Section 2 through Eq. (19.2): the mean x_0 becomes $x_0 e^{-kt}$, and the variance $D\theta$ becomes $D\theta e^{-2kt} = \frac{1}{2}\ell^2(1 - e^{-2kt})$. That is the whole solution; the Mehler sum is the same function written in eigenfunctions (they agree to 10^{-15}).

20.2 Gap and moments

- The gap is k, independent of D. In the Rayleigh quotient Eq. (1.15) the slowest function is $f = x$, and its cost is set by the curvature of the well. Noise sets only the width ℓ.
- $x \propto H_1$ has overlap with mode 1 only, so $\langle x(t)\rangle = x_0 e^{-kt}$ exactly, and $C(t) \propto e^{-k|t|}$ is a single exponential. Compare the reflecting interval, where the slowest mode carried 98.6 % (Section 5).
- x^2 is a combination of H_0 and H_2, so $\langle x^2\rangle - \frac{1}{2}\ell^2 = (x_0^2 - \frac{1}{2}\ell^2)e^{-2kt}$: it relaxes at $2k$, faster than the gap. This is the symmetry selection of Section 1.6 (Fig. 21b).

21 Ornstein–Uhlenbeck on the half-line, reflecting at the bottom

Setup and key results

$A = -kx$, $B = D$ on $(0, \infty)$, $J = 0$ at $x = 0$, the bottom of the well. With $m_t = x_0 e^{-t/\tau}$ and $\sigma_t^2 = \frac{1}{2}\ell^2(1 - e^{-2t/\tau})$:

$$p(x, t \,|\, x_0, 0) = G_{\sigma_t}(x - m_t) + G_{\sigma_t}(x + m_t), \qquad G_\sigma(y) = \frac{e^{-y^2/2\sigma^2}}{\sqrt{2\pi}\,\sigma} \tag{21.1}$$

$$p_s(x) = \frac{2}{\ell\sqrt{\pi}}\, e^{-x^2/\ell^2} \text{ (half-Gaussian)}, \qquad \langle x \rangle_s = \frac{\ell}{\sqrt{\pi}} \tag{21.2}$$

$$\lambda_m = 2mk, \quad Q_m = \frac{H_{2m}(\xi)}{\sqrt{2^{2m}(2m)!}}, \quad P_m = p_s Q_m \quad \text{(even Hermite)}, \quad \text{gap } 2k \tag{21.3}$$

$$\langle x^2(t) \rangle = x_0^2 e^{-2t/\tau} + \tfrac{1}{2}\ell^2(1 - e^{-2t/\tau}) \quad \text{(exact, as on the line)} \tag{21.4}$$

Physical picture. A trapped bead whose trap centre sits exactly on a hard surface, such as a coverslip at the focus of an optical tweezer. It explores only one half of the trap, so its density is a half-Gaussian.

From Gardiner. The wall is [G (5.2.13)] at $x = 0$. Of the Hermite eigenfunctions [G (5.2.118)], only the even ones satisfy the reflecting backward condition [G (5.2.69)], $H_n'(0) = 0$, which gives the spectrum. p_s is Eq. (1.5) restricted to $x > 0$.

21.1 From the free half-line

The wall sits at the fixed point of the map, so Eq. (19.2) applied to the image pair of Section 3 gives Eq. (21.1) directly: the source and its mirror image each turn into the OU Gaussian. Where the free half-line spread forever with no stationary state, the well freezes the spreading into the half-Gaussian.

21.2 Gap

The reflecting wall removes the odd modes, including the line's slowest one, H_1. The surviving Q_m are the line's, normalisation included; $P_m = p_s Q_m$ is twice the line's P_{2m}, because p_s is. The gap doubles to $2k$, and now even $\langle x \rangle$ relaxes at $2k$ (Fig. 22b). The second moment is unchanged by the wall, for the reason given in Section 3: x^2 is even.

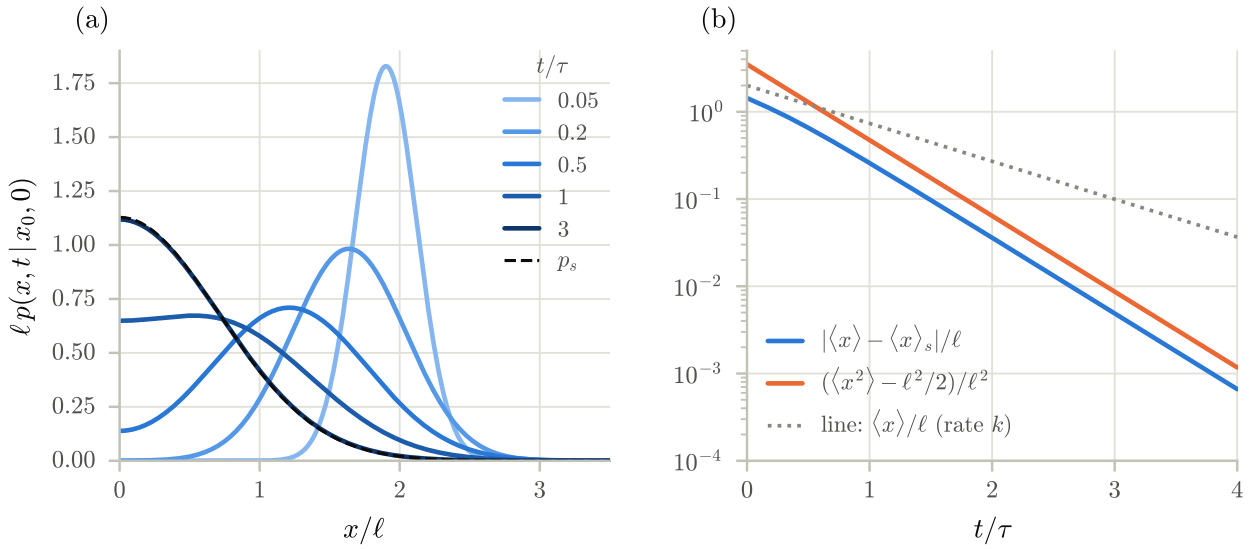


Figure 22: OU with a reflecting wall at the bottom of the well, from $x_0 = 2\ell$. (a) Relaxation to the half-Gaussian. (b) Both moments relax at $2k$; the dotted line is the line's $\langle x\rangle$, which relaxes at k.

22 Ornstein–Uhlenbeck on the half-line, absorbing at the bottom

Setup and key results

$A = -kx$, $B = D$ on $(0,\infty)$, $p = 0$ at $x = 0$. With m_t, σ_t as in Section 21:

$$p(x,t\,|\,x_0,0) = G_{\sigma_t}(x - m_t) - G_{\sigma_t}(x + m_t) \tag{22.1}$$

$$S(t\,|\,x_0) = \operatorname{erf}\frac{x_0/\ell}{\sqrt{e^{2t/\tau}-1}} \xrightarrow{t\gg\tau} \frac{2}{\sqrt{\pi}}\frac{x_0}{\ell}\,e^{-t/\tau} \tag{22.2}$$

$$\lambda_m = (2m+1)k \quad \text{(odd Hermite): bottom } k,\ \text{gap } 2k \tag{22.3}$$

$$Q_m = \frac{H_{2m+1}(\xi)}{\sqrt{2^{2m+1}(2m+1)!}}, \qquad P_m = p_s Q_m \tag{22.4}$$

$$q_\nu(x) = \frac{2\nu}{\ell}\,\xi\, e^{-\xi^2} M\!\left(\tfrac{1-\nu}{2}, \tfrac{3}{2}, \xi^2\right) \sim x^{-1-\nu}, \qquad \lambda = \nu k, \quad 0 < \nu \le 1 \tag{22.5}$$

$$q(x,t) \to q_1 = \frac{2x}{\ell^2}\,e^{-x^2/\ell^2}\ \text{(Rayleigh)}, \quad \kappa \to k \qquad \text{if } p(x,0) \text{ has a finite mean} \tag{22.6}$$

$$q(x,t) \to q_\nu, \quad S \propto e^{-\nu t/\tau} \qquad \text{if } p(x,0) \sim x^{-1-\nu},\ 0 < \nu < 1 \tag{22.7}$$

$$T(x_0) = \sqrt{\pi}\,\tau \int_0^{x_0/\ell} \operatorname{erfcx}(u)\,du \ \simeq\ \tau\left[\ln\frac{2x_0}{\ell} + \frac{\gamma_{\mathrm{E}}}{2}\right] \quad (x_0 \gg \ell) \tag{22.8}$$

with $\operatorname{erfcx}(u) = e^{u^2}\operatorname{erfc} u$ and γ_{E} Euler's constant. In P_m, $p_s = 2e^{-\xi^2}/\ell\sqrt{\pi}$ is the half-Gaussian of Section 21 [G (5.2.91)]. The q_ν are a continuum of QSDs; the tail of the initial density decides which one is reached.

Physical picture. Any mean-reverting quantity that is stopped the first time it returns to its mean. Examples are a trapped particle removed by a sticky surface through the trap centre, or a leaky integrate-and-fire neuron whose threshold sits at its resting potential. The logarithmic T

says that the return from far away is fast and deterministic, and only the last stretch, within ℓ, is diffusive.

From Gardiner. S is $G(x_0,t)$ of [G (5.2.140)]. The mean exit time is [G (5.2.161)] with the reflecting wall b moved to infinity and $\psi = e^{-x^2/\ell^2}$: $T = \frac{2}{D}\int_0^{x_0} \frac{\mathrm{d}y}{\psi(y)} \int_y^\infty \psi(z)\,\mathrm{d}z$, which becomes the erfcx integral. The odd Hermite functions satisfy the absorbing backward condition [G (5.2.67)], $H_n(0) = 0$. The other QSDs are the odd Kummer functions Eq. (19.4) with non-integer ν, which the book does not consider.

22.1 From the free half-line

Apply Eq. (19.2) to Section 4. The free survival $\mathrm{erf}(x_0/\sqrt{2D\theta})$ becomes the boxed S, because $2D\theta = \ell^2(e^{2kt}-1)$. Two of the free half-line's results change character:

- **QSDs appear.** Mapped back, the free Rayleigh profile $(x/D\theta)\,e^{-x^2/2D\theta}$ has width $\sqrt{D\theta}\,e^{-kt}$, which tends to $\ell/\sqrt{2}$. It stops spreading and freezes into the Rayleigh QSD $q_1 = (2x/\ell^2)e^{-x^2/\ell^2}$. In spectral terms, the bottom of the spectrum is no longer a continuum at 0 but the isolated level H_1 at k (Section 1.6). This is the QSD reached from a point start, but it is not the only one (below).
- **The mean exit time is finite and logarithmic.** From far away the particle relaxes deterministically, $x_0e^{-t/\tau}$, until it is within about ℓ of the wall, which takes $\tau\ln(x_0/\ell)$. Diffusion then finishes the job in a time of order τ. The constant $\ln 2 + \gamma_{\mathrm{E}}/2 \approx 0.98$ is checked numerically.

22.2 A continuum of QSDs, and which one is reached

- **The family.** For any ν, the forward eigenfunctions are $e^{-\xi^2}$ times the Kummer functions Eq. (19.4), and the odd one vanishes at the wall.
  - For $0 < \nu < 1$ it is positive. By Kummer's transformation $e^{-z}M(a,b,z) = M(b-a,b,-z)$ it decays as $\xi^{-1-\nu}$, so it is integrable: a QSD q_ν with decay rate νk.
  - The normalisation in the box follows from $\int_0^\infty M(a,b,-z)\,\mathrm{d}z = (b-1)/(a-1)$.
  - At $\nu = 1$ it is the Rayleigh density; for $\nu > 1$ it changes sign.

  None of these is in L^2 with the weight e^{ξ^2}, which is why the spectrum $(2m+1)k$ does not show them. In Gardiner's terms $q_\nu \propto p_sQ$ with $Q = \xi M(\frac{1-\nu}{2}, \frac{3}{2}, \xi^2)$, the odd solution of [G (5.2.117)] for non-integer ν; this Q grows like $e^{\xi^2}\xi^{-1-\nu}$, so $\int PQ$ [G (5.2.79)] diverges. The same happened for the continuum below the edge in Section 13. The script checks that q_ν evolves exactly as $e^{-\nu kt}q_\nu$.
- **Point start, or any start with a finite mean.** Through the map, the free half-line's density near the wall is $\propto x\int y\,p(y,0)\,\mathrm{d}y$ (Section 4). That maps back to the Rayleigh QSD, which a point start approaches at the gap $2k$.
- **Power-law starts.** If $p(y,0) \sim y^{-1-\nu}$ with $0 < \nu < 1$, the mean is infinite and the far tail dominates. The map turns the integral over the tail into $\int_0^\infty z^{-1-\nu}\sinh(wz)\,e^{-z^2/2}\,\mathrm{d}z$, with $w = \sqrt{2}\,\xi$. That integral is w times a Kummer function of $w^2/2 = \xi^2$, so the conditioned density converges to q_ν and $S \propto e^{-\nu kt}$, exactly [18] (Fig. 5b).
- **Why.** A particle starting at $y \gg \ell$ needs about $\tau\ln(y/\ell)$ to come down; this is the logarithm in $T(x_0)$ above. A tail $y^{-\nu}$ of starting points therefore arrives as $e^{-\nu kt}$, which beats the bulk's e^{-kt} when $\nu < 1$. Here the rule is exact, because the map is. A tail heavier than every power gives no limit: the conditioned density keeps following the incoming tail outward.
- The unbounded $T(x_0)$ is also what rules out a unique QSD (Section 1.8).

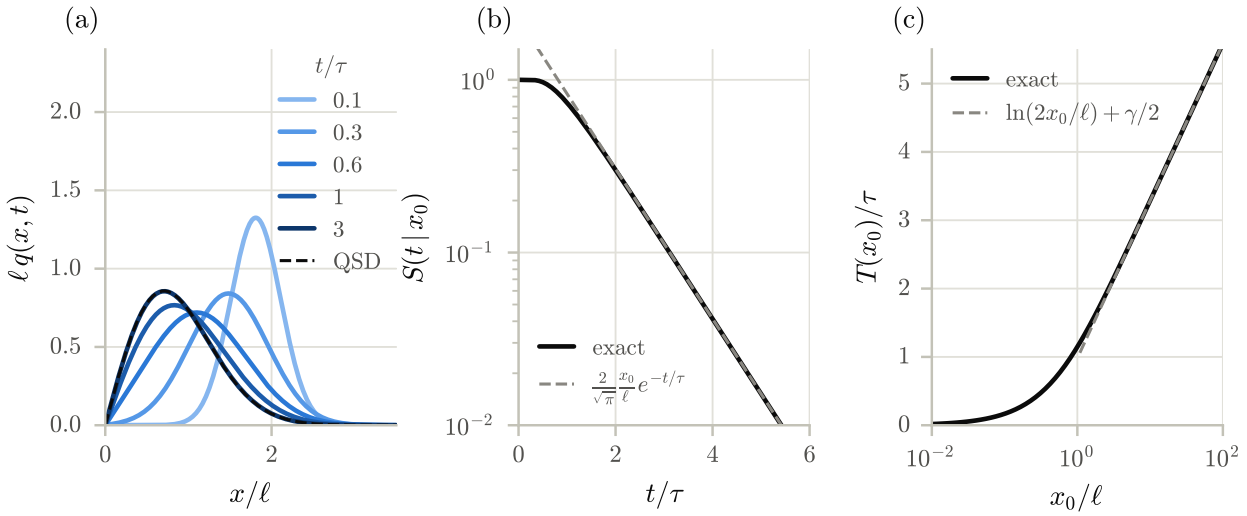


Figure 23: OU with an absorbing wall at the bottom of the well, from $x_0 = 2\ell$. (a) From a point start the conditioned density converges to the Rayleigh QSD; compare Fig. 7b, where the same shape kept spreading. (b) Survival decays at $\lambda_{\min} = k$. (c) Mean exit time, with its logarithmic asymptote.

23 Ornstein–Uhlenbeck on an interval, both walls reflecting

Setup and key results

$A = -kx$, $B = D$ on $(0, L)$ with $x = 0$ at the bottom of the well, $J = 0$ at both walls.

$$p_s(x) = \frac{2\,e^{-x^2/\ell^2}}{\ell\sqrt{\pi}\,\operatorname{erf}(L/\ell)} \quad \text{(truncated half-Gaussian)} \tag{23.1}$$

$$Q_n \propto M\!\left(-\tfrac{\nu_n}{2}, \tfrac{1}{2}, \xi^2\right), \quad Q_n'(L/\ell) = 0, \quad P_n = p_s Q_n, \quad \lambda_n = \nu_n k \tag{23.2}$$

$$p(x, t\,|\,x_0, 0) = \sum_n P_n(x)\, Q_n(x_0)\, e^{-\nu_n t/\tau}, \quad \int_0^L p_s Q_n^2\, \mathrm{d}x = 1, \quad \nu_0 = 0,\ Q_0 = 1 \tag{23.3}$$

$$\text{gap } \lambda_1 \to \frac{\pi^2 D}{2L^2}\ (L \ll \ell), \qquad \lambda_1 \to 2k\ (L \gg \ell) \tag{23.4}$$

Physical picture. A trapped bead confined to a channel, with one hard wall at the trap centre and another at distance L. Widening the channel takes it from a box ($L \ll \ell$, where the trap does not matter) to a trap ($L \gg \ell$, where the far wall does not matter).

From Gardiner. p_s is Eq. (1.5) [G (5.2.39)]. The eigenfunctions solve [G (5.2.116)]–[G (5.2.117)] for non-integer ν, with the reflecting backward condition [G (5.2.69)] at both walls. The expansion is [G (5.2.84)]–[G (5.2.88)] with $P = p_s Q$ ([G (5.2.71)], [G (5.2.73)]–[G (5.2.74)]).

23.1 Adding a second wall

This is Section 21 with a second wall at L, as Section 5 was Section 3 with a second mirror. The map no longer helps, because the wall at L moves under it. The even Kummer function Eq. (19.4) already satisfies the wall at 0, and the wall at L quantises ν; the script finds the roots numerically.

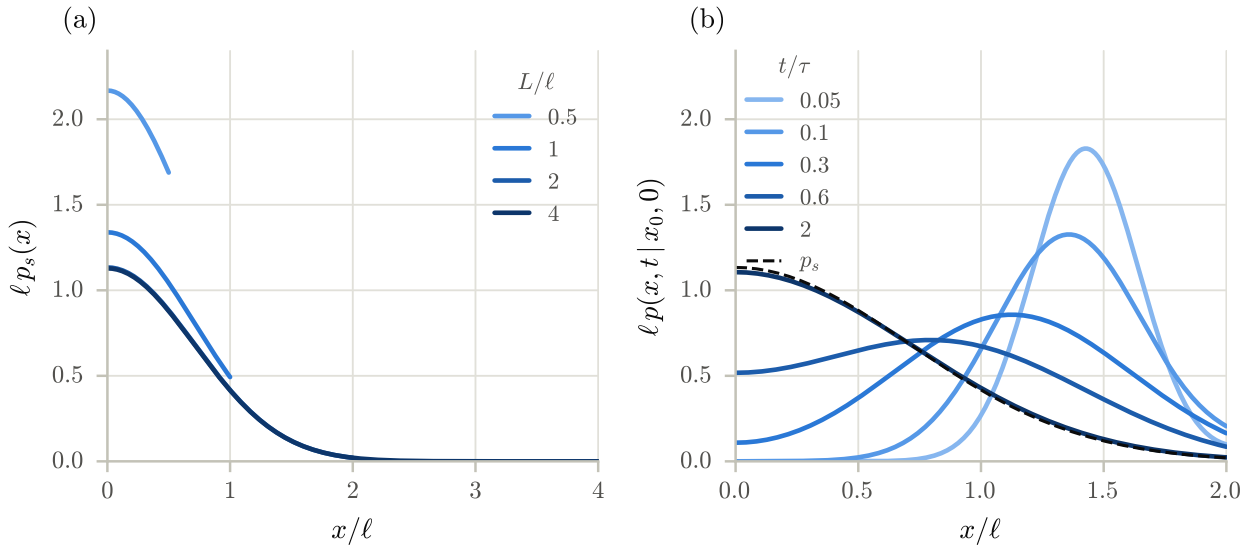


Figure 24: OU on $(0, L)$ with reflecting walls, $x = 0$ at the bottom of the well. (a) Stationary densities, the half-Gaussian truncated at L. (b) Relaxation for $L = 2\ell$ from $x_0 = 1.5\ell$.

23.2 Crossover

- For $L \ll \ell$ the potential barely changes across the interval, and the problem is Section 5: $\lambda_1 \approx \pi^2 D/2L^2$.
- For $L \gg \ell$ the far wall sits where p_s is negligible, and the problem is Section 21: $\lambda_1 \to 2k$.

In terms of the Rayleigh quotient, the slowest function varies over whichever is shorter, L or ℓ (Fig. 29).

24 Ornstein–Uhlenbeck on an interval, both walls absorbing

Setup and key results

$A = -kx$, $B = D$ on $(0, L)$ with $x = 0$ at the bottom of the well, $p = 0$ at both walls.

$$Q_n \propto \xi\, M\big(\tfrac{1-\nu_n}{2}, \tfrac{3}{2}, \xi^2\big), \quad Q_n(L/\ell) = 0, \quad P_n = p_s Q_n \tag{24.1}$$

$$\lambda_n = \nu_n k, \qquad \nu_{n+1}^{\text{refl}} = \nu_n^{\text{abs}} + 1 \tag{24.2}$$

$$\pi_L(x_0) = \frac{\operatorname{erfi}(x_0/\ell)}{\operatorname{erfi}(L/\ell)}, \qquad T(x_0) = \tau\big[\pi_L(x_0)\, G(L/\ell) - G(x_0/\ell)\big] \tag{24.3}$$

$$G(\xi) = \sqrt{\pi} \int_0^{\xi} e^{u^2} \operatorname{erf}(u)\, \mathrm{d}u \tag{24.4}$$

$$\lambda_{\min} \to \frac{\pi^2 D}{2L^2}\ (L \ll \ell), \qquad \lambda_{\min} \to k\ (L \gg \ell); \qquad q_\infty \propto P_0 \propto e^{-\xi^2} Q_0(\xi) \tag{24.5}$$

p_s and the normalisation of Q_n are those of Section 23 [G (5.2.91)].

Physical picture. The same channel with sticky ends, one of them at the trap centre. π_L is the probability of reaching the far end against the restoring force: climbing out of a well on a small scale.

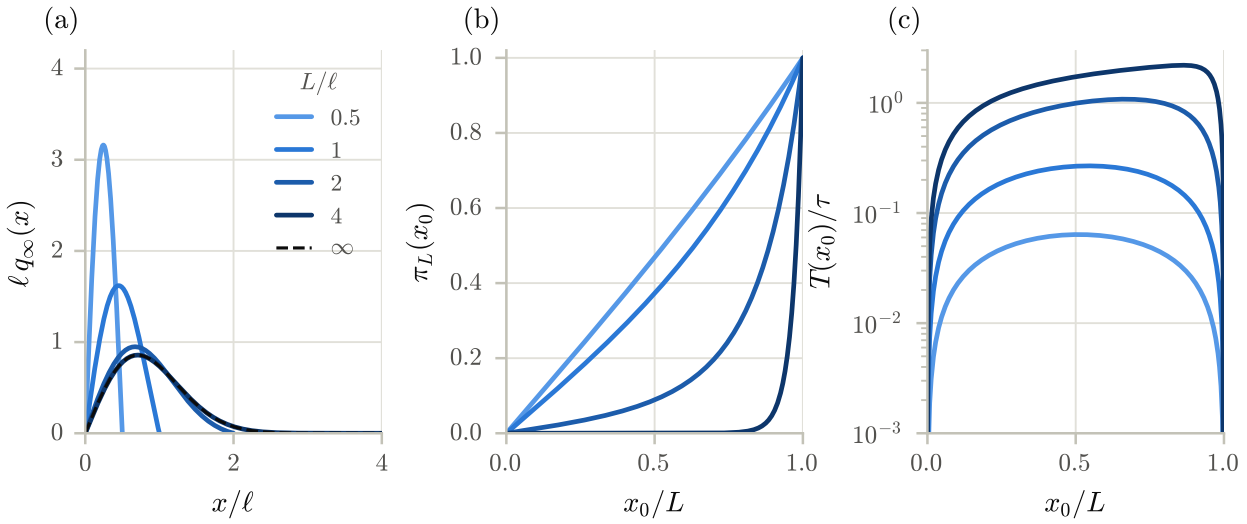


Figure 25: OU on $(0, L)$ with absorbing walls; colours as in (a). (a) Quasi-stationary densities, from the sine of Part I ($L \ll \ell$) to the Rayleigh QSD of Section 22 (dashed). (b) Probability of leaving through the upper wall. (c) Mean exit time.

From Gardiner. The absorbing backward condition [G (5.2.67)] at both walls selects the odd Kummer function and quantises ν. With $1/\psi = e^{x^2/\ell^2}$ the corrected [G (5.2.189)]–[G (5.2.190)] give π_L as a ratio of $\int_0^x e^{y^2/\ell^2}\mathrm{d}y \propto \mathrm{erfi}(x/\ell)$. T is the compact form of [G (5.2.154)]/[G (5.2.158)] from Section 15, $T = \pi_L F(L) - F(x_0)$, with $F = \tau G$.

24.1 Building on the earlier sections

- The odd Kummer function vanishes at the bottom wall, as the odd Hermite functions did in Section 22; the wall at L quantises ν.
- The ladder of Section 19 gives the whole spectrum from Section 23: the derivative of each reflecting eigenfunction is an absorbing one, with ν lower by exactly 1. The script checks this to 10^{-15}.
- Crossover: for $L \ll \ell$, Section 6 (sine QSD, $\lambda = \pi^2 D/2L^2$); for $L \gg \ell$, Section 22 (Rayleigh QSD, $\lambda = k$) (Fig. 25a, Fig. 29).
- Climbing is hard. For $x_0, L \gg \ell$, $\pi_L \approx (L/x_0)\, e^{-(L^2 - x_0^2)/\ell^2}$: to leave through L the particle must climb the potential, and it almost always falls back to the bottom wall (Fig. 25b).

25 Ornstein–Uhlenbeck: reflecting at the bottom, absorbing at L (escape from a well)

Setup and key results

$A = -kx$, $B = D$ on $(0, L)$, $J = 0$ at the bottom $x = 0$, $p = 0$ at $x = L$.

$$Q_n \propto M(-\tfrac{\nu_n}{2}, \tfrac{1}{2}, \xi^2), \quad Q_n(L/\ell) = 0, \quad P_n = p_s Q_n, \quad \lambda_n = \nu_n k \tag{25.1}$$

$$T(x_0) = \tau\,[G(L/\ell) - G(x_0/\ell)] \tag{25.2}$$

$$L \gg \ell: \quad \lambda_0 \simeq \frac{1}{T} \simeq \frac{2}{\sqrt{\pi}\,\tau}\,\frac{L}{\ell}\Big(1 - \frac{\ell^2}{2L^2}\Big)e^{-L^2/\ell^2} \quad \text{(Kramers)} \tag{25.3}$$

$$L \ll \ell: \quad \lambda_0 \to \frac{\pi^2 D}{8L^2} \quad \text{(Section 7)} \tag{25.4}$$

p_s and the normalisation of Q_n are those of Section 23 [G (5.2.91)].

Physical picture. Escape from a harmonic well whose rim is a cliff: a bead escaping an optical trap of finite extent, or a harmonic bond that breaks when stretched beyond L, the simplest model of thermally activated bond rupture. The Arrhenius factor $e^{L^2/\ell^2} = e^{\mathcal{K}L^2/2k_BT}$ is the Boltzmann factor of the rim.

From Gardiner. [G (5.2.160)] ("a reflecting, b absorbing") with $\psi = e^{-x^2/\ell^2}$ gives T exactly: $T = \frac{2}{D}\int_{x_0}^{L}\frac{\mathrm{d}y}{\psi(y)}\int_0^y \psi(z)\,\mathrm{d}z = \tau[G(L/\ell) - G(x_0/\ell)]$. [G §5.2.7c] is the asymptotic analysis of the same integrals, [G (5.2.165)]–[G (5.2.174)]. The inner integral is Gaussian about the minimum, as here. Gardiner's barrier is a smooth maximum, which makes the outer integral Gaussian too and gives the prefactor of [G (5.2.174)]. Here the barrier top is the absorbing wall itself, so the outer integral is cut off at L and contributes $e^{L^2/\ell^2}/(2L/\ell)$ instead. That is the origin of the L/ℓ in the rate.

25.1 Escape

- This is the mixed interval of Part I in a harmonic well, and the harmonic version of Section 16. Both have Arrhenius factor $e^{2\Delta U/D}$: e^{2L/ℓ_v} for the linear potential, with $\ell_v = D/|v|$ the length ℓ of Part II, and e^{L^2/ℓ^2} here.
- Unfolding, as in Section 7: a reflecting wall at the bottom of a symmetric well is the full well $(-L, L)$ with absorbing walls at both ends. The half-well escape rate is therefore the rate of a full well with two exits, twice that of a full well with one.
- For $L \gg \ell$, T is the same from every start that is not within about ℓ of the exit (Fig. 26b). The QSD is the half-Gaussian p_s of Section 21, cut off near L (Fig. 26a): $Q_0 \approx 1$ except within about ℓ of the exit, so $P_0 = p_s Q_0 \approx p_s$. The gap to the next level, about $2k$, is huge compared with λ_0: the particle equilibrates in the well long before it escapes (Fig. 4).
- The Kramers formula with its first correction matches λ_0 to 0.1 % at $L = 5\ell$, and $\lambda_0 T(0) = 1.0005$ already at $L = 3\ell$ (Fig. 26c).

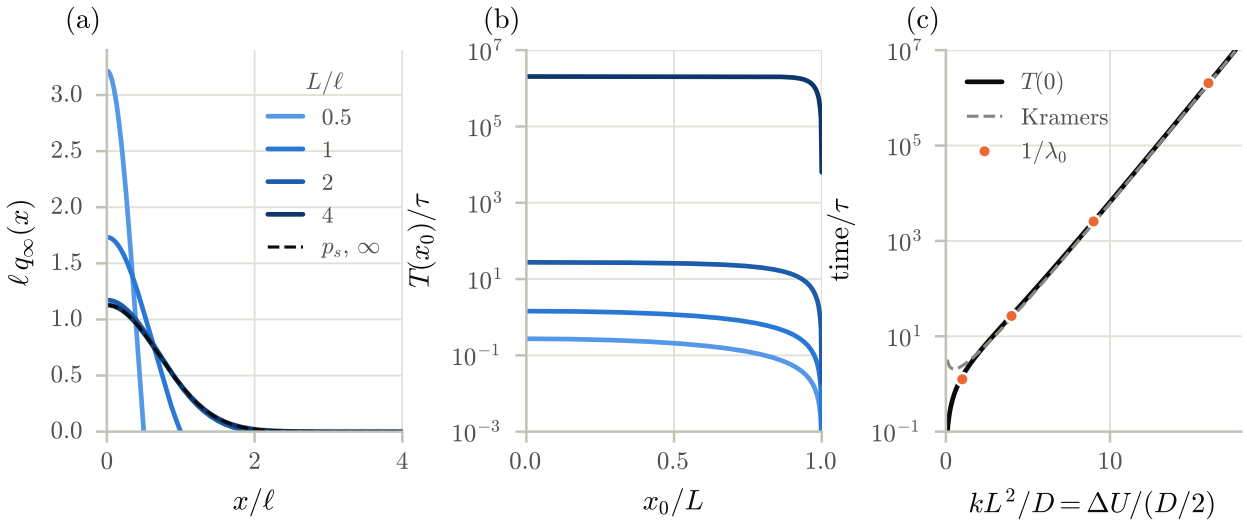


Figure 26: Escape from a harmonic well: reflecting at the bottom, absorbing at L. (a) Quasi-stationary densities; dashed, the half-Gaussian. (b) Mean exit time (log scale); colours as in (a). (c) Arrhenius plot: $T(0)$ and $1/\lambda_0$ against the barrier $\Delta U/(D/2) = L^2/\ell^2$, with the leading Kramers asymptote.

26 Ornstein–Uhlenbeck on the circle

$A = -kx$ is not a function on a circle, so OU cannot simply be put on one. The nearest thing is to repeat it: measure x from the bottom of the well, take $A = -kx$ for $|x| < L/2$, and repeat with period L. This is a sawtooth. At the antipode $x = \pm L/2$ the drift jumps from $-kL/2$ to $+kL/2$, pointing away on both sides, and the potential $U = \frac{1}{2}kx^2$ has a cusp. That point is special, but not because of the topology: A is discontinuous there, and [G (5.2.15)]–[G (5.2.16)] hold there as they would at a discontinuity anywhere. Cut at the antipode, this is OU on $(-L/2, L/2)$ with Gardiner's periodic conditions but *without* [G (5.2.19)]; cut anywhere else, it is a drift with an interior jump, and [G (5.2.19)] holds. It is one problem either way.

Setup and key results

$A = -kx$ for $|x| < L/2$, repeated with period L (sawtooth); $B = D$; $\ell = \sqrt{D/k}$, $\tau = 1/k$ as in Eq. (19.1). The cusp at $x = \pm L/2$ is the only special point.

$$p_s(x) = \frac{e^{-x^2/\ell^2}}{\ell\sqrt{\pi}\,\mathrm{erf}(L/2\ell)} \quad (|x| \le L/2), \qquad J_s = 0 \;\; (\Phi = 0) \tag{26.1}$$

$$p(x,t\,|\,x_0,0) = \tfrac{1}{2}\big[p_{\mathrm{R}}(|x|,t\,|\,|x_0|) + \mathrm{sgn}(xx_0)\,p_{\mathrm{A}}(|x|,t\,|\,|x_0|)\big] \tag{26.2}$$

$$\text{spectrum} = \{\nu_n^{\mathrm{R}} k\} \cup \{\nu_n^{\mathrm{A}} k\}, \qquad \nu_{n+1}^{\mathrm{R}} = \nu_n^{\mathrm{A}} + 1 \text{ exactly} \tag{26.3}$$

$$Q_n = Q_n^{\mathrm{R}}(|x|) \text{ or } \mathrm{sgn}(x)\,Q_n^{\mathrm{A}}(|x|), \qquad P_n = p_s Q_n = \tfrac{1}{2}P_n^{\mathrm{R}}(|x|) \text{ or } \tfrac{1}{2}\mathrm{sgn}(x)\,P_n^{\mathrm{A}}(|x|) \tag{26.4}$$

$$D_{\mathrm{eff}} = \frac{D\,(L/\ell)^2}{\pi\,\mathrm{erf}(L/2\ell)\,\mathrm{erfi}(L/2\ell)} \;\to\; \begin{cases} D, & L \ll \ell \\ \tfrac{1}{2}L^2\lambda_0^{\mathrm{mix}}, & L \gg \ell \end{cases} \tag{26.5}$$

Here p_{R}, p_{A}, ν_n^{R}, ν_n^{A} and $P_n^{\mathrm{R,A}}$, $Q_n^{\mathrm{R,A}}$ are the propagators, eigenvalues and eigenfunctions of Sections 23 and 24 on $(0, L/2)$, and λ_0^{mix} is the escape rate of Section 25 with its absorbing wall at the cusp, $L \to L/2$.

Physical picture. A periodic row of identical parabolic traps that touch at sharp ridges, such as a one-dimensional lattice of optical traps with no gaps between them. The circle is one lattice cell, and winding is hopping from trap to trap.

From Gardiner. The cusp is a discontinuity in the sense of [G §5.2.1c]: p and J are continuous there [G (5.2.15)]–[G (5.2.16)], while "the derivatives of p are not necessarily continuous", and here $\partial_x p$ does jump. Since $\Phi = 0$, p_s is the potential solution [G (5.2.39)]. The two sectors are the Kummer problems of Sections 23 and 24, that is, [G (5.2.116)]–[G (5.2.117)] with walls. D_{eff} is [G (5.2.160)] applied on the cover, as shown below.

26.1 Folding into two intervals

A is odd under $x \to -x$, which fixes the bottom and the cusp. By Section 1.9 the even sector is OU with reflecting walls at the bottom and at $L/2$, which is Section 23 with $L \to L/2$, and the odd sector is Section 24 with $L \to L/2$. Gluing the two gives the propagator in the box. The script checks that the glued density solves the equation, stays normalised, and has continuous J at the cusp; its slope jumps there (Fig. 27a). The eigenfunctions glue the same way: the circle's p_s is half the interval's, so the interval's Q_n, extended evenly or oddly, keep $\int p_s Q_n^2 = 1$ over the whole circle, and the P_n are halved. At the cusp Q_n and Q_n' are continuous (the wall conditions $Q' = 0$ and $Q = 0$ hold from both sides), and only Q_n'' jumps, with A. P_n inherits the kink of p_s, while P_n and $J(P_n)$ stay continuous, as [G (5.2.15)]–[G (5.2.16)] require.

The ladder of Section 19 then fixes the whole spectrum. Each reflecting level is an absorbing one plus exactly k, so each doublet of the free circle (Section 8), where cos and sin had equal rates, is split by exactly k, at every L (Fig. 27b):

- for $L \ll \ell$ both members sit near the free doublet $2\pi^2 n^2 D/L^2$;
- for $L \gg \ell$ the pairs tend to $(1,2)k$, $(3,4)k$, …: the odd and even Hermite levels of Sections 21 and 22, which together make the OU ladder nk of Section 20, each level once.

In the symmetrised form of Section 1.5, the jump of A by kL at the cusp puts a repulsive $\frac{1}{2}kL\,\delta$ into V on top of the harmonic well (Fig. 3c). Unlike the smooth barrier top of Section 27, it supports no levels of its own, which is why the levels here are single.

26.2 Winding over the cusp

[G (5.2.160)] on the cover gives D_{eff} for any potential symmetric about the start. The first full winding from the bottom is the exit from $(-L, L)$. By symmetry this is the interval $(0, L)$, reflecting at 0 and absorbing at L, so with $\psi = e^{-2U/D}$

$$T_{\text{w}} = \frac{2}{D}\int_0^L \mathrm{d}y\, e^{2U(y)/D} \int_0^y \mathrm{d}z\, e^{-2U(z)/D} = \frac{L^2}{D}\,\langle e^{2U/D}\rangle\langle e^{-2U/D}\rangle, \tag{26.6}$$

where $\langle\cdot\rangle$ averages over one period. The second form uses $U(L-y) = U(y)$: substituting $y \to L - y$ turns the inner integral into the rest of the period, so the double integral is half the product of the single ones. With $D_{\text{eff}} = L^2/T_{\text{w}}$ (Section 1.9) this is the Lifson–Jackson formula [4],

$$D_{\text{eff}} = \frac{D}{\langle e^{2U/D}\rangle\langle e^{-2U/D}\rangle}. \tag{26.7}$$

For the sawtooth the two averages are $\ell\sqrt{\pi}\,\mathrm{erfi}(L/2\ell)/L$ and $\ell\sqrt{\pi}\,\mathrm{erf}(L/2\ell)/L$, which gives the box.

- For $L \ll \ell$ the well hardly matters and $D_{\text{eff}} \to D$ (Section 8).
- For $L \gg \ell$ the particle hops between copies of the well. To get over, it must reach the cusp, which happens at the rate λ_0^{mix} of Section 25 with the absorbing wall at $L/2$; from the cusp it falls either way with probability $\frac{1}{2}$. So it hops to each neighbour at rate $\lambda_0^{\text{mix}}/4$, and a random walk with steps $\pm L$ at total rate $\lambda_0^{\text{mix}}/2$ has $D_{\text{eff}} = \frac{1}{2}L^2\lambda_0^{\text{mix}}$. At $L = 8\ell$ this matches Eq. (26.7) to 2×10^{-6} (Fig. 27c).
- T_{w} is the same from every starting point, not just the bottom; the script checks this from [G (5.2.158)]. For winding, no point of the circle is special, not even the cusp.

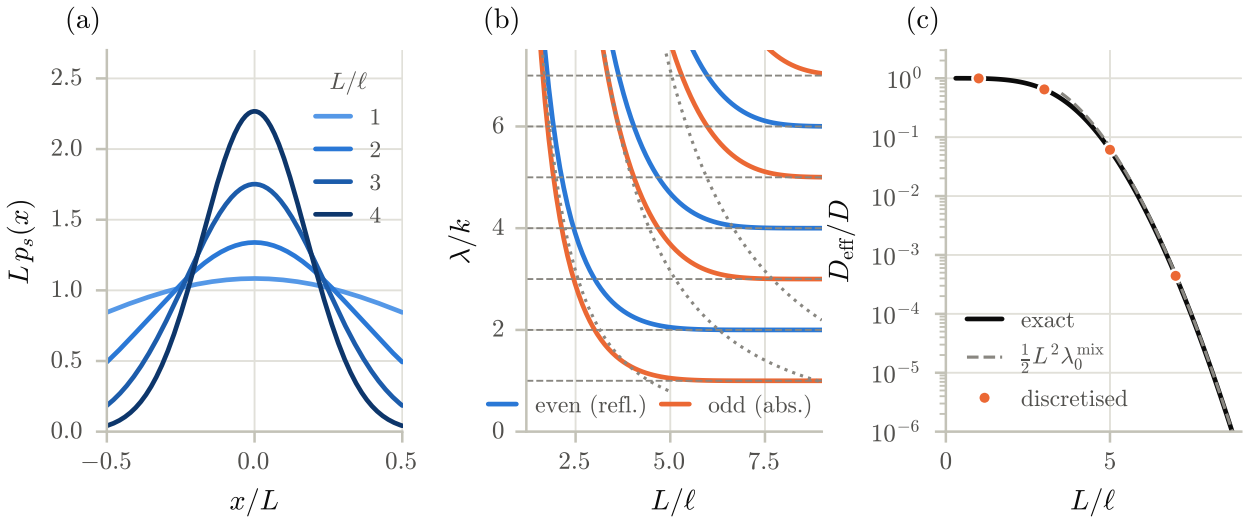


Figure 27: OU on the circle (sawtooth drift), x measured from the bottom of the well. (a) Stationary densities: truncated Gaussians, repeated, with a kink at the cusp $x = \pm L/2$. (b) Spectrum: even sector (blue; reflecting walls at the bottom and the cusp) and odd sector (orange; absorbing). Each pair is split by exactly k. Dotted, the free doublets $2\pi^2 n^2 D/L^2$; dashed, the OU ladder. (c) Effective diffusion of the lift: Eq. (26.7) (black), the hopping limit $\frac{1}{2}L^2\lambda_0^{\mathrm{mix}}$ with the leading Kramers rate of Section 25 (dashed), and an independent discretised computation (dots).

27 A smooth well on the circle

The cusp disappears if the harmonic well is replaced by the smooth periodic potential with the same curvature at the bottom, $U = -kR^2\cos(x/R)$ with $R = L/2\pi$. Its drift $A = -kR\sin(x/R)$ is OU near the bottom, has a smooth barrier top at the antipode, and has no special point at all. Adding a constant drift v tilts the potential and makes $\Phi = 2vL/D \neq 0$. That gives the one example in these notes of a NESS whose density is not uniform. The model is the overdamped pendulum, or Adler's equation, with noise; Risken [5] (ch. 11) treats it in detail.

Setup and key results

$A = v - kR\sin(x/R)$, $B = D$, $R = L/2\pi$. Two numbers set everything: the depth $\kappa = 2R^2/\ell^2 = (L/\ell)^2/2\pi^2$, so that the barrier is $\Delta U/(D/2) = 2\kappa$, and the tilt v/kR.

$$v = 0: \quad p_s = \frac{e^{\kappa\cos(x/R)}}{L\,I_0(\kappa)} \text{ (von Mises)}, \qquad J_s = 0 \tag{27.1}$$

$$\text{spectrum: exact doublets;} \qquad \lambda \to \frac{m^2 k}{\kappa}\ (\kappa \ll 1), \qquad \lambda \to nk\ (\kappa \gg 1) \tag{27.2}$$

$$P_\lambda,\ Q_\lambda: \text{right, left eigenvectors of Eq. (27.7)}; \quad v = 0: \ P_\lambda = p_s Q_\lambda \tag{27.3}$$

$$D_{\mathrm{eff}} = \frac{D}{I_0(\kappa)^2} \simeq \frac{kL^2}{\pi}\,e^{-2\kappa} \quad \text{(Kramers, [G (5.2.174)])} \tag{27.4}$$

$$v \neq 0: \quad p_s,\ J_s \text{ from Eq. (1.19)}, \qquad v_{\mathrm{w}} = LJ_s \simeq \frac{D_{\mathrm{eff}}}{D}\,v \quad (v \to 0) \tag{27.5}$$

$$\kappa \to \infty: \quad v_{\mathrm{w}} \to 0 \ \ (|v| < kR), \qquad v_{\mathrm{w}} \to \mathrm{sgn}(v)\sqrt{v^2 - k^2R^2} \ \ (|v| > kR) \tag{27.6}$$

Physical picture. $\mathcal{U} = -\mathcal{U}_0 \cos(x/R) - Fx$, with $\kappa = \mathcal{U}_0/k_BT$.

- Without tilt: a planar dipole in a field, whose orientation follows the von Mises (Langevin–Debye) distribution, or an overdamped pendulum.
- With tilt, the tilted washboard: a Josephson junction in the overdamped (RSJ) limit, where x/R is the phase difference, the tilt is the bias current, and the winding velocity v_{w} is the dc voltage. Figure 28c is then the noisy current–voltage curve of Ambegaokar and Halperin [8].
- Colloids driven around a corrugated ring trap realise it directly, including the giant diffusion near threshold [9] and the breakdown of the equilibrium Einstein relation [10].

From Gardiner. With $v = 0$, $\Phi = 0$ and [G (5.2.39)] gives the von Mises density. The eigenfunctions obey [G §5.2.5] with periodic Q, but the equation is no longer Hermite's or Kummer's; a Fourier series turns it into a three-term recurrence instead. D_{eff} is [G (5.2.160)] on the cover, exactly as in Section 26, and its large-κ limit is Gardiner's Kramers formula [G (5.2.174)]. With $v \neq 0$, p_s and J_s are [G (5.2.47)] and the corrected [G (5.2.46)], or Eq. (1.19).

27.1 No special point

A is smooth and, without tilt, odd under $x \to -x$. So the circle folds into the cosine well on $(0, L/2)$, with walls at the bottom and at the barrier top, reflecting (even sector) or absorbing (odd sector). Now $A' = -k\cos(x/R)$ is not constant, and the ladder of Section 19 does not apply. In Fourier space, with $p = \sum_m c_m e^{imx/R}$,

$$\dot{c}_m = -\Big(\frac{Dm^2}{2R^2} + \frac{ivm}{R}\Big)c_m + \frac{km}{2}\,(c_{m-1} - c_{m+1}), \tag{27.7}$$

a three-term recurrence whose truncations converge like Bessel functions. The script solves it, and checks it against an independent discretisation. Its eigenvectors $c^{(\lambda)}$ are Gardiner's $P_\lambda = \sum_m c_m^{(\lambda)} e^{imx/R}$. Writing $Q_\lambda = \sum_m b_m^{(\lambda)} e^{-imx/R}$ turns the pairing [G (5.2.79)] into $2\pi R \sum_m b_m^{(\lambda)} c_m^{(\lambda')}$, so the $b^{(\lambda)}$ are the left eigenvectors of the same matrix, normalised against the right ones. Without tilt $P_\lambda = p_s Q_\lambda$, and each doublet has one even Q_λ (a cosine series, the reflecting sector of the fold) and one odd (a sine series, the absorbing sector). With tilt $J_s \neq 0$ and $P_\lambda \neq p_s Q_\lambda$: the script checks that P_λ/p_s then fails the backward equation, while the left eigenvectors satisfy it.

27.2 Exact doublets

Every nonzero level of the cosine well is exactly doubly degenerate, for every κ (Fig. 28b), so the reflecting and absorbing intervals of the fold have identical nonzero spectra. The reason is visible in Eq. (27.7). A mode with $\lambda \neq 0$ must have $c_0 = 0$, because c_0 is the total probability, which does not decay. With $c_0 = 0$ the equations for $m > 0$ and for $m < 0$ no longer talk to each other, and the two chains are mirror images. (A $\cos(2x/R)$ term in U would couple $m = \pm 1$ directly and split the doublets.)

In a deep well the two members of a doublet have different physics. For $\kappa \gg 1$ the levels tend to nk, $n \geq 1$, each twice. One copy is the OU ladder at the bottom of the well (Section 20). The other lives at the barrier top, where a density relaxes by sliding off the hilltop, also at rates $k, 2k, \ldots$, because the curvature there is $-k$. In the symmetrised form of Section 1.5 these are the levels of the two wells of V, at the bottom and at the top of U (Fig. 3b). Compare Section 26: a cusp sheds probability at once, has no ladder of its own, and leaves the levels single. For $\kappa \ll 1$ the levels tend to the free doublets $m^2k/\kappa = Dm^2/2R^2$ of Section 8.

27.3 Winding: Kramers with a smooth barrier

Equation (26.7) with $\langle e^{\pm\kappa\cos(x/R)}\rangle = I_0(\kappa)$ gives $D_{\text{eff}} = D/I_0(\kappa)^2$ exactly. For a deep well this is hopping over a smooth barrier, the case Gardiner treats in [G §5.2.7c]. With his D replaced by our $D/2$, and $U'' = k$ at the bottom and $-k$ at the top, [G (5.2.174)] gives $2\pi\tau e^{2\kappa}$ for the time to cross in one direction. On the circle the barrier can be crossed either way, so $T_{\text{w}} \simeq \pi\tau e^{2\kappa}$ and $D_{\text{eff}} = L^2/T_{\text{w}} \simeq (kL^2/\pi)e^{-2\kappa}$. This is the leading term of $D/I_0(\kappa)^2$; the next is a factor $1 - 1/4\kappa$. Unlike Section 25, where the barrier top is a wall, and Section 26, where it is a cusp, Gardiner's prefactor applies here unchanged.

27.4 Tilt: a current, depinning, giant diffusion

With $v \neq 0$, $\Phi = 2vL/D \neq 0$ and a current circulates (Fig. 28a).

- **Small tilt.** Expanding Eq. (1.19) to first order in v gives $v_{\text{w}} = v/\langle e^{2U/D}\rangle\langle e^{-2U/D}\rangle = v\,D_{\text{eff}}/D$. The mobility of the winding equals its diffusion constant over D: the Einstein relation, which here combines the topology (the current) with Lifson–Jackson (the diffusion).
- **Depinning.** Without noise, $\dot{x} = v - kR\sin(x/R)$ has a fixed point while $|v| < kR$, and the rotor is locked. For $|v| > kR$ it runs, with period $\oint \mathrm{d}x/(v - kR\sin(x/R)) = L/\sqrt{v^2 - k^2R^2}$. Noise smooths the transition: below threshold the rotor slips by thermally activated hops, and above it the flow is nearly deterministic (Fig. 28c).
- **Giant diffusion.** Near threshold D_{eff} exceeds the bare D, by a factor that grows as the noise decreases [19] (Fig. 28d). There, each revolution is a race between lingering near the almost-fixed point and running, so the time per revolution is broadly distributed, and that spreads the winding. The script computes $D_{\text{eff}} = \Lambda''(0)$ from Eq. (27.7) with $\partial_x \to \partial_x - s$, and checks it against a discretisation and a simulation.
- The NESS density is not the Boltzmann factor of any potential. For large tilt it approaches the deterministic dwell-time density $p_s \propto 1/|v - kR\sin(x/R)|$, highest where the rotor is slowest.

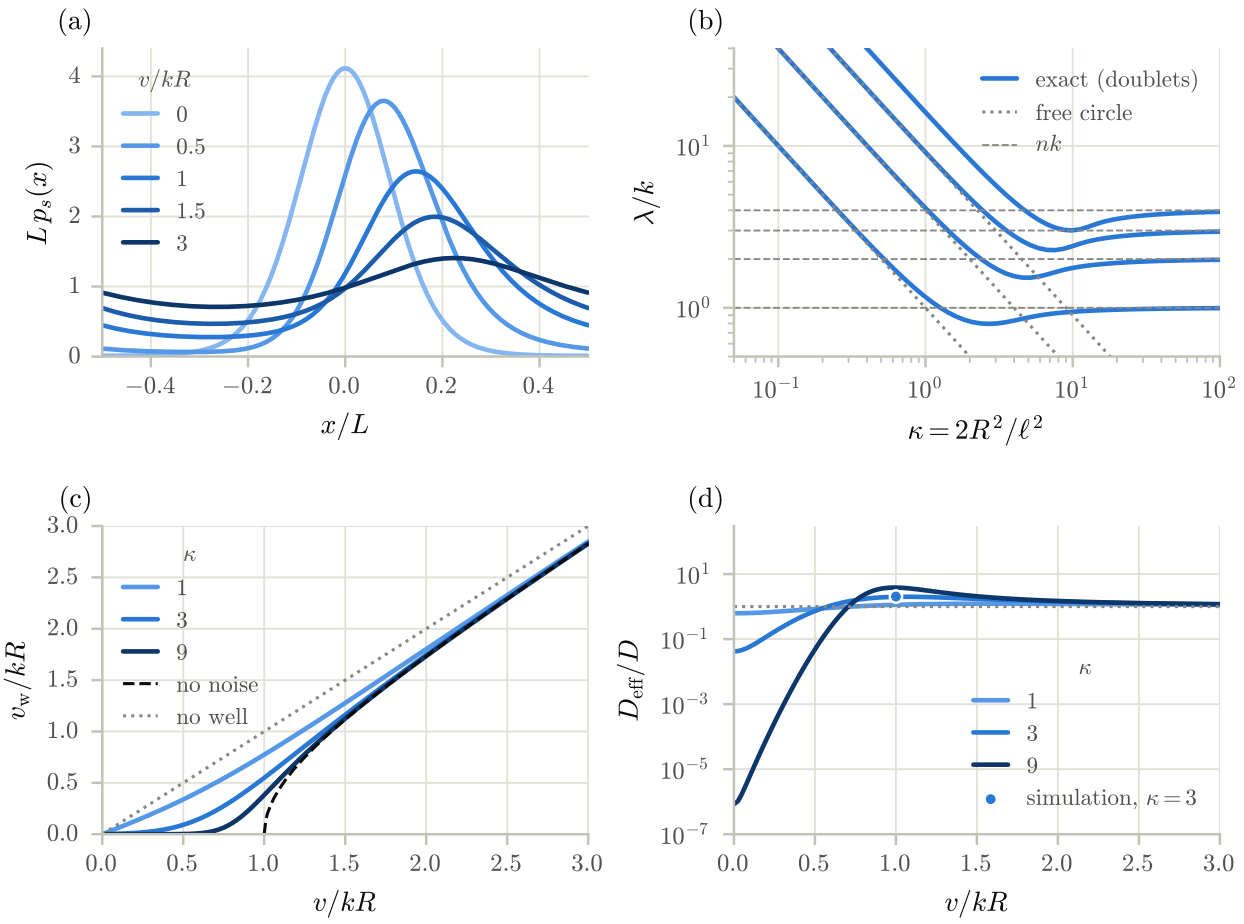


Figure 28: The cosine well on the circle, x measured from the bottom. (a) Stationary densities at $\kappa = 3$ for several tilts: von Mises at $v = 0$, a current-carrying NESS otherwise. (b) Spectrum without tilt: every level is a doublet. Dotted, the free circle m^2k/κ; dashed, the OU ladder nk. (c) Winding velocity against tilt for three depths; dashed, the noiseless rotor; dotted, no well. (d) Effective diffusion of the lift; dotted, the bare D. The dot is a simulation of 4000 paths over $200\,\tau$.

28 Summary of Ornstein–Uhlenbeck

Walls (bottom of well at 0)	Slowest rate	Long times
none	gap k	$p_s = e^{-x^2/\ell^2}/\ell\sqrt{\pi}$
reflecting at 0	gap $2k$	half-Gaussian
absorbing at 0	bottom k, gap $2k$	QSDs for $\lambda \in (0, k]$; Rayleigh from point starts
reflecting 0, L	$\frac{\pi^2 D}{2L^2} \to 2k$	truncated half-Gaussian
absorbing 0, L	$\frac{\pi^2 D}{2L^2} \to k$	QSD: sine $\to$ Rayleigh
refl. 0, abs. L	$\frac{\pi^2 D}{8L^2} \to$ Kramers	$T \simeq \frac{\sqrt{\pi}\,\tau\ell}{2L} e^{L^2/\ell^2}$
circle, sawtooth	$\frac{2\pi^2 D}{L^2} \to k$; pairs split by k	cusp; $D_{\text{eff}} \to \frac{1}{2}L^2\lambda_0^{\text{mix}}$
circle, cosine well	$\frac{k}{\kappa} \to k$; exact doublets	von Mises, $D_{\text{eff}} = D/I_0(\kappa)^2$; tilt: NESS

Table 7: Ornstein–Uhlenbeck in every geometry; $\ell = \sqrt{D/k}$, $\tau = 1/k$. Arrows go from $L \ll \ell$ to $L \gg \ell$.

- With walls at the bottom of the well, Gardiner's substitution $y = xe^{kt}$ turns every Part I result into an OU result. Free spreading freezes into stationary and quasi-stationary states, and algebraic decay becomes exponential.
- With an absorbing wall at the bottom there is a continuum of QSDs, $\lambda \in (0, k]$. A start with a finite mean reaches the Rayleigh one; a power-law tail $x^{-1-\nu}$, $\nu < 1$, reaches q_ν.
- Walls at the bottom select from the Hermite ladder: even modes for reflecting, odd for absorbing. The nonzero reflecting levels are the absorbing ones shifted up by exactly k.
- On an interval, the length to compare with is ℓ. For $L \ll \ell$ the results are those of Part I; for $L \gg \ell$ they are the half-line results, or Kramers escape (Fig. 29).
- Across the three parts, escape against a force costs $e^{2\Delta U/D}$: nothing for a flat potential ($T \sim L^2/D$), e^{2L/ℓ_v} for a linear one, and e^{L^2/ℓ^2} for a harmonic one.
- OU fits on a circle only with a special point, the cusp where the sawtooth drift jumps. Folding and the ladder then solve it with no new work, and every free doublet is split by exactly k. The smooth cosine well has no special point, and its doublets stay exactly degenerate.
- The prefactor of the escape rate depends on the shape of the barrier top: a wall (Section 25); a cusp, crossed with probability $\frac{1}{2}$ once reached (Section 26); a smooth maximum, Gardiner's [G (5.2.174)] (Section 27).

Part IV
More exactly solvable potentials

Parts I–III are the free particle, the linear potential and the harmonic potential. One mechanism explains why they are solvable, and it also identifies the other solvable potentials: the factorisation of the symmetrised operator (Section 29). Three worked sections follow, each chosen to connect back to Parts I–III: the logarithmic potential, which is radial motion in d dimensions (Section 30); a soft V-shaped potential (Section 31); and a well that builds its own walls (Section 32). Throughout, $A = -U'$ and $B = D$, so $p_s \propto e^{-2U/D}$ and the symmetrised potential Eq. (1.11) is $V = U'^2/2D - U''/2$.

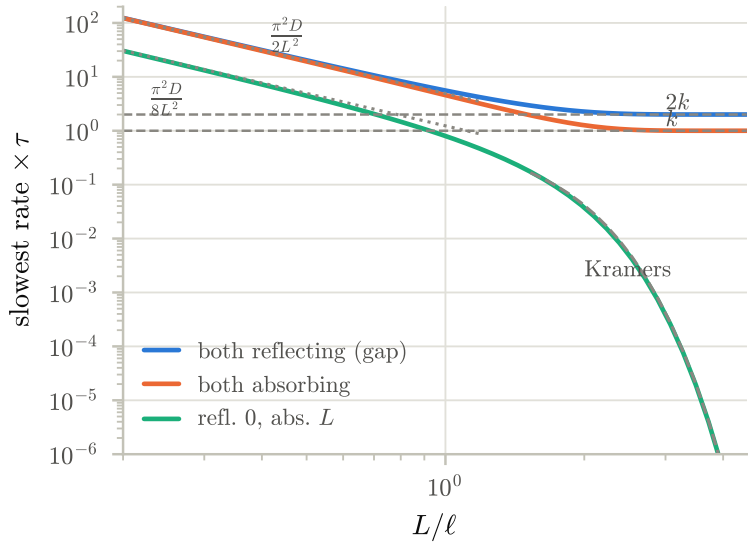


Figure 29: Slowest rates of the three OU intervals against L/ℓ, with their Part I asymptotes (dotted), half-line limits (dashed horizontal) and the Kramers rate (dashed).

29 Factorisation, partners and shape invariance

From Gardiner. Not in the book. The section builds on the symmetrised form of Section 1.5, that is, on Gardiner's $p = p_s q$ [G (5.2.71)], and on the ladder of Section 19.

Factorisation. With $B = D$ the Schrödinger operator Eq. (1.11) factorises, as multiplying out shows:

$$H = \mathcal{A}^\dagger \mathcal{A}, \qquad \mathcal{A} = \sqrt{\tfrac{D}{2}}\Big(\partial_x - \frac{A}{D}\Big), \qquad \mathcal{A}^\dagger = \sqrt{\tfrac{D}{2}}\Big(-\partial_x - \frac{A}{D}\Big). \tag{29.1}$$

Since $\partial_x \sqrt{p_s} = (A/D)\sqrt{p_s}$, $\mathcal{A}\sqrt{p_s} = 0$: the ground state at $\lambda = 0$. And $\int \psi H \psi = \int (\mathcal{A}\psi)^2 \geq 0$, which is the Dirichlet form Eq. (1.10) again.

The partner is the inverted potential. Reversing the order gives

$$\tilde{H} = \mathcal{A}\mathcal{A}^\dagger = -\tfrac{1}{2} D\, \partial_x^2 + \tilde{V}, \qquad \tilde{V} = \frac{A^2}{2D} - \frac{A'}{2}, \tag{29.2}$$

the Schrödinger operator of the drift $-A$, that is, of the inverted potential $-U$. The two are intertwined, $\mathcal{A}H = \tilde{H}\mathcal{A}$. So $\mathcal{A}$ carries every eigenfunction of H with $\lambda \neq 0$ to one of $\tilde{H}$ with the same λ, and $\mathcal{A}^\dagger$ carries it back. In Fokker–Planck terms, with $\psi = \sqrt{p_s}\, Q$, one has $\mathcal{A}\psi = \sqrt{D/2}\,\sqrt{p_s}\, Q'$: *the derivative of a backward eigenfunction is, up to the weight, an eigenfunction of the inverted problem, with the same rate.* A potential and its inverse therefore share every nonzero rate, as long as the wall conditions match.

- This is the ladder of Section 19 in general form. For OU the inverted potential gives $\tilde{V} = V + k$, so Q' is again an OU eigenfunction, with the rate lowered by k.
- It is also the origin of the exact doublets of Section 27. There $-U$ is U rotated by half a turn S, so $S\mathcal{A}S = -\mathcal{A}^\dagger$, and $S\mathcal{A}$ commutes with H while squaring to $-H$. On a real eigenfunction with $\lambda > 0$ it cannot act as a number, since that number would square to $-\lambda$. So every nonzero level has at least two eigenfunctions.

Shape invariance. Sometimes the partner belongs to the same family: $\tilde{V}(x;s) = V(x;s') + R(s)$, with a shifted parameter s' and a constant R. Then the ground state of the partner, at $R(s)$, is the first excited state of the original problem, and repeating the step gives the whole spectrum by algebra:

$$\lambda_n(s) = R(s_0) + R(s_1) + \cdots + R(s_{n-1}), \qquad s_0 = s, \quad s_{j+1} = s'_j. \tag{29.3}$$

The eigenfunctions follow by repeated $\mathcal{A}^\dagger$, which is where the Rodrigues formulas of the classical polynomials come from. The ladder stops when the partner's ground state stops being normalisable; that is where a finite set of discrete levels ends and a continuum begins. The script checks each of these partner identities:

- OU: $s' = s$, $R = k$, so $\lambda_n = nk$ (Part III).
- Harmonic plus logarithmic: $c \to c + D$, which raises the dimension d by two, and $R = 2k$, so $\lambda_n = 2nk$ (Section 30).
- $a \ln\cosh(x/w)$: $s \to s-1$, $R = (D/2w^2)(2s-1)$, so $\lambda_n = (D/2w^2)\, n(2s-n)$ for $n < s$ (Section 31).
- $-a \ln\cos(x/w)$: $s \to s+1$, $R = (D/2w^2)(2s+1)$, so $\lambda_n = (D/2w^2)\, n(n+2s)$ (Section 32).
- Free motion and constant drift: $\tilde{V} = V$ and $R = 0$. There is no ladder, and the spectrum is a continuum.

The solvable potentials. Table 8 lists the shape-invariant potentials of supersymmetric quantum mechanics [14], rewritten as Fokker–Planck potentials (Fig. 30). Two of them are not worked out in these notes, but their spectra are checked numerically.

- The linear-plus-logarithmic potential is a hydrogen atom in disguise: infinitely many discrete rates accumulate at the continuum edge $v^2/2D$, and p_s is a gamma distribution. Physically it is the distance from the origin of a particle in d dimensions pulled toward it by a force of constant magnitude; the logarithm is the entropic term of Section 30.
- The Morse potential is a soft exponential wall with gravity, the soft version of Section 12. Physically it is a colloid hovering above a like-charged surface: screened electrostatic repulsion holds it up and gravity pulls it down. Total internal reflection microscopy measures exactly this potential [12].

The cosine well on a circle (Section 27) is not shape-invariant: its symmetrised potential has two cosine harmonics, which is why it needs a recurrence instead.

$U(x)$	Domain	p_s	λ_n	Section
0	line	—	continuum from 0	2
vx	$x > 0$, reflecting	$e^{-2vx/D}$	0; continuum from $v^2/2D$	12
$\frac{1}{2}kx^2$	line	$e^{-kx^2/D}$	nk (Hermite)	20
$\frac{1}{2}kx^2 - c\ln x$	$x > 0$	$x^{2c/D}e^{-kx^2/D}$	$2nk$ (Laguerre)	30
$vx - c\ln x$	$x > 0$	$x^{2c/D}e^{-2vx/D}$	$\frac{v^2}{2D}\left[1 - \frac{\sigma^2}{(n+\sigma)^2}\right]$, $\sigma = \frac{c}{D}$ (Laguerre)	—
$vx + U_0 e^{-x/w}$	line	$e^{-2vx/D - (2U_0/D)e^{-x/w}}$	$\frac{D}{2w^2}n(2s-n)$, $n < s = \frac{vw}{D}$ (Laguerre)	—
$a\ln\cosh(x/w)$	line	$\mathrm{sech}^{2s}(x/w)$	$\frac{D}{2w^2}n(2s-n)$, $n < s = \frac{a}{D}$ (Jacobi)	31
$-a\ln\cos(x/w)$	$\lvert x\rvert < \frac{\pi w}{2}$	$\cos^{2s}(x/w)$	$\frac{D}{2w^2}n(n+2s)$ (Gegenbauer)	32

Table 8: Fokker–Planck potentials with exactly known spectra ($A = -U'$, $B = D$). Where a continuum exists it starts at $v^2/2D$, the value of V at infinity.

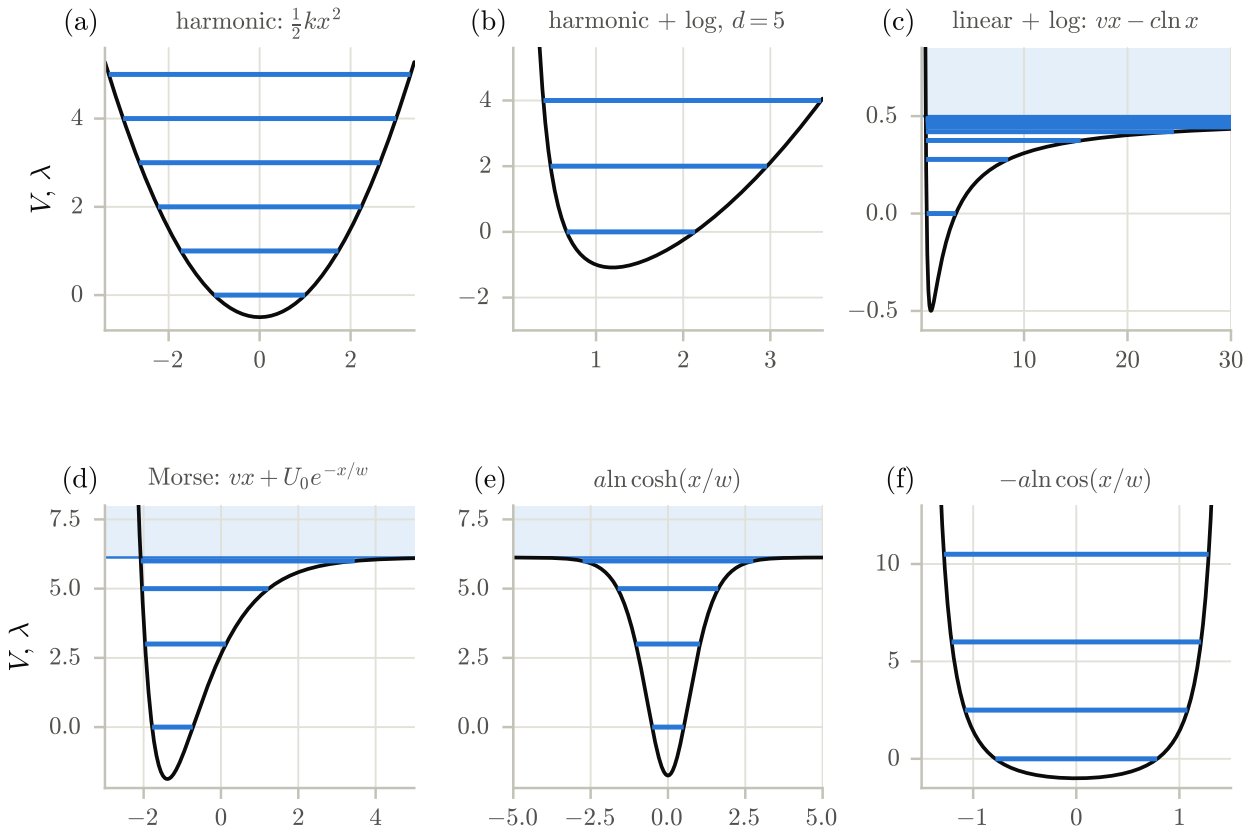


Figure 30: The solvable potentials of Table 8, in the symmetrised picture: $V = U'^2/2D - U''/2$ (black), the discrete rates drawn where $V < \lambda$ (blue), and the continuum (shaded) where there is one. $D = 1$ and (a) $k = 1$; (b) $d = 5$; (c) $v = 1$, $\sigma = 2$; (d) $s = 3.5$, $w = U_0 = 1$; (e) $s = 3.5$, $w = 1$; (f) $s = 2$, $w = 1$. Discrete-only spectra (a, b, f) come from potentials that grow faster than linearly. The others level off at $V = v^2/2D$, with finitely many levels below it (d, e), or infinitely many crowding up to it (c).

30 Logarithmic potentials: radial motion in d dimensions

Setup and key results

$U = -c\ln x$, optionally $+\frac{1}{2}kx^2$, on $x > 0$ with $B = D$. Write $c = (d-1)D/2$ and $\mu = d/2 - 1$. For integer d, $x = r$ is the distance from the origin of a d-dimensional free ($k = 0$) or OU motion; the formulas hold for every real $d > 0$. With $\ell = \sqrt{D/k}$, $m_t = r_0 e^{-kt}$ and $\sigma_t^2 = \frac{1}{2}\ell^2(1 - e^{-2kt})$:

$$k = 0: \quad p(r,t\,|\,r_0,0) = \frac{r}{Dt}\Big(\frac{r}{r_0}\Big)^{\mu} e^{-(r^2+r_0^2)/2Dt}\, I_\mu\Big(\frac{rr_0}{Dt}\Big) \qquad \text{(Bessel process)} \quad (30.1)$$

$$k > 0: \quad p(r,t\,|\,r_0,0) = \frac{r}{\sigma_t^2}\Big(\frac{r}{m_t}\Big)^{\mu} e^{-(r^2+m_t^2)/2\sigma_t^2}\, I_\mu\Big(\frac{rm_t}{\sigma_t^2}\Big) \quad (30.2)$$

$$p_s(r) = \frac{2\, r^{d-1} e^{-r^2/\ell^2}}{\ell^d\, \Gamma(d/2)} \qquad \text{(chi)} \quad (30.3)$$

$$\lambda_n = 2nk, \quad Q_n = \sqrt{\frac{n!\,\Gamma(d/2)}{\Gamma(n+d/2)}}\, L_n^{(\mu)}(r^2/\ell^2), \quad P_n = p_s Q_n \quad (30.4)$$

$$k = 0: \quad P_\lambda = \frac{r^{\mu+1}}{\sqrt{D}}\, J_\mu\Big(r\sqrt{\tfrac{2\lambda}{D}}\Big), \quad Q_\lambda = \frac{r^{-\mu}}{\sqrt{D}}\, J_\mu\Big(r\sqrt{\tfrac{2\lambda}{D}}\Big), \quad \lambda > 0 \quad (30.5)$$

$$k = 0: \quad \Pr(\text{ever reach } a < r_0) = 1 \ \ (d \le 2), \qquad (a/r_0)^{d-2} \ \ (d > 2) \quad (30.6)$$

For $d = 1$, 2 and 3 these are Section 21, Gardiner's Rayleigh process, and $e^{kt} r/r_0$ times Section 22. For $d \ge 2$ the origin is never reached and needs no condition; for $d < 2$ the box assumes it reflects. Absorption at the origin, and $d \le 0$, are in Section 37.

Physical picture. The distance from a fixed point of a particle diffusing in d dimensions, freely or in an isotropic trap. The logarithmic potential is entropic: $\mathcal{U} = -(d-1)k_BT\ln r$ is $-T$ times the entropy $k_B \ln r^{d-1}$ of the sphere of radius r, so it pushes outward because there is more room further out.

- With an absorbing sphere of radius a in three dimensions, the reach probability a/r_0 underlies Smoluchowski's rate for diffusion-limited reactions.
- With a trap in two dimensions, r is the amplitude of a noisy oscillator or optical field, Rayleigh distributed; this is Gardiner's motivation in [G §4.4.5], for example a laser below threshold.
- The squared radius is the CIR or Feller process of interest-rate models and of populations with immigration.

From Gardiner. [G §4.4.5] converts a pair of OU processes [G (4.4.32)] to polar coordinates. Itô's formula gives the amplitude equation [G (4.4.41b)], $\mathrm{d}a = (-\gamma a + \varepsilon^2/2a)\,\mathrm{d}t + \varepsilon\,\mathrm{d}W_a$, which is $d = 2$ with $c = D/2$; the whole logarithmic potential is the Itô term. [G §5.2.6d] then solves this Rayleigh process, [G (5.2.127)]–[G (5.2.138)]: p_s is [G (5.2.129)], the eigenfunctions are Laguerre polynomials [G (5.2.132)]–[G (5.2.134)] with $\lambda = 2n\gamma$ [G (5.2.133)], and the propagator is the series [G (5.2.135)]. His γ is our k and his $\mu = \varepsilon^2/2$ is our $D/2$. General d changes only the Laguerre index, to $d/2 - 1$, and the normalisation: his $Q_\lambda = L_n(\gamma x^2/2\mu)$ [G (5.2.134)] is already normalised for $d = 2$, where the factor in the box is 1. The reach probability is the corrected splitting probability [G (5.2.190)] with $\psi = r^{2c/D} = r^{d-1}$.

30.1 From the line, in d dimensions

The radius $r = |\mathbf{x}|$ of a d-dimensional Gaussian has as its density the Gaussian of Section 2 (or of Section 20) integrated over the sphere of radius r. The angular integral is $\int \mathrm{d}\Omega\, e^{(rm/\sigma^2)\cos\theta} \propto (rm/\sigma^2)^{-\mu} I_\mu(rm/\sigma^2)$, which gives the box. The Itô drift of r is $(d-1)D/2r$: a repulsion from the origin by the potential $-c\ln r$. It is purely geometric: in more than one dimension there are more directions leading away from the origin than toward it.

- $d = 1$: $I_{-1/2}$ is a cosh, and the box is the image pair of Sections 3 and 21. The origin acts as a mirror.
- $d = 3$: $I_{1/2}$ is a sinh, and the box is $e^{kt} r/r_0$ times the absorbing image pair of Sections 4 and 22. That is the Doob transform with $h = x$. Radial motion in three dimensions is one-dimensional motion conditioned never to reach the origin, as anticipated in Section 4. With OU, the conditioned process of Section 22 has stationary density $\propto x^2 e^{-x^2/\ell^2}$, the chi density with $d = 3$.
- $d = 2$: Gardiner's Rayleigh process. Its $p_s = (2r/\ell^2)e^{-r^2/\ell^2}$ is the same Rayleigh density that is the QSD of Section 22.
- The origin: for $d \geq 2$ ($c \geq D/2$) the repulsion is strong enough that the origin is never reached. In Feller's classification [G §5.2.1e] it is an entrance boundary, so the potential supplies the wall condition and none may be imposed. For $0 < d < 2$ the origin is reached and a condition has to be chosen; for $d \leq 0$ the logarithm attracts and the origin is an exit (Section 37).

30.2 Spectrum and recurrence

- By shape invariance (Section 29) the partner in dimension d is the problem in dimension $d + 2$, shifted by $2k$. So $\lambda_n = 2nk$ in every dimension: the even OU levels of Section 21. Only the eigenfunctions, the Laguerre polynomials $L_n^{(\mu)}$, know d.
- For $k = 0$ the spectrum is a continuum from 0, as on the line. Its eigenfunctions are Bessel functions, P_λ and Q_λ of the box, normalised to $\delta(\lambda - \lambda')$ with the weight $p_s = r^{d-1}$; integrating $P_\lambda(r)Q_\lambda(r_0)e^{-\lambda t}$ over λ (Weber's integral) gives the Bessel-process propagator. For $d = 1$ they are the cosines of Section 3, for $d = 3$ the sines of Section 4 times r (forward) or $1/r$ (backward), the Doob transform with $h = r$. Whether the particle comes back is decided by the splitting probability (Section 1.7) with $\psi = r^{d-1}$: $\Pr(\text{reach } a \text{ before } R) = \int_{r_0}^{R} y^{1-d}\mathrm{d}y \,/\, \int_a^R y^{1-d}\mathrm{d}y$. As $R \to \infty$ this tends to 1 for $d \leq 2$ and to $(a/r_0)^{d-2}$ for $d > 2$ (Fig. 31c). Brownian motion is recurrent in one and two dimensions and transient in more (Pólya). In this language, recurrence holds exactly when the logarithmic repulsion is $c \leq D/2$.
- The squared radius $y = r^2$ obeys $\mathrm{d}y = (dD - 2ky)\,\mathrm{d}t + 2\sqrt{Dy}\,\mathrm{d}W$ by Itô's formula. That is the Cox–Ingersoll–Ross, or Feller, process of interest-rate and population models, with the same spectrum $2nk$.

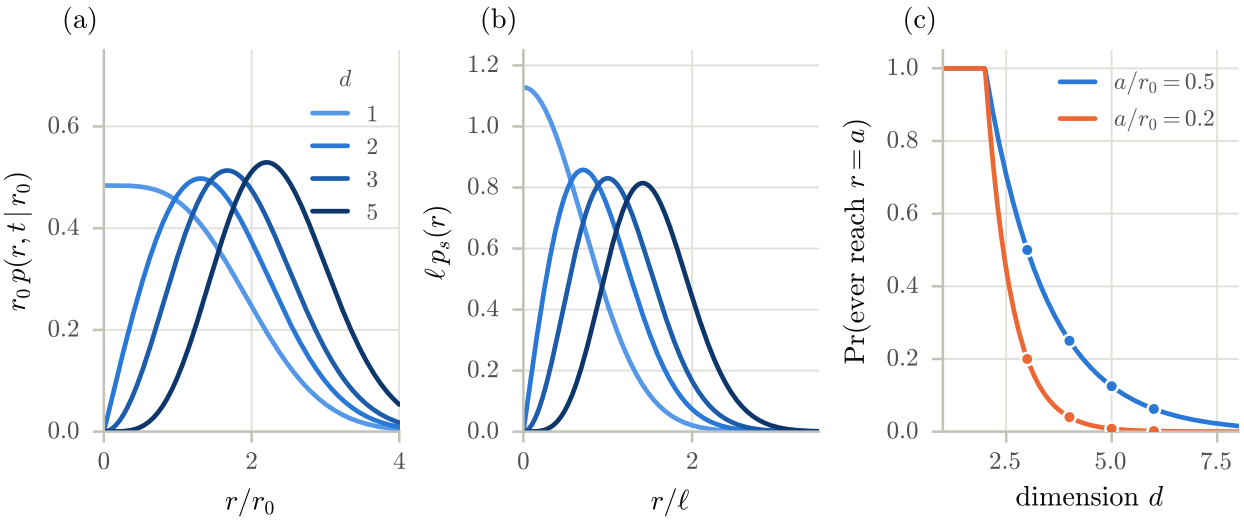


Figure 31: Radial motion; colours give the dimension d in (a) and (b). (a) Free motion (Bessel process) from r_0 at $Dt = r_0^2$. The repulsion $(d-1)D/2r$ empties the region near the origin. (b) Stationary chi densities of the radius of a d-dimensional OU motion; $d = 2$ is the Rayleigh density. (c) Probability that free motion ever comes within a of the origin. Curves are $(a/r_0)^{d-2}$; dots are the splitting-probability quadrature.

31 A soft V: $U = a \ln\cosh(x/w)$

Setup and key results

$A = -v\tanh(x/w)$ with $v = a/w$, $B = D$, on the line. For $|x| \gg w$ the potential is linear, $U \approx v|x|$; for $|x| \ll w$ it is harmonic, $U \approx \frac{1}{2}kx^2$ with $k = a/w^2$. The drift scales are those of Part II, $\ell = D/v$ and $\tau = D/v^2$, and the shape is set by $s = a/D = w/\ell$.

$$p_s(x) = \frac{\Gamma(s+\frac{1}{2})}{\sqrt{\pi}\,\Gamma(s)\,w}\,\operatorname{sech}^{2s}(x/w) \tag{31.1}$$

$$V = \frac{1}{2\tau} - \frac{D}{2w^2}\,\frac{s(s+1)}{\cosh^2(x/w)} \quad \text{(Pöschl–Teller)} \tag{31.2}$$

$$\lambda_n = \frac{D}{2w^2}\,n(2s-n), \quad n = 0, 1, \ldots,\ n < s; \qquad \text{continuum from } \frac{1}{2\tau} \tag{31.3}$$

$$Q_n \propto \cosh^n(x/w)\,P_n^{(s-n,\,s-n)}(\tanh(x/w)) \quad \text{(Jacobi)} \tag{31.4}$$

$$P_n = p_s Q_n \propto \operatorname{sech}^{2s-n}(x/w)\,P_n^{(s-n,\,s-n)}(\tanh(x/w)) \tag{31.5}$$

$$s \to 0\ (v \text{ fixed}):\ U \to v|x|,\ \text{only } \lambda_0; \qquad s \to \infty\ (k \text{ fixed}):\ \lambda_n \to nk \tag{31.6}$$

$P_n^{(a,b)}$ is the Jacobi polynomial, not one of Gardiner's P_λ; each Q_n is normalised by $\int p_s Q_n^2 = 1$.

Physical picture. $\mathcal{U} = 2sk_BT\ln\cosh(x/w)$: a restoring force that is linear near the centre and saturates at the magnitude $F = \gamma v$ far away. One example is a bead sliding under gravity in a V-shaped groove with a rounded bottom of width w. Another is a feedback trap that caps its restoring force.

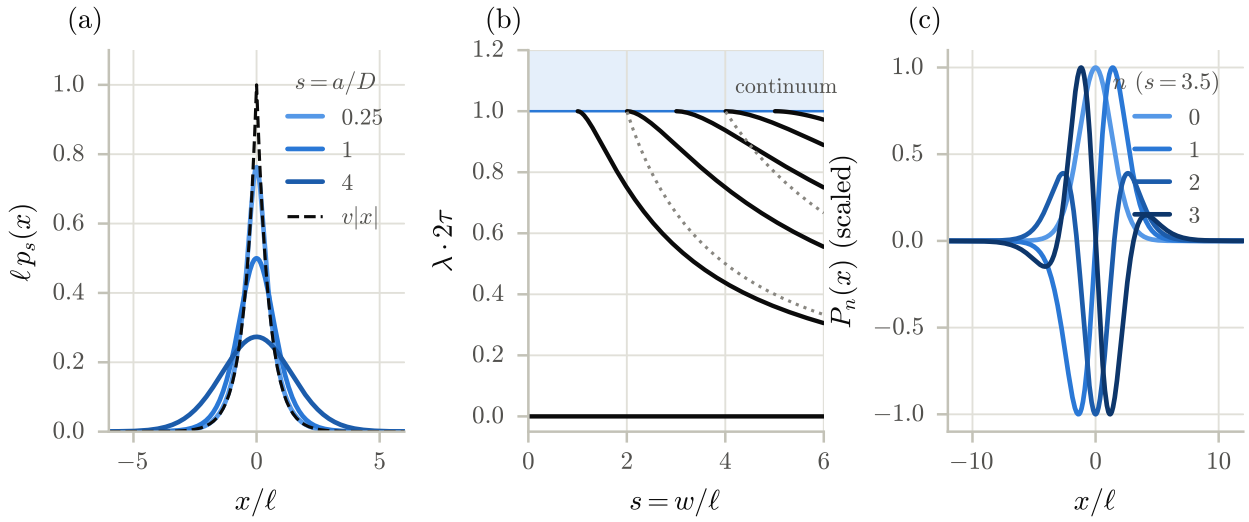


Figure 32: The soft V, $U = a \ln\cosh(x/w)$, with lengths in $\ell = D/v$. (a) Stationary densities; dashed, the sharp V ($s \to 0$). (b) Discrete rates in units of the continuum edge $1/2\tau$ against $s = w/\ell$. A new level leaves the edge at each integer s; dotted, the OU values $nk = 2n/s$. (c) The four bound states at $s = 3.5$, with alternating parity.

From Gardiner. p_s is [G (5.2.39)]; the rest is not in the book. First-passage quantities are the quadratures of Section 1.7, as for any potential (Section 34).

31.1 Between a V and a parabola

- Both limits are earlier sections. As $s \to 0$ at fixed v, $U \to v|x|$: the drift toward a reflecting wall of Section 12, unfolded at the wall. It has the stationary density $e^{-2|x|/\ell}/\ell$, a single level at 0, and a continuum from $1/2\tau$. As $s \to \infty$ at fixed curvature k the problem becomes OU (Section 20); exactly, $\lambda_1 = k(1 - 1/2s)$.
- In between, p_s has a Gaussian core of width $\sqrt{D/k} = \ell\sqrt{s}$ and exponential tails $e^{-2|x|/\ell}$ (Fig. 32a).
- The continuum starts at $V(\infty) = 1/2\tau$, by the rule of Section 1.5. Below it there are $\lceil s \rceil$ discrete levels, and a new one detaches from the edge each time s passes an integer (Fig. 32b). In Schrödinger language these are the bound states of the Pöschl–Teller well. Here they follow from the ladder of Section 29, which stops once $s - n < 0$. In Gardiner's terms, $P_n \sim e^{-(2s-n)|x|/w}$ and $Q_n \sim e^{n|x|/w}$, so $\int P_n Q_n$ [G (5.2.79)] converges exactly when $n < s$.
- Above the edge, P_λ and Q_λ are scattering states. Far from the centre they are the gauge modes of Part II on each side, $P_\lambda \propto e^{-|x|/\ell}\cos(k|x| + \delta_k)$ and $Q_\lambda \propto e^{|x|/\ell}\cos(k|x| + \delta_k)$, with $\lambda = 1/2\tau + \frac{1}{2}Dk^2$.
- Relaxation: for $s > 1$ the slowest rate is the discrete $\lambda_1 < 1/2\tau$, approached exponentially, as in OU. For $s \le 1$ there is no level between 0 and the edge, so the approach is set by the continuum edge, with an algebraic prefactor, as in Section 12. A start at the centre excites only even modes, so it relaxes at λ_2 if $s > 2$ and at the edge otherwise.
- At $s = 1$, $p_s \propto \mathrm{sech}^2(x/w)$ is the logistic distribution.

32 A well with its own walls: $U = -a \ln\cos(x/w)$

Setup and key results

$A = -(a/w)\tan(x/w)$, $B = D$ on $|x| < \pi w/2$. The potential diverges at the walls $x = \pm L/2$, $L = \pi w$. The curvature at the bottom is $k = a/w^2$, and $s = a/D$.

$$p_s(x) = \frac{\Gamma(s+1)}{\sqrt{\pi}\,\Gamma(s+\frac{1}{2})\,w}\cos^{2s}(x/w) \tag{32.1}$$

$$Q_n = \sqrt{\frac{n!\,(n+s)\,\Gamma(2s)}{s\,\Gamma(n+2s)}}\,C_n^{(s)}(\sin(x/w)) \quad \text{(Gegenbauer)}, \qquad P_n = p_s Q_n \tag{32.2}$$

$$\lambda_n = \frac{D}{2w^2}\,n(n+2s) = nk + \frac{n^2\pi^2 D}{2L^2} \quad \text{(exactly)} \tag{32.3}$$

$$\langle \sin(x(t)/w)\rangle = \sin(x_0/w)\,e^{-\lambda_1 t}, \qquad \lambda_1 = k + \frac{\pi^2 D}{2L^2} \tag{32.4}$$

For $s \geq \frac{1}{2}$ the walls are never reached (they are entrance boundaries), and no wall condition is needed.

Physical picture. Two physical systems give exactly this potential.

- **Rotational diffusion in three dimensions.** The polar angle θ of a unit vector diffusing on a sphere obeys this equation with $x = \theta - \pi/2$, $w = 1$ and $s = \frac{1}{2}$. Here $\mathcal{U} = -k_B T \ln\sin\theta$ is again entropic: the circle of latitude at θ has circumference $\propto \sin\theta$. The Gegenbauer polynomials become Legendre polynomials, the rates are $D_r n(n+1)$ with the rotational diffusion coefficient $D_r = D/2$, and $\langle\cos\theta(t)\rangle = \cos\theta_0\,e^{-2D_r t}$ is Debye's dielectric relaxation (checked by the script). On a sphere in d dimensions, $s = (d-2)/2$.
- **Population genetics:** the Wright–Fisher allele frequency with symmetric mutation, below.

From Gardiner. p_s is [G (5.2.39)]. Whether the walls need a condition is decided by Feller's classification of boundaries [G §5.2.1e]. The spectrum and eigenfunctions are not in the book.

32.1 Between a box and a parabola

- As $s \to 0$ the potential vanishes except at the walls, and $\lambda_n \to n^2\pi^2 D/2L^2$: the reflecting interval of length L (Section 5). As $s \to \infty$ at fixed k the problem becomes OU (Section 20). For every s the exact spectrum is the sum of the two, $\lambda_n = nk + n^2\pi^2 D/2L^2$: the confinement by the curvature and the confinement by the walls add (Fig. 33b).
- The ladder is that of Section 29, with $s \to s+1$ and $R = (D/2w^2)(2s+1)$. The slowest backward eigenfunction is $Q_1 = \sqrt{2s+2}\,\sin(x/w)$, so $\langle\sin(x/w)\rangle$ relaxes as a single exponential, like $\langle x\rangle$ in OU. The factor in Q_n is the Gegenbauer norm against the weight $(1-y^2)^{s-1/2}$, which is p_s in $y = \sin(x/w)$. As $s \to 0$ it turns $C_n^{(s)}$ into $\sqrt{2}\,T_n$, a Chebyshev polynomial, and Q_n into the reflecting interval's $\sqrt{2}\cos(n\pi x'/L)$, $x' = x + L/2$, up to sign (Section 5).
- The walls are made by the potential. Near $x = L/2$ the drift is about $-sD/(L/2-x)$, the radial repulsion of Section 30 with $d = 2s+1$. For $s \geq \frac{1}{2}$ ($d \geq 2$) the particle never reaches the walls, and no condition is needed or possible; for $s < \frac{1}{2}$ it does reach them, and the box assumes reflection. For $s \leq -\frac{1}{2}$ ($d \leq 0$) the potential pulls the particle into the walls, which become exits; $s = -\frac{1}{2}$ is Wright–Fisher without mutation, in which an allele is lost or fixed for good (Section 39).

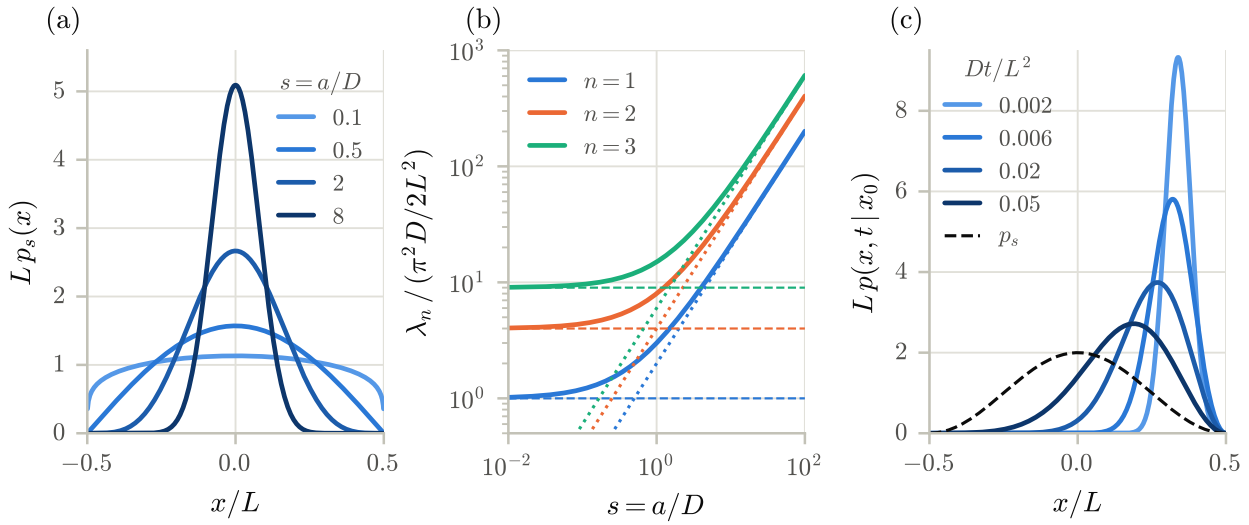


Figure 33: A well with its own walls, $U = -a \ln\cos(x/w)$, width $L = \pi w$. (a) Stationary densities, from nearly uniform ($s \to 0$, a reflecting box) to Gaussian. (b) Rates in units of the box rate $\pi^2 D/2L^2$: exactly $n^2 + 2sn$, the box (dashed) plus OU (dotted). (c) The Gegenbauer propagator at $s = 1$, from $x_0 = 0.35L$.

- A bound to come in Section 34: $U'' = k\sec^2(x/w) \geq k$ everywhere, so Bakry–Émery gives $\lambda_1 \geq k$. That bound misses by exactly the box term.
- The change of variable $y = (1 + \sin(x/w))/2 \in (0, 1)$ turns this process into the Wright–Fisher diffusion of population genetics with symmetric mutation: diffusion $(D/w^2)y(1-y)$ and drift $u(1-2y)$, with $u = (D/w^2)(2s+1)/4$ (checked by the script). Its Jacobi-polynomial spectrum (Kimura) is the λ_n above. Read backwards, this is the transformation of Section 34 that makes any diffusion coefficient constant.

33 Summary of the solvable potentials

- Every exact spectrum in these notes comes from the factorisation $H = \mathcal{A}^\dagger\mathcal{A}$: the rates are those of the inverted potential, and when that is a shifted copy of the original, the spectrum follows by algebra.
- How the potential behaves at infinity decides the type of spectrum. Faster-than-linear growth (harmonic, harmonic plus logarithmic, walls) gives infinitely many discrete levels. Linear growth gives a continuum from $v^2/2D$, with finitely many levels below it (Morse, $\ln\cosh$) or infinitely many crowding up to it (linear plus logarithmic). No growth gives a continuum from 0 (free motion, the logarithm alone).
- A logarithmic singularity of strength $c \geq D/2$ makes a boundary unreachable. The potential then replaces the wall condition (Sections 30 and 32). An attracting one of strength $D/2$ or more makes it an exit (Sections 37 and 39).
- Each new potential interpolates between earlier sections: the logarithm adds dimensions to Parts I and III, $\ln\cosh$ joins a V to OU, and $-\ln\cos$ joins a box to OU.

Part V
Arbitrary potentials

34 What holds for any potential

For a general U (with $A = -U'$ and $B = D$), some results are exact, some are bounds, and the rest need asymptotics or numerics. This section collects them. In applications $\mathcal{U} = \gamma U$ is rarely a mechanical potential. More often it is a free energy along a reaction coordinate (a potential of mean force), for example the extension of a molecule, a dihedral angle, or the progress of a reaction.

From Gardiner. The exact results are [G §5.2.2], [G §5.2.5] and [G §5.2.7]–[G §5.2.8] for general A; the change of variables is Itô's formula [G §4.3.3]; the asymptotics are [G §5.2.7c] and Chapter 9.

Exact for every potential.

- The stationary state $p_s \propto e^{-2U/D}$ [G (5.2.39)], if it is normalisable [G (5.2.40)]; on a circle with a tilt, Eq. (1.19).
- Every first-passage quantity in one dimension is a quadrature (Section 1.7): mean exit times [G (5.2.158)], [G (5.2.160)], splitting probabilities (the corrected [G (5.2.190)]), and conditional exit times [G (5.2.183)]. So is the effective diffusion on a circle, Eq. (26.7).
- The spectrum is that of $H = -\frac{1}{2}D\partial_x^2 + V$ with $V = U'^2/2D - U''/2$ and the wall conditions Eq. (1.13). It is discrete if the domain is bounded or $V \to \infty$, and otherwise has a continuum from the limit of V at infinity (Section 1.5).
- Any diffusion coefficient can be made constant. In one dimension, $y = \sqrt{D}\int^x \mathrm{d}x'/\sqrt{B(x')}$ gives, by Itô's formula, $\mathrm{d}y = \sqrt{D}[A/\sqrt{B} - B'/(4\sqrt{B})]\,\mathrm{d}t + \sqrt{D}\,\mathrm{d}W$ (the Lamperti transformation). So everything here covers every one-dimensional diffusion with $B > 0$; Section 32 is an example, run backwards from Wright–Fisher.

Bounds on the gap.

- **Upper bound.** Any trial function f in the Rayleigh quotient Eq. (1.15) gives an upper bound, $\lambda_1 \le \frac{1}{2}D\int p_s f'^2 / \int p_s f^2$. With $f = x - \langle x\rangle_s$ this is $\lambda_1 \le D/2\,\mathrm{Var}_s(x)$, with equality for OU.
- **Lower bound.** If $U'' \ge k_{\min} > 0$ everywhere, then $\lambda_1 \ge k_{\min}$ (Bakry–Émery [15]), with equality for OU. For Section 32 both bounds hold: $k \le \lambda_1$ and $\lambda_1 \le D/2\,\mathrm{Var}_s(x)$, with $\lambda_1 = k + \pi^2 D/2L^2$.
- Neither bound helps when U is not convex. In a double well λ_1 is exponentially small (Section 35).

Small noise: harmonic pieces and hops. When the barriers are large compared with the temperature $D/2$, V is close to a harmonic well near every extremum x_* of U: $V \approx U''(x_*)^2(x - x_*)^2/2D - U''(x_*)/2$. The low spectrum is then a union of OU ladders, one per extremum (the semiclassical limit).

- Each minimum contributes nU'', $n = 0, 1, 2, \ldots$
- Each maximum contributes $(n+1)|U''|$: the barrier-top modes of Section 27, in which a density slides off the hilltop.
- The $n = 0$ levels of the minima, one per well, split into 0 and exponentially small hopping rates. These are given by Kramers' formula [G (5.2.174)] for a smooth barrier top, modified for a wall (Section 25) or a cusp (Section 26).

The cosine well on a circle and the double well of Section 35 both follow this pattern.

Numerics. The symmetrised form turns the operator into a symmetric matrix.

- Discretise with jump rates $(D/2h^2)e^{-(U_j-U_i)/D}$ between neighbouring sites i and j. Detailed balance with $\pi_i \propto e^{-2U_i/D}$ then holds exactly, so the zero mode is exact and exponentially small rates come out with relative accuracy. The symmetrised matrix is tridiagonal, with off-diagonal $-D/2h^2$ and diagonal the total rate of leaving each site. The script uses this for Section 35.
- With orthonormal eigenfunctions ψ_n of H, the propagator is

$$p(x,t\,|\,x_0,0) = \sqrt{\frac{p_s(x)}{p_s(x_0)}}\,\sum_n \psi_n(x)\,\psi_n(x_0)\,e^{-\lambda_n t}, \tag{34.1}$$

  which is [G (5.2.88)] with Gardiner's $P_n = \sqrt{p_s}\,\psi_n$ and $Q_n = \psi_n/\sqrt{p_s}$ Eq. (1.12), and the general version of Mehler's formula (Section 20). The discrete version is exact as it stands: the eigenvectors of the symmetrised matrix, divided or multiplied by $\sqrt{\pi_i}$, are Q_n and P_n on the grid.

35 Worked example: the double well

Setup and key results

$U = \Delta U\,[(x/a)^2-1]^2$, $B = D$ on the line. The minima at $\pm a$ have $U''(\pm a) = 8\Delta U/a^2$; the barrier top at 0 has $|U''(0)| = 4\Delta U/a^2$. The barrier in units of the temperature is $\beta = 2\Delta U/D$.

$$\lambda_1 \simeq \frac{D/2}{\frac{1}{4}\int_{-a}^{a} \mathrm{d}x/p_s} \quad \text{[G (9.2.18)]} \tag{35.1}$$

$$\simeq 2\,\frac{\sqrt{U''(a)\,|U''(0)|}}{2\pi}\,e^{-\beta} \quad \text{(Kramers)} \tag{35.2}$$

$$\lambda_2 \to |U''(0)| \ \text{(barrier top)}, \qquad \lambda_3 \to U''(a) \ \text{(each well)} \tag{35.3}$$

$$P_0 = p_s, \quad Q_0 = 1; \quad Q_1 \simeq 2\pi_c(x) - 1 \ \text{(committor)}, \quad P_1 = p_s Q_1 \tag{35.4}$$

$$T(-a \to a) \simeq \frac{2}{\lambda_1} \tag{35.5}$$

$$\Pr(x(t) > 0\,|\,x_0 = -a) \simeq \tfrac{1}{2}(1 - e^{-\lambda_1 t}) \tag{35.6}$$

Here $\pi_c(x)$ is the probability of reaching the right minimum before the left one (Gardiner labels the minima a and c and the barrier top b). The limits of λ_2 and λ_3 are for $\beta \to \infty$; the last line holds after a short intrawell transient.

Physical picture. $\mathcal{U} = \Delta\mathcal{U}\,[(x/a)^2-1]^2$: any two-state system with a barrier. A bead in a dual-beam optical trap hops between the two foci; Kramers' theory has been tested directly this way [11]. More often x is a reaction coordinate and $\mathcal{U}$ a free energy: an isomerisation, a protein that folds and unfolds (Gardiner's example in [G §9.2.2] is the reversible denaturation of chymotrypsinogen), or a bistable switch in a cell.

From Gardiner. Chapter 9 is this problem; throughout it, Gardiner's D is our $D/2$. [G §9.1] gives p_s [G (9.1.2)], its two-peak approximation [G (9.1.5)]–[G (9.1.6)], the mean time to reach the barrier top [G (9.1.9)], and the splitting probabilities [G (9.1.10)]. Kramers' method [G §9.2.1] reduces the dynamics to rate equations for the two populations [G (9.2.8)], whose relaxation rate [G (9.2.18)] is λ_1 and whose two-state form [G (9.2.16)] is the last line of the box. The mean exit time is [G (5.2.160)], or [G (9.2.20)].

35.1 Spectrum: hops, wells and the hilltop

- $\lambda_0 = 0$ belongs to p_s, λ_1 is exponentially small, and $\lambda_2, \lambda_3, \dots$ are of order U''. The ratio $\lambda_2/\lambda_1 \sim e^{\beta}$ is the metastability: the particle equilibrates within a well long before it changes wells (Fig. 34b).
- The small-noise rule of Section 34: two OU ladders at the minima ($nU''(a)$, each twice) and one at the maximum ($(n+1)|U''(0)|$), with the two ground levels split into 0 and λ_1. For the quartic, $|U''(0)| = U''(a)/2$, so the second rate is the hilltop mode, $\lambda_2 \to |U''(0)|$, and the intrawell rate comes next, at $U''(a)$ (checked at $\beta = 80$).
- λ_1: Kramers' formula [G (5.2.174)], with $D \to D/2$, gives the one-way hop rate

$$r = \frac{\sqrt{U''(a)\,|U''(0)|}}{2\pi}\, e^{-\beta},$$

  and the flip mode relaxes at $\lambda_1 = 2r$, since hops go both ways. Gardiner's [G (9.2.18)] keeps the integral $\int \mathrm{d}x/p_s$ unexpanded; it matches the exact λ_1 to 2×10^{-5} already at $\beta = 12$, while the Kramers formula has $1/\beta$ corrections (2.5 % at $\beta = 16$).
- From a minimum, the mean time to reach the other minimum is $T = 2/\lambda_1$, one hop at rate r (checked by quadrature of [G (5.2.160)]).

35.2 The slow mode is the committor

- The backward eigenfunction Q_1 is a step: about -1 in the left well and $+1$ in the right, switching across the barrier (Fig. 34d). It is $2\pi_c - 1$, with π_c the splitting probability [G (9.1.10)] of the box. The reason is that $\lambda_1 \approx 0$ makes the eigen-equation $\mathcal{L}^\dagger Q_1 = -\lambda_1 Q_1$ nearly the equation $\mathcal{L}^\dagger \pi_c = 0$ of the splitting probability (checked to 3×10^{-4} at $\beta = 8$).
- $2\pi_c - 1$ is ± 1 wherever p_s is not negligible, so $\int p_s (2\pi_c - 1)^2\, \mathrm{d}x \approx 1$: the committor is already normalised as [G (5.2.80)] asks (to 10^{-3} at $\beta = 8$). $P_1 = p_s Q_1$ is p_s with the sign of the left peak reversed.
- So the slowest observable is which well the particle is committed to. Keeping $P_0 Q_0$ and $P_1 Q_1$ in [G (5.2.88)],

$$p(x, t \,|\, x_0, 0) \approx p_s(x)\big[1 + Q_1(x)\, Q_1(x_0)\, e^{-\lambda_1 t}\big].$$

  From $x_0 = -a$, $Q_1(x_0) = -1$, and the probability of being on the right is $\frac{1}{2}(1 - e^{-\lambda_1 t})$ after a brief intrawell transient (Fig. 34c): a two-state system, left well $\rightleftharpoons$ right well, with rate r each way, as in [G (9.2.16)].
- The faster modes are local. By the small-noise rule of Section 34, the ψ_n of Eq. (1.12) are harmonic-oscillator states centred on an extremum: for λ_2 on the hilltop (at $\beta = 80$, all but 2×10^{-5} of ψ_2^2 lies within $a/4$ of it), for λ_3 in each well. So $P_2 = \sqrt{p_s}\,\psi_2$ is a small density bump on the barrier top, and Q_2 is large there.

35.3 From this potential to any

The structure carries over to any potential with several wells: one slow rate per extra well, with a Kramers prefactor at each saddle; fast rates from local OU ladders at every extremum; and slow eigenfunctions that are committors. The solvable examples of these notes fit inside this picture. OU has one well and no hops. The cosine well on a circle has one well on a loop, so its hops change the winding instead of the well (Section 27). The double well has two wells, and its hops change the well.

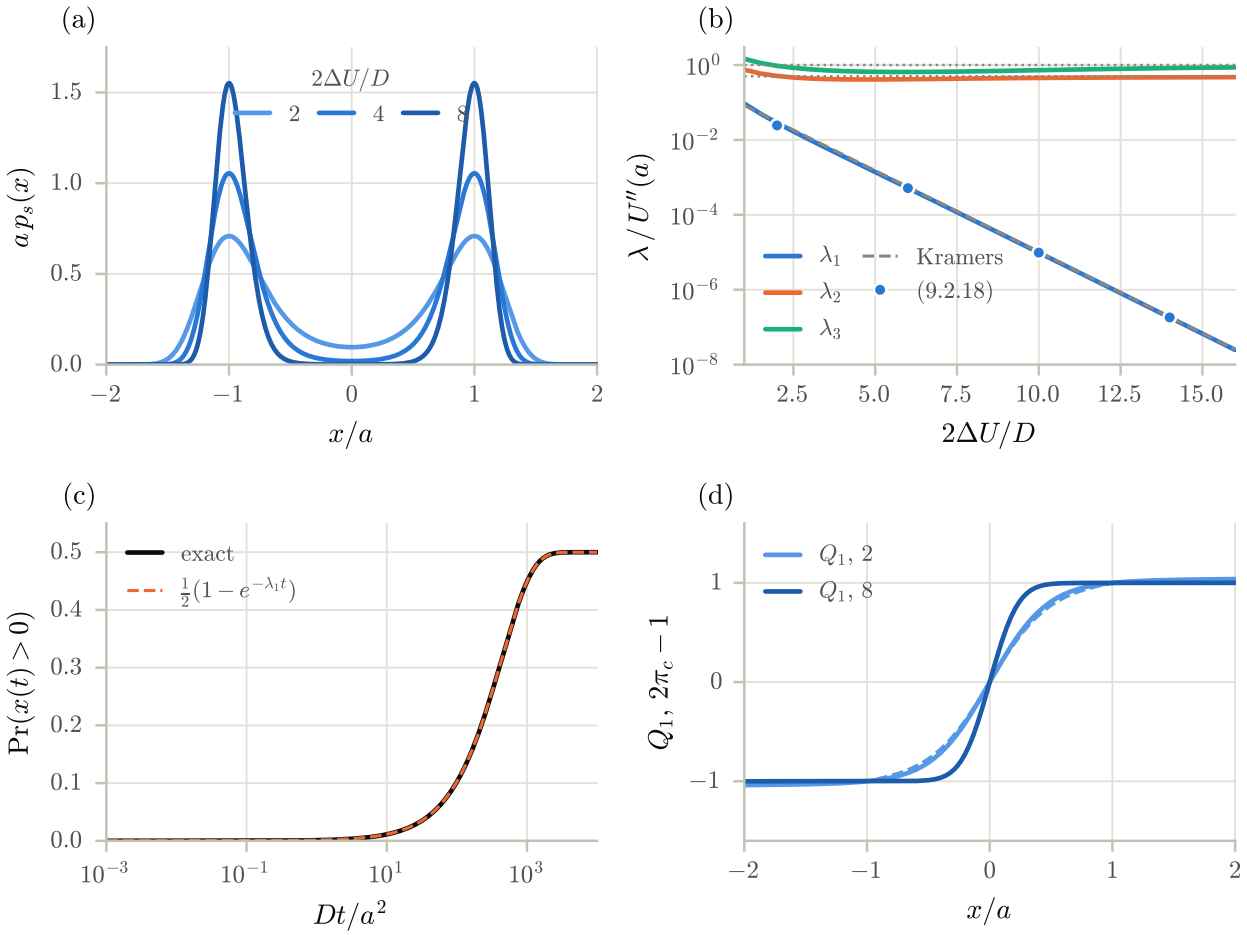


Figure 34: The quartic double well, $\beta = 2\Delta U/D$. (a) Stationary densities. (b) The three slowest rates against the barrier, in units of $U''(a)$: λ_1 (exact; dashed, Kramers; dots, Gardiner's [G (9.2.18)]) falls exponentially. λ_2 tends to the hilltop value $|U''(0)|/U''(a) = \frac{1}{2}$, and λ_3 to the intrawell value 1 (dotted lines). (c) From the left minimum at $\beta = 8$: probability of being on the right, exact against the two-state form. (d) The slowest backward eigenfunction (solid) and $2\pi_c - 1$ (dashed) at $\beta = 2$ and 8.

Part VI
Interfaces and singular boundaries

Parts I–III use two kinds of wall, reflecting and absorbing [G (5.2.13)]–[G (5.2.14)], plus the circle, which has none. Part IV met walls that a diverging potential makes for itself (Sections 30 and 32). Gardiner's list in [G §5.2.1] has three more items: conditions at a discontinuity of the coefficients [G §5.2.1c]; boundaries whose type the equation itself prescribes, because B vanishes there [G §5.2.1e]; and boundaries at infinity [G §5.2.1f], including a current that leaves at $+\infty$ and comes back at $-\infty$ [G (5.2.33)]. This part works one exactly solvable example of each (Sections 36 to 38), then sorts every boundary in the notes into Feller's four types (Section 39).

36 A step in the diffusion coefficient

Setup and key results

$A = 0$ and $B = D_\pm$ for $\pm x > 0$, on the line: free diffusion (Section 2) on each side of an interface at $x = 0$. There the current is continuous and the density jumps by a fixed factor [G §5.2.1c],

$$J(0^+) = J(0^-), \qquad p(0^+) = K\,p(0^-), \qquad K = \Big(\frac{D_-}{D_+}\Big)^{1-\chi}, \tag{36.1}$$

with $\chi = 1$ for a thermal particle (Gardiner's [G (5.2.16)]), $\chi = \frac{1}{2}$ for Stratonovich noise and $\chi = 0$ for Itô noise. With $g_\pm$ the Gaussian of variance $D_\pm t$, a start at $x_0 > 0$, $\eta = K\sqrt{D_+/D_-}$ and $r = (\eta - 1)/(\eta + 1)$:

$$p(x, t \,|\, x_0, 0) = \begin{cases} g_+(x - x_0) + r\, g_+(x + x_0), & x > 0 \\ (1 - r)\, g_-\big(x - x_0\sqrt{D_-/D_+}\big), & x < 0 \end{cases} \tag{36.2}$$

$$\text{box } (-a, a), \text{ reflecting}: \quad p_s = \frac{1}{a(1+K)} \ \ (x < 0), \qquad \frac{K}{a(1+K)} \ \ (x > 0) \tag{36.3}$$

$$Q_\lambda \propto \frac{\cos k_\pm(a - |x|)}{\cos k_\pm a} \ (\pm x > 0), \quad k_\pm = \sqrt{2\lambda/D_\pm}, \quad P_\lambda = p_s Q_\lambda$$

$$\text{box } (-a, a), \text{ absorbing, from } 0: \quad \pi_+ = \frac{KD_+}{D_- + KD_+}, \qquad T = \frac{a^2(1+K)}{D_- + KD_+} = \frac{a^2}{\langle B \rangle_s} \tag{36.4}$$

From the interface, T is a^2 divided by the arithmetic (thermal), geometric (Stratonovich) or harmonic (Itô) mean of D_+ and D_-. A potential step $\Delta U = U(0^+) - U(0^-)$ at fixed D gives $K = e^{-2\Delta U/D}$ and $r = -\tanh(\Delta U/D)$.

Physical picture. A tracer crossing from one medium into another at the same temperature: a colloid moving from water into a more viscous layer, a protein entering a crowded compartment, a molecule passing between a gel and free solution. The friction, and so $D = 2k_BT/\gamma$, jumps at the interface. Which density is continuous depends on where the noise comes from (Section 36.2).

- A thermal particle has Boltzmann's density at equilibrium, which does not depend on the friction, so p is continuous ($K = 1$). Colloids near a wall, where hydrodynamics lowers D, behave this way [20].

- A run-and-tumble bacterium whose swimming speed u varies, tumbling at a fixed rate, piles up where it is slow: $p_s \propto 1/u$ [21]. Its B is $\propto u^2$, so $p_s \propto 1/\sqrt{B}$, the Stratonovich case.
- A hopper that waits at each site for a time set by that site, then jumps left or right with equal probability, spends its time where hopping is slow: $p_s \propto 1/B$, the Itô case.

If the two media also differ in free energy, as when a drug dissolves better in a membrane than in water, K is the equilibrium partition coefficient [22]. A finite step in $\mathcal{U}$ interpolates between the walls of Fig. 1: an infinitely high step is a hard wall, an infinitely deep one a cliff. With $K = 1$ the same equations describe heat flowing along two rods of different conductivity, joined end to end: temperature and heat flux are continuous.

From Gardiner. [G (5.2.15)]–[G (5.2.16)] (and [G (5.3.18)] in several dimensions) ask for continuous current and continuous density. Both are right for a jump in A, as at the cusp of Section 26. For a jump in B the second is an extra physical input, not a consequence of [G (5.2.6)]. If A is bounded, $J = Ap - \frac{1}{2}\partial_x(Bp)$ stays finite only if Bp is continuous, which is the Itô case. p is continuous only when A contains the noise-induced drift $\frac{1}{2}B'$, which at a step is a δ-function of weight $\frac{1}{2}(D_+ - D_-)$. That is the drift of a thermal particle: eliminating the velocity when the friction depends on position gives $A = -\mathcal{U}'/\gamma + \frac{1}{2}B'$ [23], and with it Gardiner's conditions. The stationary density is [G (5.2.39)]. The splitting probability and the exit time are the corrected [G (5.2.190)] and [G (5.2.158)], with $\psi = Bp_s$ piecewise constant. The backward conditions at such a surface are among the "other boundaries" that [G §5.2.4c] sets aside; they are derived below.

36.1 A partial image

On each side the equation is free. So try the image construction of Sections 3 and 4 with an image of unknown weight r, and on the far side a transmitted Gaussian of unknown weight. Away from the interface $J = -\frac{1}{2}D_\pm\partial_x p$, so the conditions Eq. (36.1) read $p(0^+) = Kp(0^-)$ and $D_+\partial_x p(0^+) = D_-\partial_x p(0^-)$. A transmitted Gaussian centred at $x_0\sqrt{D_-/D_+}$ has, at the interface, the same exponent $(x - \text{centre})^2/2D_\pm t$ as the incident one. Both conditions then hold at all times at once, and they fix the transmitted weight at $1 - r$ and r at the value in the box.

- $r = +1$ is a reflecting wall (Section 3), $r = -1$ an absorbing one (Section 4), and $r = 0$ no interface. Every interface lies in between.
- For a thermal particle, a medium in which it can hardly move ($D_- \to 0$) acts as a hard wall, $r \to 1$, and one that sweeps it away ($D_- \to \infty$) acts as a cliff, $r \to -1$. In the Itô case the signs are reversed: the slow side traps.
- In the Stratonovich case $\eta = 1$, so $r = 0$ for every ratio D_-/D_+: nothing is reflected.
- The slope of p jumps by the factor D_-/D_+ (Fig. 35a). This is Gardiner's remark that "the derivatives of p are not necessarily continuous".
- In $y = \int \mathrm{d}x/\sqrt{B}$, which is $x/\sqrt{D_\pm}$ on each side (Lamperti, Section 34), the noise is the same everywhere and the interface is a semi-permeable mirror. This is *skew Brownian motion*: from the interface the particle goes right with probability $(1 + r)/2 = \eta/(1 + \eta)$ [24].

A start at $x_0 < 0$ follows by exchanging the sides, $D_+ \leftrightarrow D_-$ and $K \to 1/K$. In the box $(-a, a)$ with reflecting walls the modes are cos on each side (the Q_λ of the box, Section 36.3), and the rates solve $K\sqrt{D_+}\tan\bigl(a\sqrt{2\lambda/D_+}\bigr) + \sqrt{D_-}\tan\bigl(a\sqrt{2\lambda/D_-}\bigr) = 0$. Like the root condition of Section 16, this is where the images stop giving a closed form.

36.2 Which density is continuous

Smooth the step over a width ε. Then B is smooth, every reading of the noise gives an ordinary equation, and the readings differ only by a drift $\chi B'/2$:

- Itô, $\mathrm{d}x = \sqrt{B}\,\mathrm{d}W$ ($\chi = 0$): x is a martingale, and $p_s \propto 1/B$;
- Stratonovich, $\mathrm{d}x = \sqrt{B} \circ \mathrm{d}W$ ($\chi = \frac{1}{2}$): $p_s \propto 1/\sqrt{B}$;
- thermal, also called Hänggi–Klimontovich or isothermal ($\chi = 1$): p_s is uniform.

In each case [G (5.2.39)] gives $\psi = B^\chi$ and $p_s = \psi/B \propto B^{\chi-1}$ (Fig. 35b). As $\varepsilon \to 0$ the extra drift becomes a δ-function of weight $\chi(D_+ - D_-)/2$. It is too thin to change the motion on either side, but strong enough to set the jump of p, to $K = (D_-/D_+)^{1-\chi}$. So a sharp interface remembers how the step was smoothed. The script solves the smoothed equation by finite differences and recovers the closed form Eq. (36.2), with an error proportional to ε; simulations of the smoothed SDE give the exit probabilities (Fig. 35c).

36.3 Backward conditions and exit from a box

Pair the forward and backward equations as in Eq. (1.9), treating the interface as two walls. What is left at $x = 0$ is $J\,[f]_{0^-}^{0^+} + \frac{1}{2}p(0^-)\big[KD_+f'(0^+) - D_-f'(0^-)\big]$. So a backward function is continuous at the interface, with $KD_+f'(0^+) = D_-f'(0^-)$.

These are the conditions on Gardiner's Q_λ. In the reflecting box, $Q_\lambda = \cos k_\pm(a - |x|)/\cos k_\pm a$ on each side has $Q' = 0$ at the walls and $Q(0^\pm) = 1$, so it is continuous; the derivative condition is the root condition above. $P_\lambda = p_s Q_\lambda$ then jumps by K, like p_s, and carries a continuous current: it meets the forward conditions Eq. (36.1), which pair with the backward ones as the wall conditions do. The script checks that these P_λ, Q_λ are biorthonormal for $\chi = 1$ and 0.

With absorbing walls at $\pm a$:

- π_+ is piecewise linear, with slopes in the ratio $D_- : KD_+$ (Fig. 35c). In the Itô case x is a martingale, so $\pi_+(x_0) = (x_0 + a)/2a$ whatever B does.
- T is piecewise quadratic. From the interface it is $a^2/\langle B\rangle_s$, where $\langle B\rangle_s = (D_- + KD_+)/(1 + K)$ is the average of B over the stationary density of the reflecting box. The reason: in all three readings $x^2 - \int_0^t B\,\mathrm{d}t'$ is a martingale (the extra drift acts only where $x = 0$), so $a^2 = \langle \int_0^T B\,\mathrm{d}t\rangle$, and from $x_0 = 0$ the times spent on the two sides before exit are in the ratio $K : 1$.
- A thermal particle released at the interface leaves through the fast side with probability $D_+/(D_+ + D_-)$, after a mean time $2a^2/(D_+ + D_-)$.

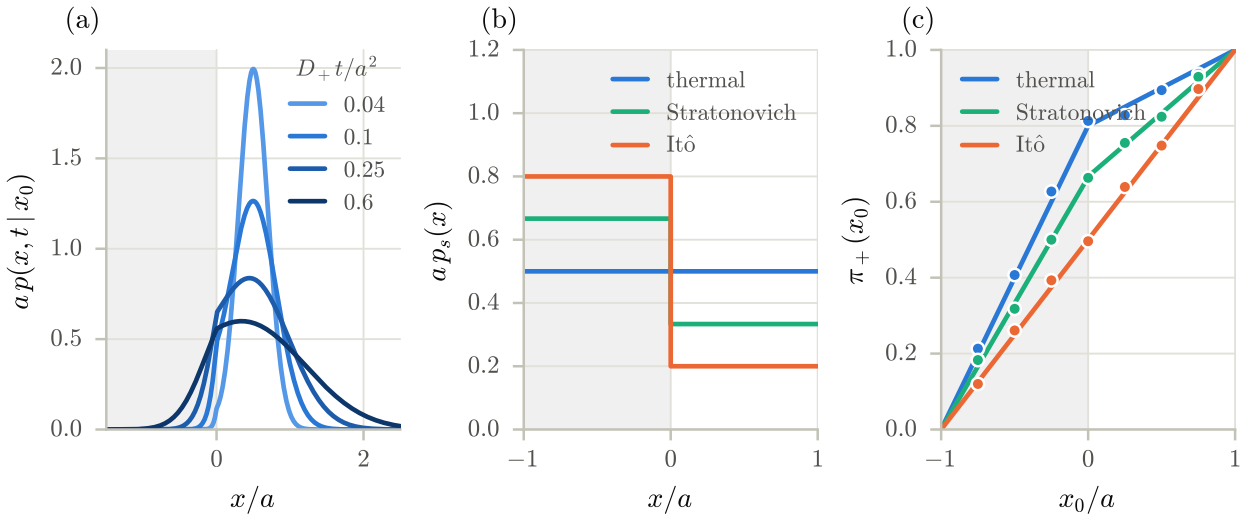


Figure 35: A step in the diffusion coefficient, $D_- = D_+/4$; the slow side is shaded. (a) A thermal particle from $x_0 = a/2$. p is continuous at the interface, and its slope jumps by the factor 4. (b) Stationary densities in the box $(-a, a)$ with reflecting walls, for the three readings of the noise. (c) Probability of leaving the box through $+a$. Lines are the closed forms; dots are simulations of the smoothed equation ($\varepsilon = 0.03a$, 1000 paths per point).

37 Where B vanishes: the square-root process

Setup and key results

$A = dD - 2ky$, $B = 4Dy$ on $y > 0$, for any real d: the squared radius $y = r^2$ of Section 30, and the equation Feller studied in 1951 [25]. Lengths in $\ell^2 = D/k$, times in $1/k$. With $\mu = d/2 - 1$, $\bar{y}_t = y_0 e^{-2kt}$, $\sigma_t^2 = \frac{1}{2}\ell^2(1 - e^{-2kt})$ and $z_t = \bar{y}_t/2\sigma_t^2 = (y_0/\ell^2)/(e^{2kt} - 1)$. The origin is an exit for $d \le 0$, regular for $0 < d < 2$ and an entrance for $d \ge 2$; infinity is natural. The propagator is p_+ when the process is reflected ($d > 0$) and p_- when it is absorbed ($d < 2$):

$$p_\pm(y, t\,|\,y_0, 0) = \frac{1}{2\sigma_t^2}\Big(\frac{y}{\bar{y}_t}\Big)^{\mu/2} e^{-(y+\bar{y}_t)/2\sigma_t^2}\, I_{\pm\mu}\Big(\frac{\sqrt{y\bar{y}_t}}{\sigma_t^2}\Big) \tag{37.1}$$

$$\text{reflected}: \quad p_s = \frac{y^{d/2-1}e^{-y/\ell^2}}{\Gamma(d/2)\,\ell^d}, \quad \lambda_n = 2nk, \quad Q_n \propto L_n^{(\mu)}(y/\ell^2), \quad P_n = p_s Q_n \tag{37.2}$$

$$\text{absorbed}: \quad S = P(1 - \tfrac{d}{2}, z_t), \quad \lambda_n = (2n + 2 - d)k \tag{37.3}$$

$$Q_n \propto y^{1-d/2} L_n^{(1-d/2)}(y/\ell^2), \quad P_n \propto e^{-y/\ell^2} L_n^{(1-d/2)}(y/\ell^2) \tag{37.4}$$

$$\text{QSDs}: \quad q_\lambda \propto M(1 + \tfrac{\lambda}{2k}, 2 - \tfrac{d}{2}, -y/\ell^2), \quad 0 < \lambda \le (2 - d)k \tag{37.5}$$

$P(a, z) = \gamma(a, z)/\Gamma(a)$ is the regularised incomplete gamma function and M is Kummer's function. Point starts reach the QSD $e^{-y/\ell^2}/\ell^2$. For $d = 0$, $\Pr(y_t = 0) = e^{-z_t}$: the extinction probability of Feller's branching diffusion.

Physical picture. Four systems lead to this equation.

- The squared distance from the centre of an overdamped particle in an isotropic harmonic trap in d dimensions (Section 30), or its potential energy $E = \frac{1}{2}\mathcal{K}r^2$. E/k_BT is gamma distributed

with mean $d/2$, which is equipartition. For $d = 2$, y is the intensity of a noisy optical field, exponentially distributed.

- A population with demographic noise. Births and deaths at nearly equal per-capita rates make the variance of the change proportional to the size, $B \propto y$; $2k$ is the net death rate and dD a rate of immigration. With no immigration ($d = 0$) this is Feller's branching diffusion, which dies out for sure. Gardiner's reaction $X + A \rightleftharpoons 2X$ [G (5.2.57)] behaves the same way near $x = 0$: with no X there is no reaction.
- The interest rate of Cox, Ingersoll and Ross [26], $\mathrm{d}y = (a - by)\,\mathrm{d}t + \sigma\sqrt{y}\,\mathrm{d}W$, which is this equation with $d = 4a/\sigma^2$. Their condition $2a \geq \sigma^2$, that the rate never reaches zero, is $d \geq 2$.
- In the radius $r = \sqrt{y}$, the logarithmic potential $-(d-1)k_BT \ln r$ of Section 30 attracts when $d < 1$. A counterion near a thin charged rod, moving in the plane perpendicular to it, feels $\mathcal{U} = 2\xi_{\mathrm{M}} k_BT \ln r$, where ξ_{M} is the Manning parameter. With the entropic term of the plane, this is $d = 2 - 2\xi_{\mathrm{M}}$. For $\xi_{\mathrm{M}} \geq 1$ the rod is an exit boundary and the counterions collapse onto it: Manning condensation [27], for a rod of zero radius.

From Gardiner. This is a prescribed boundary [G §5.2.1e]: $B(0) = 0$, and the equation decides whether a condition may be imposed there. Gardiner's quick rule [G (5.2.22)]–[G (5.2.24)], by the sign of $A(0)$, assumes that $\sqrt{B}$ is Lipschitz, which forces $\partial_y B(0) = 0$ [G (5.2.20)]. Here $\partial_y B(0) = 4D$, and the rule fails: $A(0) = dD > 0$ for every $d > 0$, but only $d \geq 2$ is an entrance. Feller's criteria [G (5.2.25)]–[G (5.2.29)] decide the type (below). The paragraph after them has a misprint: its second "exit" should read "entrance".

- For $d = 2$ this is the square of the Rayleigh process of [G §5.2.6d], with the same Laguerre spectrum [G (5.2.133)].
- Gardiner's reaction model [G (5.2.57)]–[G (5.2.60)] has $A(0) = 0$ and $\partial_x B(0) > 0$, which is the $d = 0$ case: an exit. Its stationary solution [G (5.2.59)] behaves like $1/x$ at $x = 0$ and is not normalisable, like $y^{d/2-1}$ here.
- The mean time to reach 0 is [G (5.2.161)], with the reflecting wall at infinity.
- Gardiner's backward reflecting condition [G (5.2.69)] carries the proviso "unless B vanishes". Here it becomes $\psi\,\partial_y f \to 0$: a zero derivative with respect to the scale $\int \mathrm{d}y/\psi$.

37.1 Feller's test

Gardiner's functions [G (5.2.25)]–[G (5.2.28)] are, up to constant factors,

$$f = y^{-d/2} e^{y/\ell^2}, \qquad g = \frac{2}{Bf} = \frac{y^{d/2-1} e^{-y/\ell^2}}{2D}. \tag{37.6}$$

Here $f = 1/\psi$ is the density of the scale $\int \mathrm{d}y/\psi$, and $g = 2\psi/B$ is twice the zero-current solution Eq. (1.5); Feller calls them the scale and speed densities. Near $y = 0$:

- f is integrable iff $d < 2$, and g iff $d > 0$;
- $h_1 = f \int_{y_0}^{y} g$ is integrable iff $d < 2$. This is the condition for the boundary to be reachable from inside;
- $h_2 = g \int_{y_0}^{y} f$ is integrable iff $d > 0$. This is the condition for a process started at the boundary to leave it.

Gardiner's table then gives exit for $d \leq 0$, regular for $0 < d < 2$, and entrance for $d \geq 2$. At $y \to \infty$, f grows and g decays like $e^{\pm y/\ell^2}$, and neither h_1 nor h_2 is integrable: a natural boundary. The script evaluates the four integrals with shrinking cutoffs and finds the same types, in y and in r.

37.2 From the radius: two solutions

The density of $y = r^2$ is $p_r(\sqrt{y})/2\sqrt{y}$, and this turns the radial propagator of Section 30 into p_+. In r the noise is constant and the drift $(d-1)D/2r - kr$ is singular. In y the drift is smooth and the noise vanishes. The boundary has the same type in both, because Feller's integrals do not change under a change of variable: the type belongs to the process, not to the coordinate.

The second solution, p_-, has $I_{-\mu}$ in place of I_μ. It is the Doob transform (Section 4) of the process in dimension $4-d$,

$$p_-^{(d)}(y,t\,|\,y_0) = e^{-(2-d)kt}\Big(\frac{y_0}{y}\Big)^{1-d/2} p_+^{(4-d)}(y,t\,|\,y_0), \tag{37.7}$$

which for $d = 1$ is the relation found in Section 30 between OU absorbed at its bottom and the radius in three dimensions. Both functions solve the equation for every d; the type of boundary decides which one is the answer (Fig. 36).

- **Entrance, $d \geq 2$.** From $y_0 > 0$ the origin is never reached. Started at $y_0 = 0$, the process enters at once: p_+ tends to a gamma density of scale $2\sigma_t^2$. Only p_+ is a probability density, so there is nothing to choose, and no condition can be imposed.
- **Regular, $0 < d < 2$.** The origin is reached in finite time and can be left, and both solutions are legitimate, so a condition must be chosen. Reflection, $J(0) = 0$, gives p_+, which diverges at the origin like $y^{d/2-1}$. Absorption gives p_-, which tends to a nonzero constant there (Fig. 36c). So where B vanishes an absorbing boundary does not have $p = 0$ [G (5.2.14)]. What vanishes is p/p_s, the backward function, as the operator picture of Section 1.5 suggests. Partly absorbing and sticky conditions are also possible at a regular boundary; they are not needed here.
- **Exit, $d \leq 0$.** The origin is reached in finite time and never left, and p_- is the only solution. The zero-current solution $y^{d/2-1}e^{-y/\ell^2}$ is not normalisable, so nothing can be reflected, and probability collects at $y = 0$ as an atom of mass $1 - S$.

For $d = 1$ the regular case and its two choices are Sections 21 and 22, squared.

37.3 Spectrum, survival and QSDs

- The reflected process has the Laguerre ladder $2nk$ of Section 30 and the gamma stationary density. The absorbed one has the ladder of dimension $4-d$, shifted up by $(2-d)k$ through the factor $e^{-(2-d)kt}$. In the regular window both ladders exist. At an exit only the absorbed one does, at an entrance only the reflected one (Fig. 36b). They meet at $d = 2$, where the origin stops being reachable.
- **Gardiner's P_n and Q_n.** With $z = y/\ell^2$:
  - reflected: $Q_n = \sqrt{n!\,\Gamma(d/2)/\Gamma(n+d/2)}\,L_n^{(\mu)}(z)$ and $P_n = p_s Q_n$, the Q_n of Section 30 in the variable r^2;
  - absorbed: $P_n = w\,Q_n$ with the zero-current solution $w = z^{d/2-1}e^{-z}/\ell^2$, which gives $Q_n = c_n z^{1-d/2} L_n^{(1-d/2)}(z)$, $P_n = c_n e^{-z} L_n^{(1-d/2)}(z)/\ell^2$ and $c_n = \sqrt{n!/\Gamma(n+2-d/2)}$. In the regular window $w = \Gamma(d/2)\,p_s$ is the reflected p_s of [G (5.2.91)], up to a constant. At an exit there is no reflected problem to borrow it from, but w still works as the weight, although it cannot be normalised.

  At the origin the absorbed $Q_n \to 0$ while P_n tends to a constant: absorption where B vanishes sets $Q = 0$, not $P = 0$, as found for p_- above. The QSDs q_λ with $\lambda < (2-d)k$ are forward eigenfunctions off the ladder; q_λ/w grows like e^z, so they do not enter [G (5.2.88)]. At $\lambda = (2-d)k$ the Kummer function is e^{-z} and $q_\lambda \propto P_0$.

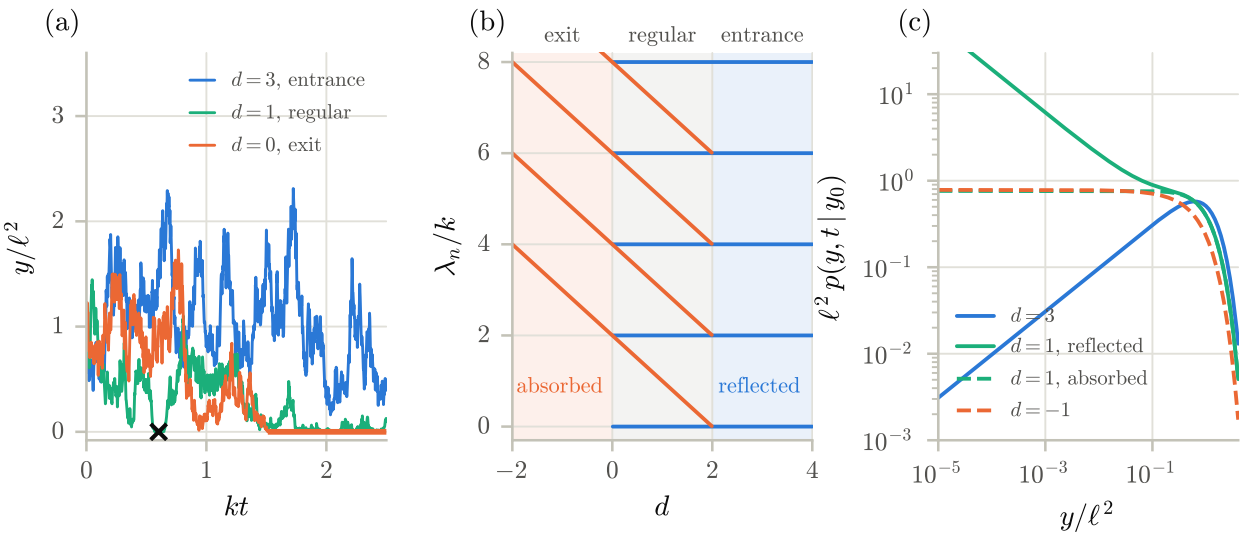


Figure 36: The square-root process; y is in units of $\ell^2 = D/k$. (a) Paths from $y_0 = \ell^2$. $d = 3$: an entrance, never reached. $d = 1$: regular, reflected each time it arrives; absorption would stop it at the cross. $d = 0$: an exit, absorbed for good. The $d = 3$ and $d = 1$ paths are squared radii of OU motions; the $d = 0$ path is sampled exactly. (b) Rates against d. The reflected ladder $2nk$ exists for $d > 0$, the absorbed one $(2n + 2 - d)k$ for $d < 2$. Only in the regular window do both exist, and there a condition must be chosen. (c) Densities near the origin at $kt = 0.25$ from $y_0 = \ell^2$. The entrance density vanishes, the reflected one diverges like $y^{-1/2}$, and the absorbed ones tend to constants.

- Survival decays at $\lambda_0 = (2-d)k$: that is k for OU absorbed at its bottom ($d = 1$, Section 22) and $2k$ for the branching diffusion. Integrating p_- term by term in the series of $I_{-\mu}$ gives the incomplete gamma function of the box. With no trap ($k = 0$), $z_t = y_0/2Dt$ and $S \simeq (y_0/2Dt)^{1-d/2}/\Gamma(2-d/2)$. This generalises the $t^{-1/2}$ of Section 4, and the mean time to reach the origin is finite only for $d < 0$.
- With the trap, the mean time $T(y_0) = \int_0^\infty S\, \mathrm{d}t$ is finite for every $d < 2$ and diverges as $d \to 2^-$. It passes smoothly through $d = 0$: seen from inside, an exit and a regular boundary look alike. They differ in what can happen after arrival.
- $y = \infty$ is natural, so the criterion of Section 1.8 predicts a continuum of QSDs, as for OU absorbed at its bottom. The Kummer functions in the box are that continuum: each is positive, with a tail $y^{-1-\lambda/2k}$. A point start reaches the minimal one, at $\lambda = (2 - d)k$, where M reduces to e^{-y/ℓ^2}: the Yaglom limit is exponential in every dimension. For $d = 1$, in $x = \sqrt{y}$, it is the Rayleigh QSD. A logistic term, $A \to A - cy^2$, makes infinity an entrance, and the QSD becomes unique [16].

38 A current through infinity

Setup and key results

$A = x^2 + I$, $B = D$ on the line, in the units of the neuron model below (x and I dimensionless, D a rate). The noise sets the scales $D^{1/3}$ for x, $D^{-1/3}$ for t and $D^{2/3}$ for I. $+\infty$ is an exit and $-\infty$ an entrance; what leaves at $+\infty$ is reinjected at $-\infty$, $J(+\infty) = J(-\infty) = J_s$ [G (5.2.33)].

$$p_s(x) = \frac{2J_s}{D}\int_x^\infty e^{\varphi(x)-\varphi(y)}\,\mathrm{d}y \;\simeq\; \frac{J_s}{x^2} \quad (x \to \pm\infty), \qquad \varphi = \frac{2}{D}\Big(\frac{x^3}{3} + Ix\Big) \tag{38.1}$$

$$\frac{1}{J_s} = \sqrt{\frac{2\pi}{D}}\int_0^\infty \frac{\mathrm{d}u}{\sqrt{u}}\,\exp\Big[-\frac{2}{D}\Big(\frac{u^3}{12} + Iu\Big)\Big] \tag{38.2}$$

$$\to\; \frac{\pi}{\sqrt{I}} \;\; (D \to 0,\; I > 0), \qquad 6.27\,D^{-1/3} \;\; (I = 0), \qquad \frac{\pi}{c}\,e^{8c^3/3D} \;\; (D \to 0,\; I = -c^2)$$

$$P_0 = p_s, \quad Q_0 = 1; \qquad Q_\lambda(+\infty) = Q_\lambda(-\infty); \qquad P_\lambda \neq p_s Q_\lambda \;\; (\lambda \neq 0) \tag{38.3}$$

The constant is $6.27 = \frac{\sqrt{2\pi}}{3}\,6^{1/6}\,\Gamma(\frac{1}{6})$. In $\theta = 2\arctan x$ the line closes into a circle,

$$\mathrm{d}\theta = \big[1 - \cos\theta + I(1+\cos\theta) - \tfrac{D}{2}\sin\theta\,(1+\cos\theta)\big]\mathrm{d}t + \sqrt{D}\,(1+\cos\theta)\,\mathrm{d}W, \tag{38.4}$$

and both infinities are the point $\theta = \pi$, where the noise vanishes and the drift is 2.

Physical picture. The quadratic integrate-and-fire (QIF) neuron. x is the membrane potential, rescaled. It runs off to $+\infty$ in finite time, which is a spike, and is reset to $-\infty$; J_s is the firing rate. The model is the normal form of a neuron that starts firing through a saddle-node bifurcation on its limit cycle, and in θ it is the theta neuron of Ermentrout and Kopell [28]. Mechanically, it is an overdamped particle sliding off the cubic potential $\mathcal{U} = -\gamma(x^3/3 + Ix)$. This is what the tilted washboard of Section 27 looks like near its critical tilt, around the point where the minimum and the maximum merge, seen through a coordinate that sends the rest of the circle to infinity. Each passage through infinity is one winding: a phase slip of the overdamped pendulum or of the Josephson junction. For $I < 0$ the particle sits in a well and escapes over a barrier, as below depinning in Section 27.

From Gardiner. [G §5.2.1f] allows every kind of boundary of [G §5.2.1] at infinity, and notes that a nonzero current there needs A or B to be infinite; here $A \sim x^2$. Of his two conditions, $J(\pm\infty) = 0$ [G (5.2.32)] and $J(+\infty) = J(-\infty)$ [G (5.2.33)], this is the second, "the limit of periodic".

- With constant J the stationary solution is [G (5.2.45)], with the seam at infinity.
- The mean time from $-\infty$ to $+\infty$ is [G (5.2.160)], with $a = -\infty$ reflecting, $b = +\infty$ absorbing and $\psi = e^{\varphi}$. Its inverse is J_s.
- The types of $\pm\infty$ follow from [G (5.2.25)]–[G (5.2.29)]. In θ they are Gardiner's sign rule [G (5.2.22)]–[G (5.2.23)].

38.1 An exit at $+\infty$, an entrance at $-\infty$

With $\psi = e^{\varphi}$, Gardiner's functions are $f = e^{-\varphi}$ and $g = (2/D)e^{\varphi}$.

- At $+\infty$, f is integrable and g is not, while $h_1 = f\int g \simeq 1/x^2$ is integrable: an exit. The drift x^2 carries the particle to infinity in the finite time $\int^\infty \mathrm{d}x/x^2$, faster than the noise can bring it back. The process explodes.

- At $-\infty$ the roles swap: g is integrable, and so is $h_2 \simeq 1/x^2$: an entrance. The particle comes down from $-\infty$ in finite time, as a particle in $A = -kx^3$ comes down from $+\infty$ (Section 1.8).
- The mirror image $x \to -x$ turns A into $-A$, that is U into $-U$: the partner of Section 29. It exchanges the two ends, and with them exit and entrance. That is true of the partner in general (Section 39).

Without reinjection the process is absorbed at $+\infty$, and $1/J_s$ is the mean time to get there from $-\infty$. It is finite, because both ends are crossed in finite time.

38.2 Through infinity: the circle again

Reinjecting at $-\infty$ whatever leaves at $+\infty$ is the condition [G (5.2.33)]. The stationary equation $J = J_s$ integrates as in [G (5.2.45)] to the box, and normalisation fixes J_s. Writing $y = s + u/2$ and $x = s - u/2$ turns $\varphi(y) - \varphi(x)$ into $\frac{2}{D}(us^2 + u^3/12 + Iu)$, and the Gaussian integral over s leaves the single integral of the box [29].

- Without noise ($D \to 0$, $I > 0$) the period is $\int \mathrm{d}x/(x^2 + I) = \pi/\sqrt{I}$: the firing rate rises as $\sqrt{I}$ from the bifurcation.
- At the bifurcation ($I = 0$) the noise alone sets the rate, $J_s \propto D^{1/3}$.
- Below it ($I = -c^2$) the particle sits in the well at $x = -c$ and must cross the barrier at $+c$, of height $\Delta U = \frac{4}{3}c^3$. Kramers' formula [G (5.2.174)], with $U'' = \pm 2c$, gives $J_s \simeq (c/\pi)e^{-2\Delta U/D}$. Once over the top, the particle falls to infinity.
- $p_s \simeq J_s/x^2$ at both ends: the density of a particle moving at speed x^2 and carrying the current J_s.
- Gardiner's eigenfunctions. Pairing as in Eq. (1.9) leaves $-J_s[f(+\infty) - f(-\infty)]$ at the two ends, so the backward condition that goes with [G (5.2.33)] is $Q_\lambda(+\infty) = Q_\lambda(-\infty)$: a function of the start must not notice the reinjection. $Q_0 = 1$ satisfies it, and $P_0 = p_s$. Because $J_s \neq 0$ the other P_λ are not $p_s Q_\lambda$, as on the tilted circle of Section 27; both have to be found separately, and the notes do not compute them.

The angle $\theta = 2\arctan x$ shows this as the circle it is. Itô's formula gives the box, with bounded coefficients. At $\theta = \pi$, the image of both infinities, the noise vanishes like $(\theta - \pi)^4$ while the drift is 2. $\sqrt{B}$ is Lipschitz there, so [G (5.2.20)] holds, and Gardiner's sign rule makes $\theta = \pi$ an exit for the arc below it [G (5.2.22)] and an entrance for the arc above [G (5.2.23)]. On the circle the two are one point, which the particle crosses without stopping, and the density there is finite, $J_s/2$ (Fig. 37c). The point is special because B vanishes there, a property of the coefficients and not of the topology (Section 1.9). The cut-invariant formula Eq. (1.19) assumes $B > 0$ and does not apply. No probability diffuses back through $\theta = \pi$, so the integral stops there, as in the box.

38.3 Regular at infinity: a choice

Infinity can also be regular. Take the free circle, or constant drift on it (Sections 8 and 17), with circumference 2π, and look at it through the same coordinate, $x = \tan(\theta/2)$. Then $B = \frac{1}{4}D(1+x^2)^2$ and $A = \frac{1}{2}v(1+x^2) + \frac{1}{4}Dx(1+x^2)$, both infinite at $\pm\infty$. Gardiner's $f \propto e^{-(4v/D)\arctan x}/(1+x^2)$ and $g \propto e^{(4v/D)\arctan x}/(1+x^2)$ are both integrable, so $\pm\infty$ are regular: reached in finite time, and left again. A condition must be chosen, and Gardiner's two are the natural ones. [G (5.2.32)], $J(\pm\infty) = 0$, is the interval $(-\pi, \pi)$ with reflecting walls (Sections 5 and 14). [G (5.2.33)] is the circle, on which p_s is the Cauchy density $1/\pi(1+x^2)$, carrying the current $J_s = v/2\pi$. Gardiner's description of the two as "the limits of reflecting and periodic boundary conditions" is exact.

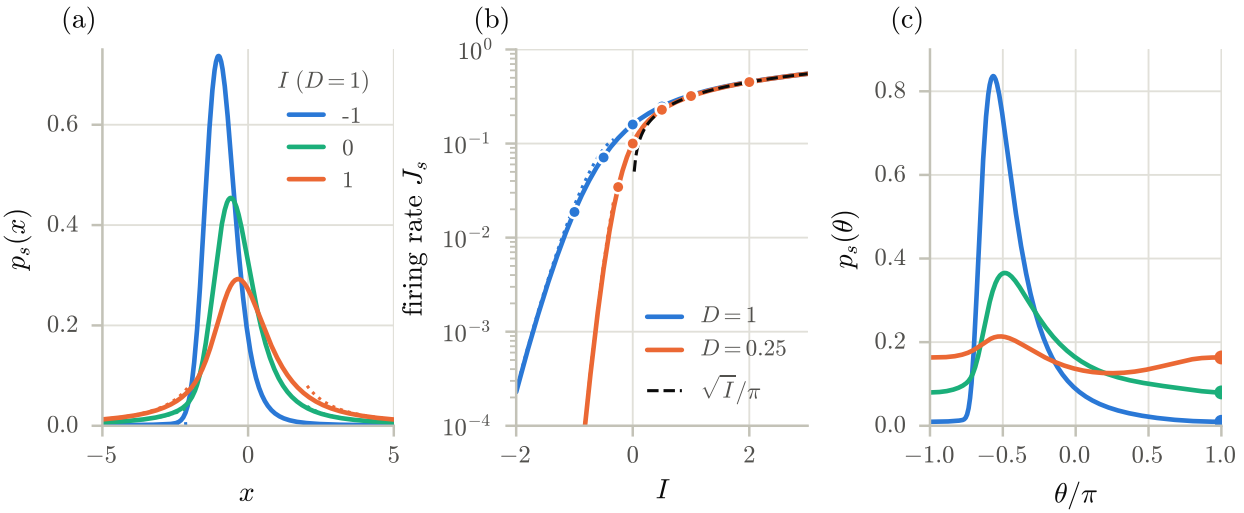


Figure 37: A current through infinity: the QIF neuron. (a) Stationary densities for $D = 1$; dotted, the tails J_s/x^2. (b) Firing rate against I: exact (solid), the noiseless $\sqrt{I}/\pi$ (dashed), and Kramers' formula for $I < 0$ (dotted). Dots are simulations of the θ equation. (c) The densities of (a) in the angle $\theta = 2\arctan x$. Both infinities are the point $\theta = \pi$, where the density is $J_s/2$ (dots).

39 Feller's classification across the notes

Every boundary of a one-dimensional diffusion, at a finite point or at infinity, is of one of four types. Two questions decide the type: can the process reach the boundary from inside, and can it start there and leave? In Gardiner's notation [G (5.2.25)]–[G (5.2.29)] the first holds when $h_1 = f\int g$ is integrable near the boundary, the second when $h_2 = g\int f$ is, with $f = 1/\psi$ and $g = 2\psi/B$ (Section 37.1). Table 9 lists what each type means and where it appears in these notes.

Type	Reached? Left?	Condition	Examples in these notes
regular	yes, yes	must be chosen: reflecting, absorbing, or gluing to another boundary [G (5.2.33)]	every wall of Parts I–III; the origin in §30 and §37 for $0 < d < 2$ (for $d = 1$, §21 and §22); the walls of §32 for $\lvert s\rvert < \frac{1}{2}$; $\pm\infty$ for a circle in the coordinate $\tan(\theta/2)$ (§38)
exit	yes, no	none possible: absorbing	the origin in §37 for $d \le 0$ (branching diffusion, Gardiner's $X + A \rightleftharpoons 2X$, Manning condensation); the walls of §32 for $s \le -\frac{1}{2}$ (neutral Wright–Fisher); $+\infty$ for the QIF neuron
entrance	no, yes	none possible	the origin in §30 and §37 for $d \ge 2$; the walls of §32 for $s \ge \frac{1}{2}$; $-\infty$ for the QIF neuron; $+\infty$ for $A = -kx^3$
natural	no, no	none possible	$\pm\infty$ for free motion, constant drift, OU, $\ln\cosh$ and the double well; ∞ in §30 and §37; 0 and ∞ for geometric Brownian motion

Table 9: Feller's four types of boundary, with the examples in these notes.

What the table says.

- **A wall condition is a choice only at a regular boundary.** Every wall of Parts I–III is a finite point where $B > 0$ and A is bounded, so it is regular, and the notes chose reflection

or absorption there. At the other three types the equation decides by itself, and imposing a condition is either redundant or impossible.

- **Stationary states and arrival times.** Near an exit the zero-current solution is not normalisable, and the mean time to arrive is finite. Near an entrance it can be normalised, and the mean time to arrive is infinite. A regular boundary allows both. This is Gardiner's remark after [G (5.2.29)], with its misprint corrected.
- **QSDs.** A unique QSD that attracts every start needs the far end to be an entrance (Section 1.8). OU and the square-root process have natural boundaries at infinity, hence their continua of QSDs.
- **A change of variable does not change the type.** $y = r^2$ (Section 37), Lamperti's transformation and $x = \tan(\theta/2)$ (Section 38) move a singularity from the drift to the noise, or out to infinity, but the integrals of [G (5.2.25)]–[G (5.2.28)] do not change. A boundary can look ordinary in one coordinate ($\theta = \pi$ on a circle) and singular in another ($x = \pm\infty$).
- **The partner swaps exit and entrance.** With $B = D$, $U \to -U$ turns f into $\frac{D}{2}g$ and g into $\frac{2}{D}f$, so h_1 and h_2 trade places; regular and natural boundaries keep their type. Examples: the logarithm in dimension d and in dimension $2 - d$; the walls of $-a \ln\cos$ with s and $-s$; the two ends of the QIF neuron. The partner also shares every nonzero rate (Section 29). So neutral Wright–Fisher ($s = -\frac{1}{2}$, absorbed at both ends) relaxes at exactly the rates of rotational diffusion on the sphere ($s = \frac{1}{2}$), $\frac{D}{2w^2}n(n+1)$ with $n \geq 1$, but has no zero mode: this is Kimura's spectrum.

Gardiner's sign rule. Where B vanishes at a but $\sqrt{B}$ stays Lipschitz, so that $B \simeq c\,(x-a)^2$, the sign of $A(a)$ decides the type [G (5.2.22)]–[G (5.2.24)]. The standard example is $\mathrm{d}x = (a_0 - bx)\,\mathrm{d}t + \sigma x\,\mathrm{d}W$ on $x > 0$. The logarithm $y = \ln x$ reduces it to earlier sections. By Itô's formula $\mathrm{d}y = (a_0 e^{-y} - b - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma\,\mathrm{d}W$: constant noise in the potential $U = a_0 e^{-y} + (b + \frac{1}{2}\sigma^2)y$.

- $a_0 = 0$: geometric Brownian motion, which is constant drift on the line (Part II). $x = 0$ is $y = -\infty$, a natural boundary [G (5.2.24)].
- $a_0 > 0$: the Morse potential of Table 8. Its exponential wall sends the particle back from $y = -\infty$ in finite time, so $x = 0$ is an entrance [G (5.2.23)]. The stationary density in x is the inverse gamma density $\propto x^{-2-2b/\sigma^2} e^{-2a_0/\sigma^2 x}$.
- $a_0 < 0$: the wall becomes a cliff, and the particle falls to $y = -\infty$, that is to $x = 0$, in finite time: an exit [G (5.2.22)].

Where $\sqrt{B}$ is not Lipschitz, as for $B \propto y$ in Section 37, only the full test decides.

Which examples to use. To demonstrate the classification, in this order:

1. Section 37. One parameter, d, moves the origin through exit, regular and entrance; the exact propagators show what each type allows; infinity is natural.
2. Section 32. The same three types at both walls at once, as s varies, each with a physical case: neutral Wright–Fisher (exit), the reflecting box as $s \to 0$ (regular), and rotational diffusion on the sphere (entrance).
3. Section 30 with $d = 1$, that is Sections 21 and 22: the two choices at a regular boundary, side by side.
4. Section 38. Exit and entrance at infinity, the reinjection [G (5.2.33)], and Gardiner's sign rule in the θ coordinate; through the same coordinate, a regular boundary at infinity.
5. Geometric Brownian motion and the Morse potential (above): Gardiner's sign rule, and natural boundaries at a finite point.

Part VII
How the problems reduce to each other

40 Reductions: a summary

Almost every exact result in these notes was obtained by transforming the problem into an earlier one, until it became the Gaussian of Section 2. This section collects the transformations (Table 10), shows how they chain together (Fig. 38), and gives a procedure for a new problem.

40.1 The idea: change variables until the problem is one you know

Every reduction is one of three kinds of move.

- **Change the unknown:** $p \to e^{\text{something}} p$. This removes drift, at the cost of a shift in the rates and a change in the wall conditions. Examples: the gauge, Doob's transform.
- **Change the variables:** $x \to f(x,t)$ and $t \to \theta(t)$. This removes drift or makes B constant, at the cost of moving walls or changing the clock. Examples: the boost, Gardiner's map, Lamperti, polar coordinates.
- **Use a symmetry of the domain** to build the solution from copies of a simpler one. This costs a sum over copies, or a restriction to one symmetry sector. Examples: images, the cover, folding, cutting.

For each move, three things matter: what it needs, what it does to the wall conditions, and what it does to the rates.

The model case: $A = v$ reduces to $A = 0$ (Section 10). Write $p = e^{\alpha(x-x_0)-t/2\tau}u$ with $\alpha = v/D$.

- *What it needs:* A and B constant.
- *What it does to the rates:* u obeys the free equation, so every rate gains the constant $1/2\tau$.
- *What it does to the walls:* an absorbing wall stays absorbing ($u = 0$). A reflecting wall becomes the Robin condition $u' = \alpha u$, which the images of Part I cannot handle. That is why the mixed interval with drift ends at an eigenvalue condition instead of an image sum.
- *Why it works:* it is the symmetrisation $p = \sqrt{p_s}\,\psi$ of Section 1.5, and for constant A the potential $V = A^2/2D + A'/2$ is the constant $1/2\tau$.

The same idea explains why OU does not reduce this way. Symmetrising OU gives a harmonic V, not a constant one, so the free problem is not reached; Gardiner's map, a change of variables, is needed instead. And on a circle the gauge factor is not periodic, so a boost does the job.

40.2 A procedure for a new problem

1. **Physics to equation.** For a particle with inertia, check that it is overdamped, $m|\mathcal{U}''| \ll \gamma^2$ (Section 1.2). Otherwise the problem lives in (x, v) and the rest of these notes does not apply directly.
2. **Classify the boundaries** (Table 9). Only a regular boundary needs, or allows, a condition; at an interface, decide which density is continuous (Section 36).
3. **Make B constant** with the Lamperti change of variable (Section 34).
4. **Use the symmetries of the domain.** A mirror symmetry means images, or folding into sectors; a periodic domain means the cover; a target point means cutting there.
5. **Symmetrise and read off** $V = U'^2/2D - U''/2$ (Section 1.5).

Move	What it does	Needs	Walls, rates	Where
Images	sum of mirrored copies: $+$ at a reflecting wall, $-$ at an absorbing one	drift odd under the mirror	keeps cos or sin modes	3–7, 21
Cover	circle = line summed over windings, all $+$	periodic A, B	—	8, 17
Folding	symmetric problem = even sector (reflecting) $\oplus$ odd sector (absorbing)	A odd, B even	halves the domain	7, 8, 26
Cutting	target on a circle = absorbing interval; first winding = interval on the cover	—	—	1.9
Gauge (symmetrise) $p = \sqrt{p_s}\,\psi$	Fokker–Planck $\to$ Schrödinger, $V = \frac{A^2}{2D} + \frac{A'}{2}$	$B = D$, $J_s = 0$	absorbing $\to \psi = 0$, reflecting $\to \psi' = \frac{A}{D}\psi$	1.5, Part II
Boost $x \to x - vt$	removes a constant drift	no walls	none	11, 17
Map $y = xe^{kt}$, clock $\theta(t)$	OU $\to$ free diffusion	walls at $x = 0$ only	none	19
Partial image	interface: image of weight r, transmitted copy $1 - r$	free motion on each side	$r = \pm 1$ are the walls	36
Change of variable (Itô)	any $B \to$ constant (Lamperti); $r^2 \to$ CIR; sin $\to$ Wright–Fisher; $\ln x$: geometric BM $\to$ drift, Morse	invertible map	walls to walls; Feller type kept	34, 30, 32, 37, 39
Chart $\theta = 2\arctan x$	line with a current through infinity = circle	—	$\pm\infty \to$ one point	38
Projection	radius of d-dim motion = motion in $-c\ln r$	isotropy	origin unreachable for $d \geq 2$	30
Doob transform	condition on survival or an exit: drift $A + B\,\partial_x \ln h$	$h > 0$ backward eigenfunction or π	rates shift by λ_h; absorbed $d \leftrightarrow$ reflected $4 - d$	4, 6, 30, 37
Partner $U \leftrightarrow -U$	same nonzero rates; ladders by algebra	$B = D$, matching walls	backward $Q \to p_s Q'$; exit $\leftrightarrow$ entrance	19, 29, 39
Adiabatic elimination	(x, v) Kramers equation $\to$ Smoluchowski	$m\lvert\mathcal{U}''\rvert \ll \gamma^2$	reflecting stays reflecting	1.2
Tilted generator	winding statistics $\to$ one eigenvalue $\Lambda(s)$	periodic coefficients	—	1.9
Backward equation	first-passage questions $\to$ an equation in the start x_0	—	$p = 0 \leftrightarrow f = 0$, $J = 0 \leftrightarrow \partial f = 0$	1.7

Table 10: The reductions used in these notes. "Where" gives the main sections.

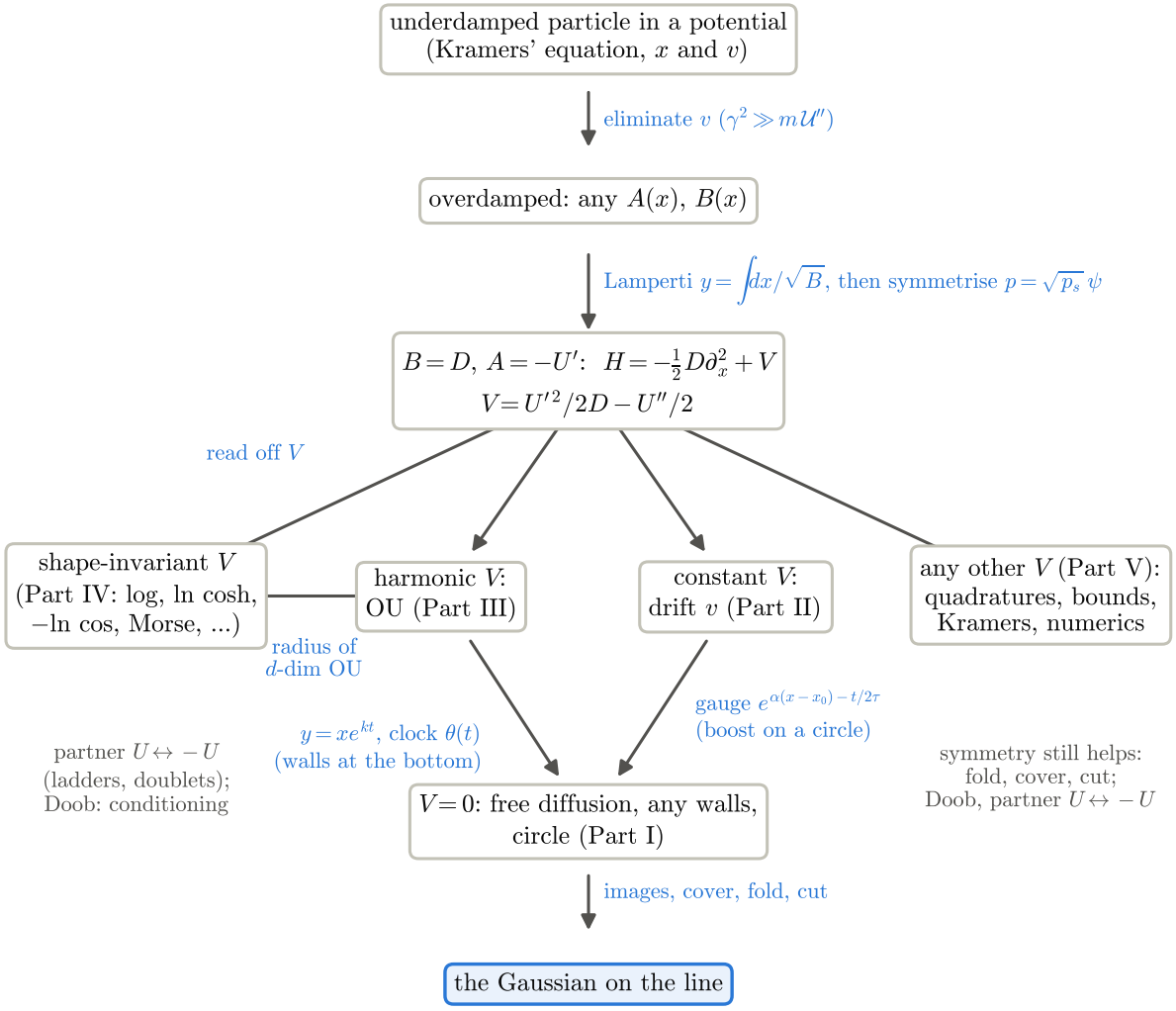


Figure 38: How the problems reduce. Each arrow points to a simpler problem and is labelled by the move that gets there. Everything that is solvable in closed form ends at the Gaussian.

- V constant: the free problem (Part I), with every rate raised by V and Robin walls. Without walls, a boost does the same with no cost.
- V harmonic: OU. With walls only at the bottom, map to the free problem; otherwise use Kummer functions.
- V shape-invariant (Table 8): the spectrum follows by algebra.
- Anything else: exact quadratures for p_s and every first-passage quantity, bounds on the gap, small-noise ladders and Kramers rates, and numerics (Section 34).

6. **Transform the question as well as the problem.** Conditioned paths call for a Doob transform. A question about $-U$ calls for the partner. First-passage questions go to the backward equation. Winding on a circle goes to the tilted generator.
7. **Track the walls and the rates** through every step, using Table 10. A wall that moves under a map, or turns into a Robin condition, is where a closed form usually stops.

40.3 Chains back to the Gaussian

Each line follows one problem down to something already solved:

- drift toward an absorbing wall (Section 13) $\xrightarrow{\text{gauge}}$ free absorbing half-line (Section 4) $\xrightarrow{\text{one image}}$ Gaussian;
- OU absorbed at the bottom (Section 22) $\xrightarrow{\text{map}}$ free absorbing half-line $\xrightarrow{\text{image}}$ Gaussian;
- radial OU in three dimensions (Section 30) $\xrightarrow{\text{Doob, } h=x}$ OU absorbed at the bottom $\xrightarrow{\text{map, image}}$ Gaussian; or directly, as the projection of the 3D Gaussian;
- square-root process absorbed at the origin (Section 37) $\xrightarrow{y=r^2}$ radial motion in dimension d $\xrightarrow{\text{Doob}}$ radial motion in dimension $4-d$ $\xrightarrow{\text{projection}}$ Gaussian (for integer d; otherwise the chain ends at the Bessel function);
- step in D (Section 36) $\xrightarrow{\text{partial image}}$ Gaussian;
- constant drift on a circle (Section 17) $\xrightarrow{\text{boost}}$ free circle (Section 8) $\xrightarrow{\text{cover}}$ Gaussian;
- free circle $\xrightarrow{\text{fold}}$ reflecting interval of half the length (Section 5) $\xrightarrow{\text{images}}$ Gaussian;
- mixed interval (Section 7) $\xrightarrow{\text{unfold}}$ absorbing interval of length $2L$ (Section 6) $\xrightarrow{\text{images}}$ Gaussian;
- OU on a circle (Section 26) $\xrightarrow{\text{fold}}$ OU reflecting and absorbing intervals (Sections 23 and 24), related by the partner ladder. The chain stops here at Kummer roots, because the far wall moves under the map;
- mixed interval with drift (Section 16) $\xrightarrow{\text{gauge}}$ free problem with a Robin and a Dirichlet wall. The chain stops at the root condition $\tan kL = -k/\alpha$;
- double well (Section 35) $\xrightarrow{\text{symmetrise}}$ a Schrödinger operator with no closed form. The chain ends in numerics, or in small-noise asymptotics: local OU ladders and Kramers rates.

A Figures and numerical checks

The script `scripts/make_figures.py` needs numpy, scipy and matplotlib (the conda env `research` has all three). Run from the `fpe-examples` folder,

```
python scripts/make_figures.py    or    python scripts/make_figures.py mixed line
```

to redo every section or just the named ones (the keys of `SECTIONS` in the script). It writes the figures to `figures/` and prints a list of checks. All of the following pass.

- Image sum = mode sum for p on every interval, at $Dt/L^2 = 0.02$ and 0.2 (to 10^{-15}).

- Normalisation, $\langle x\rangle$ and $\langle x^2\rangle$ series vs. quadrature of p; $C(0) = L^2/12$; folded-normal $\langle x\rangle$ and $\langle x^2\rangle = x_0^2 + Dt$ on the half-line.
- S series vs. quadrature of p; image forms of the exit fluxes vs. $-\partial_t S$ series; $S = \operatorname{erf}(x_0/\sqrt{2Dt})$ vs. quadrature.
- $\int f\,\mathrm{d}t = 1$ (or $1 - S$ on the half-line), $\int j_L\,\mathrm{d}t = \pi_L$, and T, $\operatorname{Var} T$, T_0, T_L from quadrature of the fluxes vs. the closed forms; mode and median of the Lévy–Smirnov density.
- $\int q_\infty T\,\mathrm{d}x_0 = 1/\lambda_{\min}$ for every QSD on an interval; the Rayleigh limit at $Dt/x_0^2 = 10^4$; the resolvent Eq. (2.4).
- Drift (Part II), for several v of both signs: the PDE residual of each propagator by finite differences (below 10^{-4} relative); $J = 0$ at reflecting walls and $p = 0$ at absorbing ones; normalisation; $\langle x(t)\rangle \approx x_0 + vt$ at $Dt/L^2 = 10^{-3}$, which checks the initial condition; mode sums vs. quadrature; π_L, T, T_0, T_L from quadrature of the exit fluxes; T_L even in v; inverse-Gaussian moments and escape probability; S against its $t^{-3/2}e^{-v^2t/2D}$ asymptote; and the eigen-equation residual of the half-line QSDs.
- Ornstein–Uhlenbeck (Part III):
  - the map Eq. (19.2) against the line and half-line propagators (to 10^{-15});
  - the Mehler sum against the Gaussian;
  - PDE residuals, wall conditions and normalisation for every geometry;
  - survival, T and π_L from the closed forms against quadrature, against $\sum_n c_n/\lambda_n$, and against the integrated exit flux;
  - $\int q_\infty T = 1/\lambda_{\min}$;
  - the reflecting-equals-absorbing-plus-k ladder (to 10^{-15});
  - the $L \to 0$ and $L \to \infty$ limits of every slowest rate;
  - $\lambda_0 T(0) \to 1$ for Kramers;
  - the constant $\ln 2 + \gamma_{\mathrm{E}}/2$;
  - the heavy-tailed QSDs q_ν: normalisation, eigen-equation residual, tail exponent $-1-\nu$, and exact evolution as $e^{-\nu kt}q_\nu$ through the map.

  The interval eigenvalues come from root-finding on the Kummer functions Eq. (19.4) with `scipy.special.hyp1f1`. They were checked against `mpmath` to machine precision.
- The physical model (Section 1.2): the closed-form position correlation of the damped oscillator against the regression equation, integrated numerically, for $\zeta = 0.25$, 1, 3 (to 10^{-11}); the slowest rate at critical damping ($2\mathcal{K}/\gamma$) and at $\zeta = 10$ against the corrected Smoluchowski value $1 + 1/4\zeta^2$; and the Maxwell–Boltzmann density as a stationary solution of Kramers' equation in a double well.
- The symmetrised operator Eq. (1.11), solved by finite differences (4000 points, Robin walls by ghost points), against the spectra found from the forward equation: the drift interval with reflecting walls; OU intervals with reflecting and absorbing walls (Kummer roots); the cosine circle at $\kappa = 3$ and 16 (Fourier recurrence); and the sawtooth circle with the δ-spike at the cusp. All agree to about 10^{-5}, relative.
- Part IV: the spectra of all six shape-invariant potentials of Table 8 from finite differences of H (the two not worked out included, to 10^{-3} or better); the partner identities $\tilde{V}(s) = V(s') + R(s)$ (to 10^{-12}); the radial propagators against the angular integral of the 2D Gaussian, the $d = 1$ images, the $d = 3$ Doob form and the Laguerre series (to 10^{-12}), their PDE residual, and the recurrence probability; the Jacobi and Gegenbauer eigenfunctions (residuals), the normalisations, the Gegenbauer propagator, $\langle \sin(x/w)\rangle$, the Bakry–Émery and variance bounds, the Wright–Fisher transformation, and the $s = \frac{1}{2}$ case as rotational diffusion on the sphere (drift, Legendre eigenfunctions, Debye rate).
- Part V, the double well, with the detailed-balance jump discretisation: convergence of λ_1; λ_1

against [G (9.2.18)] (to 2×10^{-5} at $\beta = 12$) and Kramers (to 2.5 % at $\beta = 16$); $\lambda_1 T/2 = 1$ with T from [G (5.2.160)]; $\lambda_2 \to |U''(0)|$ and $\lambda_3 \to U''(a)$ at $\beta = 80$; $Q_1 = 2\pi_c - 1$; and the two-state population law.

- Part VI:
  - the interface: normalisation, the jump by K, continuity of the flux and the free equation on each side, for four values of K; the smoothed step, solved by finite differences (Crank–Nicolson), against the partial image for all three readings, with an error proportional to ε (about 1 % at $\varepsilon = 0.01a$); π_+ and T against Gardiner's quadratures; $T(0)$ against the three means; the box rates against a jump process with exact detailed balance; and simulations of the smoothed SDE;
  - the square-root process: normalisation of p_+; $\int p_- = P(1 - \frac{d}{2}, z_t)$ for five values of d, and the extinction probability; PDE residuals; p_+ against the radial propagator, and the Doob relation (both to 10^{-15}); $J(0) = 0$ when reflected and $J(0) = \dot{S}$ when absorbed; the power laws at the origin; T against [G (5.2.161)]; both Laguerre ladders and the decay rate of S; the QSDs (residual, normalisation, positivity, tails) and the Yaglom limit; the exact sampler behind the figure;
  - the QIF neuron: $1/J_s$ against [G (5.2.160)] on a grid; its three limits; normalisation and tails of p_s; the density at $\theta = \pi$; the Itô form of the θ equation; simulations of θ at eleven parameter values; and, for the drifting circle through $\tan(\theta/2)$, the Cauchy density's constant current and the integrability of f and g;
  - Feller's test [G (5.2.25)]–[G (5.2.29)], with shrinking cutoffs, for the square-root process in y and in r, Gardiner's reaction model and geometric Brownian motion; the map from x to $\ln x$ onto the Morse drift; and the rates of neutral Wright–Fisher from finite differences.
- Gardiner's P_λ and Q_λ as named in the boxes (section `pq` of the script):
  - biorthonormality $\int P_n Q_m = \delta_{nm}$ [G (5.2.79)], with $P = p_s Q$, for the free and drift intervals, the circle (with and without drift), Hermite on the line and half-lines, the Kummer intervals, the sawtooth circle (both sectors together), the Laguerre ladder of Section 30, the bound states of $\ln\cosh$, the normalised Gegenbauer Q_n, the interface box ($\chi = 1$, 0) and both Laguerre ladders of the square-root process (also at an exit, $d = -0.5$);
  - forward and backward eigen-equation residuals on the intervals and circles, and on the drift mixed interval the wall conditions $J(P) = Q' = 0$ at 0 and $P = Q = 0$ at L;
  - $\sum P_n(x) Q_n(x_0) e^{-\lambda_n t}$, or its integral over a continuous label, against the propagators of the line (free and drifting, gauge and Fourier modes), the half-lines (free; drift, with the Robin continuum and the bound state $P_0 = p_s$), OU on the half-lines, the sawtooth circle, the Bessel process and $p_\pm$ of the square-root process, to 10^{-6} or better;
  - on the cosine circle, left and right eigenvectors of Eq. (27.7) as Q_λ and P_λ; P_λ/p_s is a backward eigenfunction without tilt and is not with tilt;
  - the double well: $\int p_s (2\pi_c - 1)^2 = 1$ to 10^{-3}, the normalised Q_1 against $2\pi_c - 1$, and the hilltop mode at $\beta = 80$.
- Which QSD is reached (Section 1.8): for drift toward the wall, an exponential start $e^{-x/2\ell}$ converges to the QSD with the same tail and decays at $\beta|v| - \frac{1}{2}D\beta^2$, while $e^{-2x/\ell}$ converges to the edge QSD with an error $\propto 1/t$. For OU, a start with tail $x^{-3/2}$ converges to $q_{1/2}$ with $S \propto e^{-kt/2}$, while a tail $x^{-5/2}$ (finite mean) converges to the Rayleigh QSD.
- Circles, in all three parts:
  - image sum against mode sum, normalisation and PDE residuals; the fold onto the reflecting interval of half the length (to 10^{-15});
  - $\langle e^{iq_1 x} \rangle$ against the complex eigenvalue; $J_s = v/L$; the twisted periodicity of the gauged u;

[G (5.2.46)] as printed equals -2 times the corrected form;
- the cut-invariant p_s Eq. (1.19) against [G (5.2.47)] with the seam at 0 and against the Fourier null vector, with and without tilt;
- v_{w} and D_{eff} three ways: LJ_s; $\Lambda'(0)$ and $\Lambda''(0)$ from Eq. (27.7); and an independent jump-process discretisation of the generator on a ring of 4000 sites;
- Lifson–Jackson against L^2/T_{w} from [G (5.2.158)] on the cover; T_{w} independent of the start; the Einstein relation; the Kramers limit;
- for the sawtooth: continuity of J at the cusp, the splitting by exactly k, and $D_{\mathrm{eff}} = \frac{1}{2}L^2\lambda_0^{\mathrm{mix}}$ at $L = 8\ell$;
- for the cosine well: the exact doublets, the $\kappa \to 0$ and $\kappa \to \infty$ limits, and a simulation of the tilted lift (v_{w} and D_{eff} within statistical error).

Simulation: Euler steps of $\mathrm{d}x = \sqrt{D}\,\mathrm{d}W$ with $\Delta t = 10^{-4}$ or $2 \times 10^{-4}\,L^2/D$, plus a Brownian-bridge test for crossings within a step (crossing probability $e^{-2x_a x_b/D\Delta t}$ for the wall at 0). A reflecting wall at 0 folds the path, $x \to |x|$, which is exact. With 4000 paths per starting point, T and π_L agree with the closed forms to within about two standard errors.

To add a figure, write a `fig_*` function in the script, register it in `SECTIONS` with its checks, and `\includegraphics` the PDF. Screenshots from `sde-explorer` can go in `figures/` too.

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